A minimal introduction to probability theory, statistical inference & hypothesis testing

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Preface

The purpose of these notes

In the following pages one will find an introductory course to the theory of probability and statistical inference, aiming to cover both foundations and basic mathematical concepts, but also practical tools to deal with real data science problems, such as bayesian probability and hypothesis testing. The text is composed by seven chapters, together with some appendix sections reviewing basic mathematical notions, and a bibliographic note. The purpose of these lecture notes is to make both probability and statistical analysis an easy, engaging and exciting topic for anyone interested, without the need for prior experience.

Both, predictive probability and descriptive statistics have deep historical roots, from ancient works on chance and divination to modern scientific topics oriented towards information theory, modelling and data analysis. As one could guess, rivers of ink have been written about such topics, and endless literature sources are available. However, after following many different courses at both bachelor and postgraduate levels, and teaching such topics myself during the last three years, I have found that most resources belong, almost certainly, to one of the next three classes. Either (*i*) deeply mathematical, and hence out of reach for most experimental or clinically oriented scientists, (*ii*) laboratory oriented, focusing on inference and experimental design, and hence missing most of the mathematical background, or (*iii*) with a direct focus towards programming and computation, relying on domain specific notebooks (Python, R, Matlab, SPSS, etc), and online resources with precompiled libraries for simulation, which again miss most of the mathematical and formal intuitions. Indeed, the misuse of statistics in experimental sciences is a critical topic in modern times, as mathematicians have extensively discussed during the last decades. The well-known article by John P. A. Ioannidis, *"Why most published research findings are false"* [1], serves as a prominent example, and it may serve as motivation for a rigorous study.

As a matter of fact, when it comes to modern statistics, data analysis or experimental design, concepts like *stochasticity, randomness, sampling, hypothesis, significance, statistic test, p-value*—just to mention some of them—are frequently used, but for most bachelor and even master's level degrees they are rarely introduced or properly defined. Indeed, for most experimental and clinically oriented degrees, they are not introduced at all, leaving the student with just a superficial knowledge relying on intuition about some particular cases. Hence, developing high-quality, simple, and accessible open source material for present and future generations, covering both probability and statistical inference from both a fundamental *and* applied level, remains an urgent task for scientists and educators.

This is intended to be a complete introductory course, and no previous mathematical background is required. By keeping the theory simple and always followed by examples, we will build the definitions and quantities from simple to more complex. All mathematical formulas will be introduced with rigorous notation, but keeping in mind that it is not the symbols or the numbers, but the intuitions and the general understanding, what we are after. Additionally, all topics will be introduced alongside with some short historical discussion and context, as we believe that a purely technical knowledge just grasps the complexity—and deep philosophical insterest—of scientific topics. As one could anticipate already, a proper understanding of ideas such as uncertainty, variation, chance, probability and inference can be applied to describing a vast amount of real-world phenomena, ranging from gambling and games of chance to data analysis and modelling in biology and clinical sciences, but also physics, quantum mechanics, and machine learning, among many others.

As mentioned, the course is organised in seven chapters, covering the topics listed below:

- Chapter 1: Descriptive statistics, sampling, data types and estimators.
- Chapter 2: Fundamentals of probability theory and random variables.
- Chapter 3: Estimation, confidence and critical regions.
- Chapter 4: Introduction to hypothesis testing.
- Chapter 5: Modelling, dependence and correlation.
- Chapter 6: Introduction to Bayesian probability.
- Chapter 7: Introduction to stochasticity and Markov processes.

At the end of each chapter there will be a series of exercises and coding examples to illustrate and demonstrate the concepts discussed. To avoid misconceptions, let us emphasize here that both, probability and statistics are just branches of mathematics dealing chance and information in random events, *much earlier* than computers, coding languages, Python, R or P-values were even conceived. The data-oriented, practical ways in which probability and statistics are usually taught, relying heavily on computation, is just a consequence of the fact that automatized measurements are nowadays available and trendy in modern times, with simulation at our reach.

Example textbooks covering introduction to probability and statistical inference in a more technical, applied, mathematical or philosophical level, for further reading:

- A simple, intuitive introduction to statistics with few mathematical concepts is provided in Spiegelhalter's *"The Art of Statistics: How to Learn from Data"* [2].
- A more foundational textbook, with more advanced mathematical approach, can be found at DeGroot and Schervish's *"Probability and Statistics"* [3].
- For a philosophical and historical perspective on probability and statistics, please find Forster and Bandyopadhyay's handbook *"Philosophy of Statistics"* [4].
- A comprehensive introduction with focus on practical applications and modern data analysis tools is can be found at Diez, Barr & Mine *"OpenIntro Statistics"* [5].
- For fundamental concepts in probability and statistics, including random variables, distributions and statistical inference, with practical examples and exercises follow Hossein Pishro-Nik's *"Probability, Statistics & Random Processes"* [6].

Introduction

Even fire obeys the laws of numbers.
— J.B. Joseph Fourier

A bit of history

As one might expect, the origins of probability and related concepts can be traced back to very ancient times. Civilizations such as the Babylonians, Egyptians, and Greeks already encountered uncertainty in various aspects of life, including commerce, games of chance, and divination. Consequently, notions of randomness and stochasticity have deep historical roots. For instance, archaeological findings suggest that the earliest known dice date back over 5,000 years, reflecting humanity's early fascination with chance and unpredictability [7]. Although these cultures had not yet developed a formal mathematical theory of probability, they recognized recurring patterns in random events and attempted to anticipate outcomes through either empirical observation or superstition. For a detailed historical overview, see F. N. David's 1962 manuscript *"Games, Gods and Gambling"* [8].

While classical Greek and Roman philosophers frequently discussed the nature of chance, necessity, and determinism, their inquiries remained primarily philosophical rather than mathematical. From Aristotle to Cicero, many have distinguished between events occurring by chance and those determined by fate, foreshadowing later developments in probability theory [9]. These early ideas, though lacking quantitative formalism, provided the intellectual foundation for later scientific inquiry into randomness and causality.

A significant shift occurred during the late medieval and early Renaissance periods, when more rigorous mathematical ideas began to shape. Italian mathematician and gambler Gerolamo Cardano (1501–1576) is normally credited as making the first substantial contributions to the mathematical analysis of chance. His work *"Liber de Ludo Aleae (Book on Games of Chance)"* [10], posthumously published in 1663, is one of the earliest known texts to explore probability through the analysis of gambling problems. However, Cardano's reasoning, while insightful, lacked the symbolic clarity and mathematical rigour of modern probability theory. Readers consulting the original manuscript will notice an ambiguous and sometimes inconsistent symbolic system, quite unlike the formal structures we use nowadays.

The formalization of probability as a mathematical discipline did not occur until the 17$^{\text{th}}$ century, most notably through the seminal correspondence between Blaise Pascal and Pierre de Fermat. Their work, motivated by problems such as finding a fair division of stakes in interrupted games of chance, introduced foundational concepts such as combinatorics, expected values, and variance [11]. These developments paved the way for later contributions by Christiaan Huygens, who in 1657 wrote the first published textbook on probability *"De Ratiociniis in Ludo Aleae (On Reasoning in Games of Chance)"* [12], and Jacob Bernoulli, whose 1713 *"Ars Conjectandi (The Art of Conjecturing)"* remains among the most influential early texts in the field. Their works, along with many others, collectively laid the groundwork for the probabilistic and statistical methods that foreshadow modern scientific reasoning [13, 14].

From the 19$^{\text{th}}$ century onwards, probability theory became increasingly intertwined with statistics and inference, giving rise to the mathematical and empirical frameworks that led to modern scientific analysis of uncertainty. Building on the analytical foundations laid by Laplace—who attempted to unify probability with inference through inverse probability—and Gauss's theory of errors, Adolphe Quetelet—Belgian astronomer and mathematician—played a decisive role in extending probabilistic reasoning to social and biological phenomena. His notion of the statistical individual, or *homme moyen*, framed variation not as noise to be eliminated, but as a fundamental object of study, helping to establish statistics as a distinct discipline concerned with populations rather than isolated events [15, 16, 17].

Within this emerging landscape, Florence Nightingale stands as a central figure in the practical and institutional adoption of statistical reasoning. Through her innovative use of graphical representations and her advocacy for quantitative evidence in public health policy, she demonstrated the power of statistical methods as tools for decision-making under uncertainty [18]. In parallel, Joseph Fourier's work on heat conduction introduced series expansions and integral transforms that, while originally developed in the context of mathematical physics, would later become indispensable in the study of random processes, signal analysis, noise, and diffusion. Although Nightingale and Fourier approached uncertainty from fundamentally different directions—one through empirical data and social reform, the other through mathematical analysis of physical systems—their contributions expanded the scope of probabilistic thinking and helped prepare the intellectual ground for the later development of stochastic processes and statistical physics [19, 20].

A further conceptual leap occurred in the early 20$^{\text{th}}$ century with the work of Russian mathematician Andrey Markov. At a time when probability theory was still largely built on the assumption of independent events, he tried to understand whether the familiar regular patterns of probability—such as averages stabilizing over time—could still arise when events influenced one another. To explore this question, he studied sequences in which the outcome of each step depends only on the immediately preceding one, rather than on the entire past.

This simple idea, later formalized as what are now called Markov chains, showed that randomness and order need not rely on complete independence. Even when successive events are linked, long-term statistical patterns can still emerge under suitable conditions. Markov's work, developed within the strong Russian tradition of rigorous analysis, opened the door to the modern study of systems that evolve over time under uncertainty, from physical processes to language, biology, and many contemporary data-driven models [21].

Markov's investigations inaugurated the systematic study of dependence in stochastic processes and laid the groundwork for much of modern probability theory. The conceptual framework he introduced has since become central to a wide range of disciplines, including statistical mechanics, linguistics, finance, and more recently, machine learning and data science, where Markovian models serve as fundamental tools for modeling sequential and temporal data [22].

As a final note, the modern axiomatic formulation of probability was introduced in the early 20$^{\text{th}}$ century by the Russian mathematician Andrey Kolmogorov. In his 1933 monograph *"Grundbegriffe der Wahrscheinlichkeitsrechnung (Foundations of the Theory of Probability)"* [23], Kolmogorov synthesized classical and frequentist ideas into a rigorous mathematical framework based on measure theory. His axioms remain the standard foundation for probability theory to this day. It may seem surprising that a concept with such ancient origins was not formally axiomatized until relatively recent times, and we will return to Kolmogorov's formulation and its implications in greater detail in further chapters. Nevertheless, philosophical discussions about the interpretation of probability and its relation to the physical sciences—especially in the context of determinism, epistemology, and modern topics such as quantum mechanics—predate Kolmogorov's formulation and continue to evolve to this day.

Chapter 1

Descriptive statistics

Statistics is the grammar of science.
— Karl Pearson

As a first approach to probability and statistics, we should properly define both topics and their main fields of study. Although deeply related, and both historically rooted in *combinatorics*—the mathematical study of counting discrete structures, and how they change—they constitute well-differentiated fields of mathematical analysis. A clear distinction often made is that probability is a *predictive* branch of mathematics, dealing with random events, and aiming to compute expected values for unknown outcomes. On the other hand, statistics can be viewed as a *descriptive* approach to uncertainty, based on sampling finite sets of observations from a given population and constructing informative quantities, called *estimators*, to explore central tendency and variation. Such distinctions have been extensively debated by mathematicians, experimental scientists, and philosophers of science [24].

As a rule of thumb, probability provides a formal language for modelling uncertainty, whereas statistics concerns the epistemic problem of learning from data. In this chapter, we introduce basic ideas of statistical inference such as population, sampling, and estimators of central tendency and variation, together with notions of representation and visualization. The foundations of probability theory, rooted in the works of Bernoulli, Laplace, and Gauss, among others, will be covered in Chapter 2.

A philosophical position often adopted is that statistics is essentially the study of uncertainty, and that the statistician's role is to assist other fields that encounter uncertainty in their work. In practice, statistics is ordinarily associated with data, and it is the link between the variability in the data and the uncertainty inherent in the phenomenon under study that has occupied statisticians [25].

As a final note, let us emphasize how these two approaches can and do coexist in science. It is often said that science proceeds by formulating hypotheses and making predictions, which are then compared and benchmarked against experimental results. While this description applies well to many areas, it is a simplification and not universally accurate. Some sciences—such as Newtonian mechanics, much of physics, and chemistry—rely on building precise models that generate quantitative predictions later tested by experiment. A clear example is the use of Newtonian mechanics to predict where and when a stone will fall when thrown, followed by experimental measurement.

By contrast, a paradigmatic example of an inference-driven scientific theory is Darwinian evolution, which does not aim primarily to predict individual outcomes but rather to reconstruct and explain observed patterns from available evidence. This distinction is worth emphasizing, as definitions of science that focus exclusively on predictive power can be misleading. Different sciences may differ substantially in methods, instrumentation, and conceptual tools, yet they are all equally legitimate in their aims and explanatory roles [26, 27].

1.1 Population and sampling

A large part of the history of science can be seen as a continuous effort to translate observations of reality into precise mathematical terms. Describing natural phenomena in numerical language requires tools that relate one or more relevant quantities—often called *variables*—and explain how they change with respect to one another. The goal of modelling may be, for example, to determine the distance from the Earth to the Sun, estimate the number of stars in the observable universe, or relate the incidence of lung cancer to environmental or behavioral factors, such as smoking.

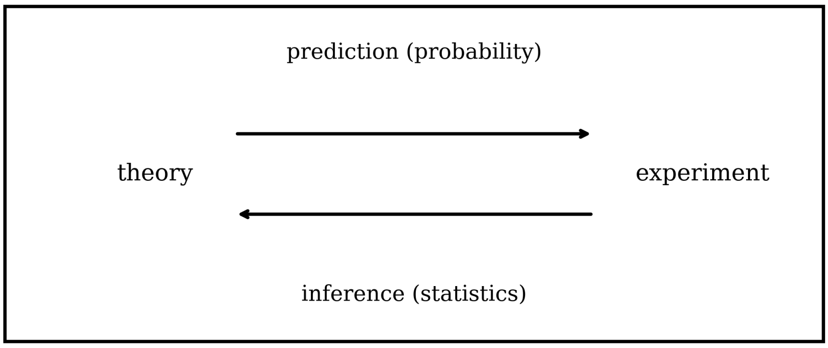


Figure 1.1: Representation of two complementary approaches to randomness and variability: a predictive, theory-driven path from model to experiment, and a descriptive, inference-based path from data to explanation. Probability and statistics are normally framed as the branches of mathematics addressing both ways.

In the same way mathematics is often summarized as the tasks related to *count*, *measure*, and *sort*, statistical problems can be grouped into three broad categories. *Sampling* concerns the selection of a finite set of observations from a larger, typically unknown population. *Estimation* involves constructing numerical summaries that describe how the data are distributed, while *visualization* addresses how observations are represented and how such representations influence interpretation. All three are fundamentally concerned with uncertainty and variability.

Hence, all statistical inquiries begin with observations and measurements, which we normally refer to as *data*. And data begins with the act of selection, or *sampling*. The natural world offers a vast abundance of phenomena, with endless opportunities for observation, but only a finite subset can ever be recorded. This distinction gives rise to two central notions: the *population*, which we denote by $\mathcal{P}$, represents the complete set of all possible observations under study. We will write it as

$$\mathcal{P} = \{x_1, x_2, \ldots, x_N\} \ . \tag{1.1}$$

The *sample* $\mathcal{S}$, on the other hand, is the finite subset actually collected. For a series of N observations x_1, x_2, ..., x_N, a sample of just n elements—less than the total, which is denoted by the upper case N—is defined as

$$\mathcal{S} = \{x_1, x_2, \ldots, x_n\}, \quad n < N \ , \tag{1.2}$$

The population represents the ideal object of inference, while the sample is the concrete, finite evidence available to us. As an example, if I want to study some disease and its relation smokers in a given country, the behavior of a specific population, or the active genes in the genome, I will never have access to the *complete population*, but only to the amount of them that I am able to question, measure, or survey. This distinction is far from trivial. A poorly chosen sample often misrepresents the population and may induce bias, whereas a carefully constructed one mirrors its essential features, and can be used to describe the underlying nature.

Equally important is the recognition that not all data is equal, neither behaves in the same way. A common distinction is to consider *categorical* and *numerical* data. Categorical—or *qualitative*—data describes qualities or labels such as the eye colour of students in a classroom (blue, brown, green), the brand of a purchased smartphone, etc. Sometimes they are further divided into *nominal* categories, with no natural order, like the eye colour or the smartphone brand, and *ordinal* categories with a meaningful order. Examples of these would be the finishing places in a race (first, second, third), survey responses ranging from *strongly disagree* to *strongly agree*, etc.

The other big family is normally referred to as numerical—or *quantitative*—data. It represents numerical quantities and is often subdivided into *discrete*, countable numbers, such as the number of books on a shelf (4, 5, 6) or the number of goals scored in a match, and *continuous* values that can take any number within a range, such as the time a sprinter takes to run 100 meters, or the height of a person measured with some arbitrary precision.

Distinguishing between these types is no mere slang; different types of observations require different mathematical tools, and will be described in different ways. For example, it would not make sense to compute a mean out of smartphone brands, but to compute the mean of their prices is informative. Similarly, the distribution of finishing places after a race might be summarized by a median position, whereas heights of athletes could be studied with averages and measures of spread. A correct classification of data is thus a safeguard against misuse and a guide toward insight.

As a summary, sampling and proper description of data establish the ground upon which statistics is built. Before calculating, summarizing, or diving into inference, one must ensure that the information collected is both *representative and properly understood.* Without these foundations, descriptive measures risk floating unmoored, detached from the reality they claim to represent. Accurate sampling and rigorous description will lead to a faithful representation of the phenomena under study and their relationships, detecting anomalies, and even building accurate predictions.

Let's end this section with a historical note. As we have mentioned, uncertainty has long been associated with games of chance and gambling, but it was not addressed as a statistical problem until much later. The Royal Statistical Society, founded in 1834, together with many other early statistical organizations, was originally established to gather and publish data, with the aim of informing social and economic questions through systematic measurement. It did not take long before statisticians began to ask how such data might best be analyzed and interpreted, and modern *statistical inference* gradually emerged. Among the Society's influential early figures were Adolphe Quetelet, whose work helped establish statistics as a tool for studying social phenomena, and Charles Babbage, who advocated quantitative approaches to scientific and administrative problems [28, 29].

Among its most famous members was Florence Nightingale, admitted in 1858 as the Society's first female member, whose work exemplified the practical use of statistical reasoning in policy and public health. In the early 20^{th} century, foundational questions concerning probability and inference were further clarified through the work of thinkers such as Frank P. Ramsey, who connected probability with rational belief and decision-making [30]. Other notable presidents of the Royal Statistical Society have included William Beveridge and Ronald Fisher, whose contributions will be discussed in Chapter 4.

The famous quote often attributed to Andrew Lang *"most people use statistics as a drunken man uses lamp-posts—for support rather than illumination"*, highlights the tendency to use statistics as a crutch, relying on them for validation rather than seeking genuine understanding. Lang's observation serves as a cautionary reminder to approach statistical data with critical thinking and not merely as a tool to bolster preconceived notions.

1.2 Central tendency and variation

Once we have drawn a clear distinction between the population under study and the selected sample, we immediately face a fundamental problem. Neither the population mean—often referred to as the *true* mean and usually denoted by μ—nor the population variance, known as the *true* variance and written as σ^2, are directly available to us. As we have seen, the only information at our disposal is the finite set of observations contained in our sample. From this limited information, we therefore seek to construct quantities that provide insight into key features of the population, such as its central value or its variability. These quantities are known as *statistical estimators*. Common examples include the *sample mean*, the *median*, the *variance* and the *standard deviation*.

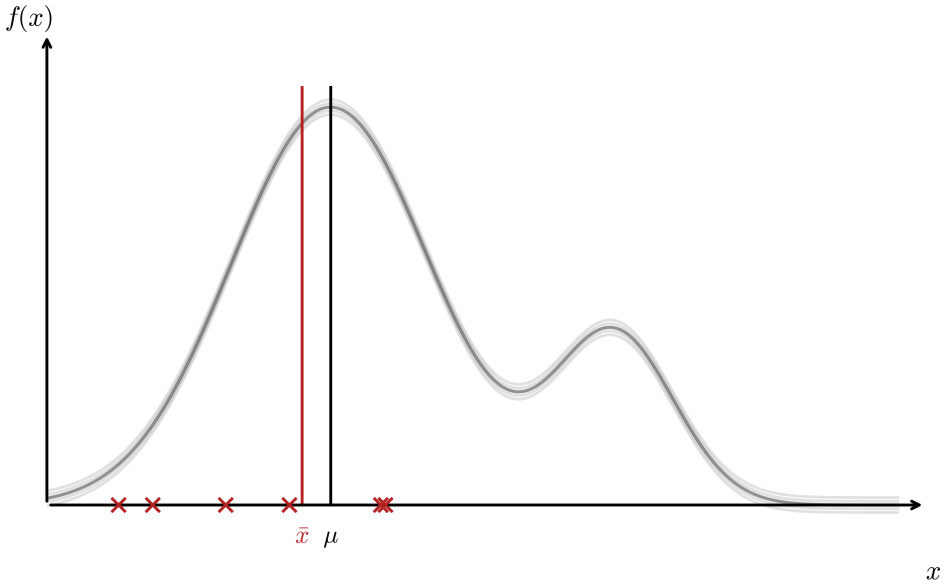


Figure 1.2: Representation of the *true* population mean μ, in black, and the observed *sample* mean $\bar{x}$. The true mean is an ideal and inaccessible quantity, while the sample mean can be computed as an estimator of the finite sample.

Once observations have been collected, a natural question arises: what is the *center*, or *typical*, value of this data set? Measures of central tendency address this question by summarizing the data with a single representative number, offering an immediate sense of where the observations are located within the distribution.

The sample mean:

The *sample mean*, or *average* is perhaps the most familiar measure of central tendency. Imagine we are doing an experiment where we measure some variable, and let's call it x for simplicity. x can be anything we could measure, like position at a given time, energy of some system, concentration of a specific substance, etc. Let's imagine we repeat the measurement n times, and we obtain the values $x_1, x_2, \ldots, x_n$. That will be our set of observations—our *sample*—$\mathcal{S}$. We could simply write it as a list—or a *vector*—in the following way:

$$\mathcal{S} = \{x_1, x_2, \ldots, x_n\} .$$

Keep in mind that from the mathematics perspective the word *vector* has a slightly different meaning, with subtleties related to algebraic operations and relations they should satisfy, but for the purpose of this course, where we prioritize above all simplicity, a vector and a list of numbers will be essentially the same thing.

We can define the sample mean of an arbitrary large sample of n observations, as the sum of all elements divided by the total. We will write it as $\bar{x}$, and define it as follows:

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + ... + x_n) \ . \tag{1.3}$$

We can write this in a slightly more compact way as a *summation*, as follows:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \ . \tag{1.4}$$

Here we denote the sum of all elements x_i with the greek letter $\sum$, starting with the first one (x_1, for $i = 1$) and until the last one (x_n, for $i = n$). The expressions (1.3) and (1.4) mean *exactly* the same thing, just written in different ways.

Let's illustrate with an example. Suppose we repeat a measurement three times, obtaining the results $x_1 = 1$, $x_2 = 2$, and $x_3 = 3$. Our sample is then $\mathcal{S} = \{1, 2, 3\}$, and the sample mean is

$$\bar{x} = \frac{1}{3}\sum_{i=1}^{3} x_i = \frac{1}{3}(1 + 2 + 3) = 2 \ .$$

As a warm-up exercise, try computing the same mean value for a second sample, let's say $\mathcal{S} = \{4, 5, 6\}$. Substituting into the general expression (1.4) gives

$$\bar{x} = \frac{1}{3}\sum_{i=1}^{3} x_i = \frac{1}{3}(4 + 5 + 6) = 5 \ .$$

As we see, the sample mean captures information about the "central" value, where most events cluster. Although useful, it is sensitive to extreme values—often called *outliers*—which motivates the definition of additional, more robust measures of central tendency.

The median:

The *median* represents similar information, as the value that splits the ordered data set in half. For an ordered sample $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$, the median M is defined as

$$M = \begin{cases} x_{(k+1)} \ , & \text{if } n = 2k + 1 \text{ (odd)} \ , \\ \dfrac{x_{(k)} + x_{(k+1)}}{2} \ , & \text{if } n = 2k \text{ (even)} \ . \end{cases} \tag{1.5}$$

Note that here k is just an integer that helps locate the middle position of an ordered data set of size n. If the sample size n is even, we write $n = 2k$, while for n odd, we write $n = 2k + 1$. In the case of an odd-sized sample, the median is just the middle-point, while for an even size, it is computed as the average of the two middle points.

The mathematical definition (1.5) may seem a bit unnatural at first, so let's navigate it with a couple of examples. Consider the sample $\mathcal{S} = \{1, 2, 3, 5, 3, 2, 7\}$. First, we order the data:

$$\mathcal{S}_{\text{ordered}} = \{1, 2, 2, 3, 3, 5, 7\} \ .$$

Since the sample has an odd number of elements ($n = 7$), the median is just the middle value:

$$M = x_{(4)} = 3 \ .$$

Now consider an even-sized sample $\mathcal{S} = \{1, 2, 3, 5, 4, 3, 2, 7\}$. Ordering the data gives

$$\mathcal{S}_{\text{ordered}} = \{1, 2, 2, 3, 3, 4, 5, 7\}.$$

Which has now an even number of elements ($n = 8$). Hence, applying such case in (1.5), the median is the average of the two middle values

$$M = \frac{x_{(4)} + x_{(5)}}{2} = \frac{3+3}{2} = 3 \ .$$

Unlike the mean, the median is robust to outliers and skewed data, capturing the central position of the dataset even with repeated values. To illustrate that, let's have a look at the following sample $\mathcal{S} = \{1, 2, 3, 3, 4, 4, 200\}$, which contains the value 200 as a huge outlier. The sample mean would be

$$\bar{x} = \frac{1}{7}(1 + 2 + 3 + 3 + 4 + 4 + 200) = \frac{217}{7} = 31 \ .$$

While the median, given a size $n = 7$ would just be the middle (4th) value

$$M = 3 \ .$$

For instance, the data represented in LHS of Figure 1.4 will be accurately described by computing the mean, given its symmetric behaviour, while the one in the RHS will be better addressed with a median, accounting for the skewness and the presence of outliers.

The sample variance:

Beyond central location, it is important to understand the *spread* of the data. We define the *sample variance* s^2 of a data set as a quantity that captures how far the elements are from the mean value:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2 \ . \tag{1.6}$$

The factor $n-1$ in the denominator of (1.6) is called the *Bessel correction factor*. A technical explanation is that this correction ensures that s^2 is an *unbiased estimator* of the population variance, a concept we will discuss in Chapter 3. In particular, it reflects the fact that the sample mean $\bar{x}$ is itself estimated from the data.

Note that the variance is obtained by summing the squared differences between each observation and the mean. The differences are squared so that positive and negative deviations do not cancel and the result is always non-negative. If all elements in the sample are very close to the mean, the variance s^2 will be small; if the elements are widely spread, the variance will be larger.

Let us illustrate this definition with an example. For the sample $\mathcal{S} = \{1, 2, 3\}$, which has $n = 3$ observations and sample mean $\bar{x} = 2$, the variance is

$$s^2 = \frac{1}{3-1} \sum_{i=1}^{3} (x_i - \bar{x})^2 = \frac{1}{2}\big((1-2)^2 + (2-2)^2 + (3-2)^2\big) = \frac{1}{2}(1 + 0 + 1) = 1 \ ,$$

which we could interpret as, on average, the elements of the list being *one unit* away from the mean.

As a warm-up exercise, try to compute the sample variance for a second sample, let's say $\mathcal{S} = \{4, 5, 6\}$. By substituting in the general expression (1.6) you should get the result

$$s^2 = \frac{1}{3-1} \sum_{i=1}^{3} (x_i - \bar{x})^2 = \frac{1}{2}\big((4-5)^2 + (5-5)^2 + (6-5)^2\big) = \frac{1}{2}(1 + 0 + 1) = 1 \ .$$

We obtain again a variance $s^2 = 1$, indicating as in the previous example, that the elements of this sample $\mathcal{S}$ are also *one unit* away from the mean.

The standard deviation:

Another useful quantity used to characterize variability is the so-called *standard deviation*, which is simply the square root of the variance,

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \ , \tag{1.7}$$

and therefore has the same units as the original data. While the variance measures spread in squared units, the standard deviation provides a more interpretable measure of how far, on average, observations lie from the mean.

At a glance, variance and standard deviation both quantify how much the elements of a data set deviate from the mean, capturing the notion of *spread*.

Quantiles:

Finally, *quantiles* provide another way to describe the distribution of data by dividing an ordered data set into equal proportions. The p-th quantile Q_p is defined as the value below which a fraction p of the data lies. Common special cases include the *first quartile* (Q_1, corresponding to the 25th percentile), the *median* (Q_2, the 50th percentile), and the *third quartile* (Q_3, the 75th percentile). Informally, they answer questions of the form: *"What value separates the lowest p% of observations from the rest?"* They are particularly useful for describing skewed data and for comparing distributions without relying solely on the mean. To illustrate this idea with a simple example, consider the ordered sample

$$\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 8\} \ .$$

Since there are $n = 8$ observations, the median Q_2 lies halfway between the fourth and fifth elements and is therefore

$$Q_2 = \frac{4+5}{2} = 4.5 \ .$$

The first quartile Q_1 separates the lowest 25% of the data and lies between the second and third elements, while the third quartile Q_3 lies between the sixth and seventh elements. In this way, quartiles summarize the distribution by marking its lower tail, central region, and upper tail.

A rigorous definition of quantiles requires the notion of a distribution function and cumulative probability, which we will introduce in the next chapter. For a continuous cumulative distribution function (CDF) F, the p-th quantile satisfies

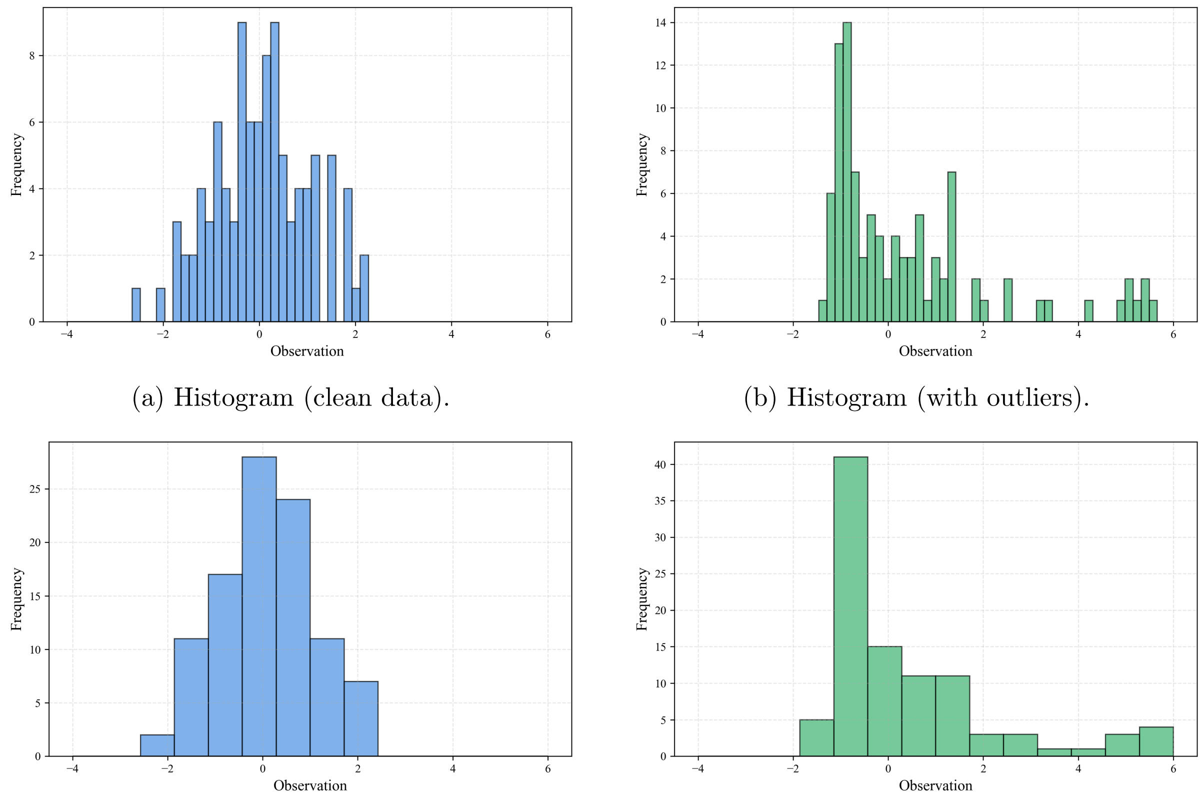
$$Q_p = \inf\{x : F(x) \geq p\} \ . \tag{1.8}$$

In summary, the mean, median, mode, variance, standard deviation, and quantiles provide a rich and complementary view of a data set's central tendency and variability. Together, they allow for both numerical and graphical summaries that capture essential features of the data.

Variation is not merely a technical detail; it is the essence of uncertainty. Without spread, probability would be trivial, as every outcome would be identical. It is precisely in the differences among observations that statistical inquiry finds its substance. Central tendency and variation thus form complementary lenses through which data become intelligible: they allow us to assess whether groups are similar or different, whether an observation is ordinary or surprising, and whether observed variation can plausibly be attributed to chance. In this sense, descriptive statistics foreshadows the inferential methods to come, hinting at deeper regularities beneath the numbers.

1.3 Data visualization

While numerical summaries are useful, the human mind often grasps patterns much more quickly through vision than through calculation. By *data visualization* we mean a whole family of techniques used to transform numbers and sequences into shapes, colours, and spatial structures that are easier to interpret and can often be understood at a glance. Visualization turns abstraction into perception and frequently reveals regularities that remain hidden when data are examined only through formulas or numerical summaries. Today, a broad range of disciplines grouped under the name of data visualization—or data *representation*—have become central pillars of scientific practice and data-driven inquiry.



(a) Histogram (clean data).

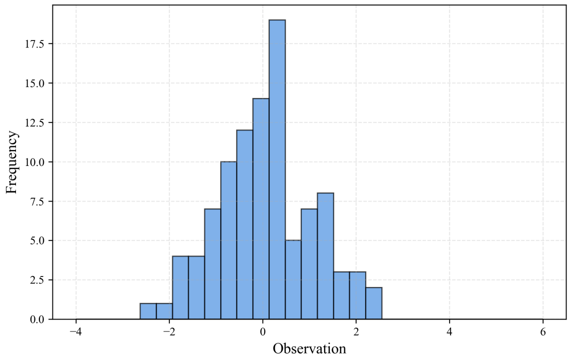
(b) Histogram (with outliers).

(c) Histogram (clean data).

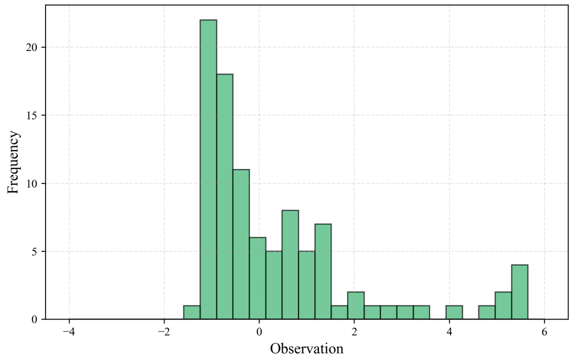
(d) Histogram (with outliers).

Figure 1.3: Comparison of graphical summaries for $n = 100$ observations drawn from a Gaussian distribution. Left column: uncontaminated data, with both small and large binning size. Right column: data affected by outliers, with both small and large binning size. Figure shows how a small bin size may lead to higher resolution, but also introduce noise.

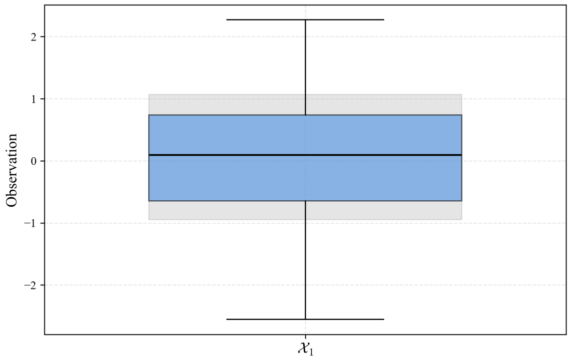
Among the oldest and most fundamental visualization tools is the *histogram*. The idea of dividing data into intervals in order to visualize frequency dates back to the late 19^{th} century, when Karl Pearson formalized the histogram as a graphical representation closely connected to probability distributions [31, 32]. A histogram divides the range of a data set into consecutive intervals, or *bins*, and represents the number—or relative *frequency*—of observations falling within each bin by the height of a bar. This simple yet powerful plot provides an immediate visual impression of the distribution, allowing one to identify symmetry, skewness, concentration of values, and potential gaps. For instance, a symmetric histogram suggests a roughly balanced distribution around the mean, whereas a right-skewed histogram indicates that large values are less frequent but may still exert a strong influence on measures such as the mean. Histograms are therefore widely used in the physical sciences and in contexts where data are compared to theoretical or mathematical predictions.



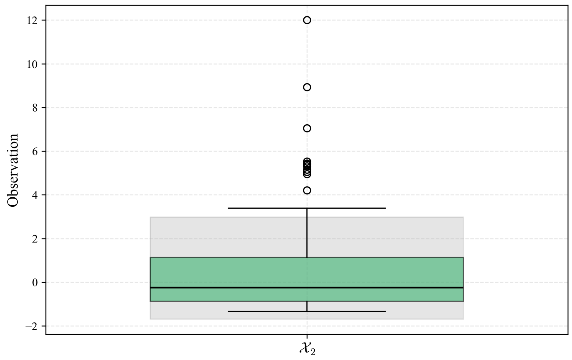
(a) Histogram (clean data).



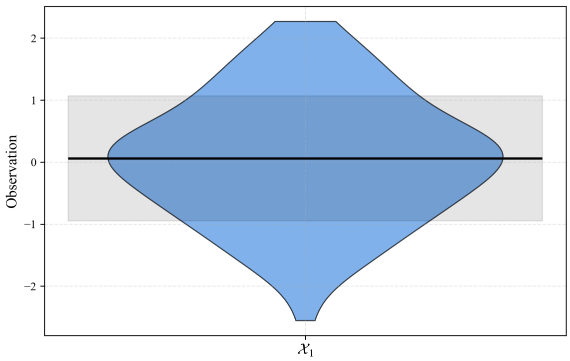
(b) Histogram (with outliers).



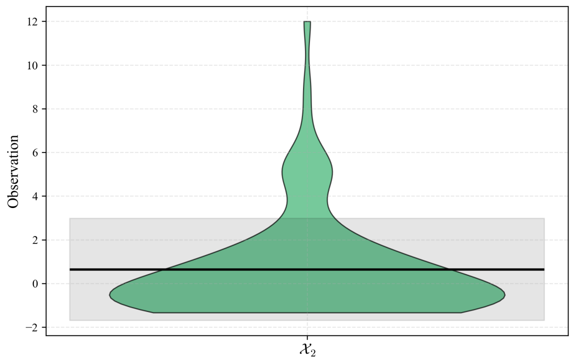
(c) Box plot (clean data).



(d) Box plot (with outliers).



(e) Violin plot (clean data).



(f) Violin plot (with outliers).

Figure 1.4: Comparison of graphical summaries for $n = 100$ observations drawn from a Gaussian distribution. Left column: uncontaminated data. Right column: data affected by outliers. The rows show, respectively, histograms, box plots, and violin plots, illustrating how different visualization techniques respond to skewness and extreme observations.

Constructing an informative histogram requires some care. As a general rule, bins should be of equal width and together cover the entire range of the data, while reflecting natural groupings when possible. The choice of bin width plays a crucial role: using many narrow bins increases resolution but can introduce spurious fluctuations, especially in the presence of outliers, whereas using a small number of wide bins produces a smoother and more robust picture at the cost of detail. Because extreme values can strongly affect the appearance of a histogram, they must be handled thoughtfully. For highly skewed distributions, numerical summaries such as the median and the interquartile range (IQR) often provide a more stable description than the mean.

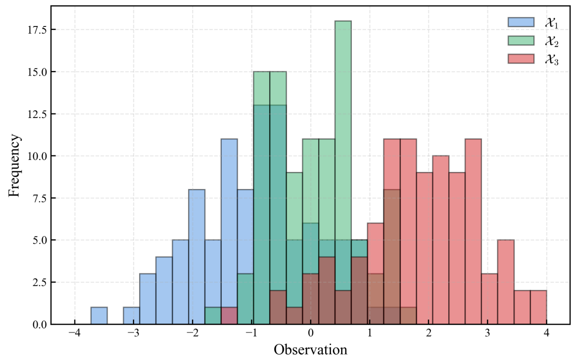
Another widely used visualization is the *box plot*, also known as the *box-and-whisker* plot, introduced by John Tukey in 1970 as part of his work on exploratory data analysis. He introduced the box plot in his 1970 unpublished course notes at Princeton, while it was formally published in his *"Exploratory Data Analysis"* in 1977 [33]. The box plot offers a compact summary of a data set's central tendency, spread, and potential outliers. It is constructed from five key statistics: the minimum, first quartile (Q_1), median (Q_2), third quartile (Q_3), and maximum. The box itself represents the interquartile range (IQR $= Q_3 - Q_1$), while observations lying more than 1.5 times the IQR from the quartiles are typically flagged as outliers. Box plots allow for rapid comparisons across multiple groups and are particularly effective at revealing asymmetry and variability without being overly influenced by extreme values. For this reason, they are especially common in biological, medical, and clinical sciences, where experiments are often repeated many times with relatively small sample sizes.

More recently, the *violin plot* has emerged as a refinement of the box plot, combining summary statistics with a smooth representation of the distribution's shape. Although its precise origin is less sharply documented, the violin plot gained prominence in the late 20^{th} century with the later development of statistical software environments such as R and Python, particularly following the work of Hintze and Nelson in the 1990s [34]. A violin plot augments the box plot by incorporating a kernel density estimate of the data, producing a mirrored shape that reflects the underlying distribution. In addition to displaying the median and quartiles, violin plots reveal features such as skewness and multimodality that may be obscured in simpler summaries, making them especially useful for comparing several groups side by side.

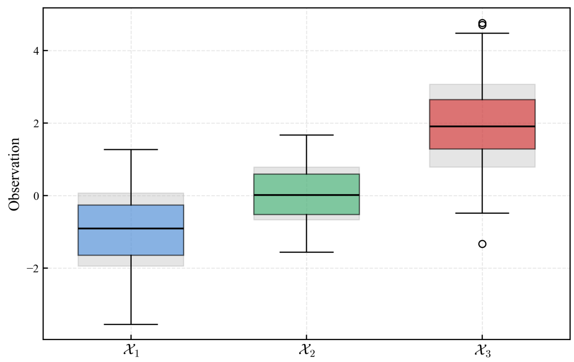
The importance of visualization extends beyond convenience. Early work by Spearman on association [35] and later demonstrations such as Anscombe's quartet [36] highlighted a crucial lesson: data sets with nearly identical numerical summaries can exhibit dramatically different graphical structures. Visualization is therefore not merely a tool for presentation, but a fundamental instrument of statistical reasoning. By complementing numerical summaries, graphical representations help guard against misinterpretation and deepen our understanding of variability, structure, and uncertainty in data.

As a note, let us emphasize that the roots of data visualization can be traced back well before modern statistics, to the early development of analytical geometry in the 17^{th} century. A decisive step was taken by René Descartes, whose introduction of coordinate systems in *"La Géométrie"* (1637) provided a systematic way to represent numerical relationships geometrically [37]. By associating quantities with positions in space, Descartes established the conceptual framework underlying graphs, curves, and plots as representations of functional relationships. Although his work was not statistical in intent, it laid the mathematical foundation upon which later visual representations of empirical data would be built. Over the following centuries, this geometric perspective was gradually adapted to observational and social data, culminating in the 19^{th}-century development of statistical graphics—such as frequency plots and histograms—as tools for exploring variability, association, and uncertainty.

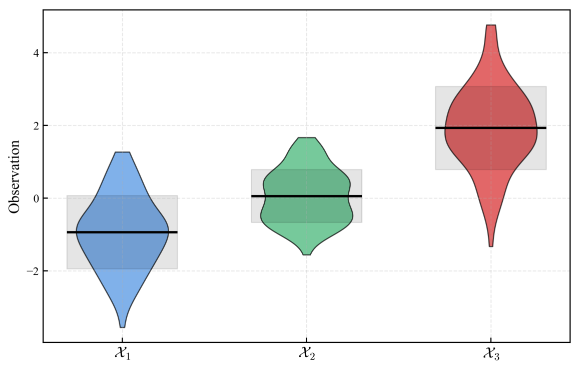
Following Descartes' geometric framework, graphical representations of data evolved through the work of figures such as William Playfair, who in the late 18^{th} century introduced line and bar charts to depict economic time series, and Adolphe Quetelet, who applied statistical regularities to social phenomena. In the 19^{th} century, visualization became both an analytical and persuasive tool, exemplified by Florence Nightingale's use of statistical graphics in public health and Charles Joseph Minard's multivariate maps. These developments established visualization as a central instrument for exploring, communicating, and reasoning about data, long before the formalization of modern statistical theory [38, 39].



(a) Three sets of observations χ_1, χ_2, χ_3 of sample size $n = 100$ drawn from a Gaussian distribution, with the mean value and standard deviation represented as a histogram.



(b) Three sets of observations χ_1, χ_2, χ_3 of sample size $n = 100$ drawn from a Gaussian distribution, with the mean value and standard deviation represented as a box plot.



(c) Three sets of observations χ_1, χ_2, χ_3 of sample size $n = 100$ drawn from a Gaussian distribution, with the mean value and standard deviation represented as a violin plot.

Figure 1.5: Comparison of the three visualization methods discussed in the chapter—the histogram, the box plot, and the violin plot—showing the mean and variability of three samples of size $n = 100$ drawn from a Gaussian distribution.

Chapter 2

Foundations of probability

It is through the calculation of probabilities that the divine order becomes visible.
— Jacob Bernoulli

At its heart, probability is nothing more—and nothing less—than a branch of mathematics developed to describe random phenomena, also referred to as *stochastic*. The word "stochastic" comes from the Greek στοχαστικός, literally meaning "to guess" or "to aim." Probability thus provides a numerical language for uncertainty, allowing us to quantify how surprising or plausible an outcome is before it is observed.

The study of probability, though having very ancient roots, began its modern development in the 17$^{\text{th}}$ century through the famous correspondence between Blaise Pascal and Pierre de Fermat. Their discussion about games of chance, and in particular the "problem of the division of stakes", laid the groundwork for a systematic mathematical analysis of uncertain events. A few decades later, Jacob Bernoulli's *"Ars Conjectandi"* provided the first sustained theoretical treatment of probability, including an early formulation of the law of large numbers [13, 14]. Subsequent refinements by Abraham de Moivre, particularly his work on normal approximation, and by Pierre-Simon Laplace transformed probability into a powerful analytical theory, while its fully axiomatic structure only crystallised in the 20$^{\text{th}}$ century, as we will see [16].

Beyond games of chance, probability rapidly became essential for the understanding of natural phenomena and human affairs. Astronomy, population studies, and various physical problems required tools to reason quantitatively about variability, error, and incomplete information. During the 19$^{\text{th}}$ century, thinkers such as Cournot emphasized the connection between probabilistic laws and empirical regularities, arguing that probability acquires meaning through its relation to observable frequencies in the world. In this sense, probability emerged not merely as a mathematical curiosity, but as a response to practical problems involving uncertainty and regularity in the empirical world.

As we have mentioned already, a decisive step toward mathematical rigor was taken by Andrey Kolmogorov in 1933. In his *"Grundbegriffe der Wahrscheinlichkeitsrechnung"* [23], Kolmogorov showed that probability could be treated as a branch of measure theory, independent of any specific interpretation. This formulation built directly on earlier developments in analysis, most notably Henri Lebesgue's theory of integration, which provided the mathematical language needed to define probabilities as measures on sets [40]. Rather than defining probability through intuition, symmetry, or frequency, Kolmogorov postulated a small set of axioms from which the entire formal theory follows.

The measure-theoretic approach introduced by Kolmogorov was later systematized and widely disseminated through modern mathematical treatments of probability, most famously in Patrick Billingsley's *Probability and Measure* [41]. These works established the now-standard framework in which random variables, expectations, and convergence are treated using the tools of modern analysis.

From a philosophical point of view, two influential strands in the interpretation of probability emerge from these debates. On the one hand, the frequentist program, most clearly articulated by Richard von Mises, grounded probability in long-run relative frequencies of repeatable experiments. On the other hand, approaches originating with Thomas Bayes and later developed in the twentieth century interpreted probability as a rational degree of belief. Both perspectives play an important role in modern probability and statistics, and their mathematical and philosophical foundations will be examined in detail in Chapter 6.

Parallel developments in the early 20^{th} century focused on statistical inference from data rather than individual decision-making. These approaches emphasized long-run frequency properties, error control, and sampling distributions, leading to the classical framework of statistical inference that still underlies much of modern applied statistics.

2.1 Probability and random events

In modern mathematics, probability is defined axiomatically following Kolmogorov's axioms [23]. Without much technicality, probability $\mathbb{P}$ is a number we associate to each event, satisfying three fundamental rules:

- Probabilities are never negative: $\mathbb{P}(A) \geq 0$ for any event A.
- The probability of a certain event is 1.
- If two events cannot occur together, the probability that one or the other occurs is the sum of their probabilities.

For a discrete set of all possible outcomes $\{x_1, x_2, \ldots, x_n\}$, these rules imply the normalization condition

$$\sum_{i=1}^{n} \mathbb{P}(x_i) = 1 \ ,$$

also referred to as *unitarity*, which simply states that *something must happen.*

The numerical value of a probability reflects how surprising an outcome would be. For a given event A, such as observing heads in a coin toss, or a specific face in a dice roll, when $\mathbb{P}(A) \to 0$, the event is almost impossible; observing it would be highly surprising. On the contrary, when $\mathbb{P}(A) \to 1$, the event is almost certain; its occurrence carries little surprise. For anything in between, there is a level of *uncertainty*, or *surprise*, where probability quantifies degrees of expectation, rather than absolute certainty or impossibility.

Why, for example, do we say that a fair coin has probability 1/2 of landing heads, or that a fair die has probability 1/6 of showing a given face? These numbers are not empirical facts but modeling assumptions based on symmetry. When all outcomes are assumed to be equally possible and indistinguishable before observation, probability assigns equal weight to each outcome. Probability theory then explores the logical consequences of these assumptions.

This idea, that half of the times we toss a coin we obtain heads, hence we assign 1/2 probability, or that one every six times we roll dice we see a specific face, hence we assign 1/6 probability, leads to common interpretation of probability known as the *frequentist* view, developed most clearly by von Mises in his *"Probability, Statistics and Truth"* [42]. In this perspective, probability is identified with the long-run relative frequency of an event in repeated, identical experiments. Saying that a coin has probability 0.5 of landing heads means that, over many tosses, roughly half will result in heads. And as we increase the number of observations, at higher *frequencies*, we expect that number to converge to a perfect half.

An alternative interpretation is the *Bayesian* view, originating with Thomas Bayes [43] and developed further by Laplace and later authors such as de Finetti [44] and Jaynes [45]. Here, probability quantifies uncertainty or degree of belief rather than long-run frequency, and probabilities are updated as new information becomes available using Bayes' theorem.

Both interpretations use the same mathematical rules and rely on Kolmogorov's axioms. The difference lies not in the calculations, but in how probability statements are interpreted. Again, Bayesian methods and their practical consequences will be introduced formally in Chapter 6.

2.2 Discrete events

With this simple definition of probability, we can now start categorizing events based on how their probabilities are *distributed*. By *discrete* events we mean those where the set of possible outcomes is finite or countably infinite. From coins and dice to counting, discrete models are particularly useful when outcomes correspond to counts, successes and failures, or categorical observations. In such cases, probability distributions will represent exact probabilities to individual outcomes and are therefore called probability *mass* functions.

Mathematically, a discrete probability distribution assigns a probability $\mathbb{P}(x_i)$ to each possible outcome x_i, such that

$$\mathbb{P}(x_i) \geq 0, \qquad \sum_{\forall i} \mathbb{P}(x_i) = 1 \ . \tag{2.1}$$

where the $\forall$ symbol is just a mathematical character meaning "for all".

2.2.1 Bernoulli trials

The Bernoulli trial was formalized by Jacob Bernoulli in *"Ars Conjectandi"* (1713) [13]. His motivation was to understand how regularity emerges from randomness when an experiment with two outcomes is repeated many times.

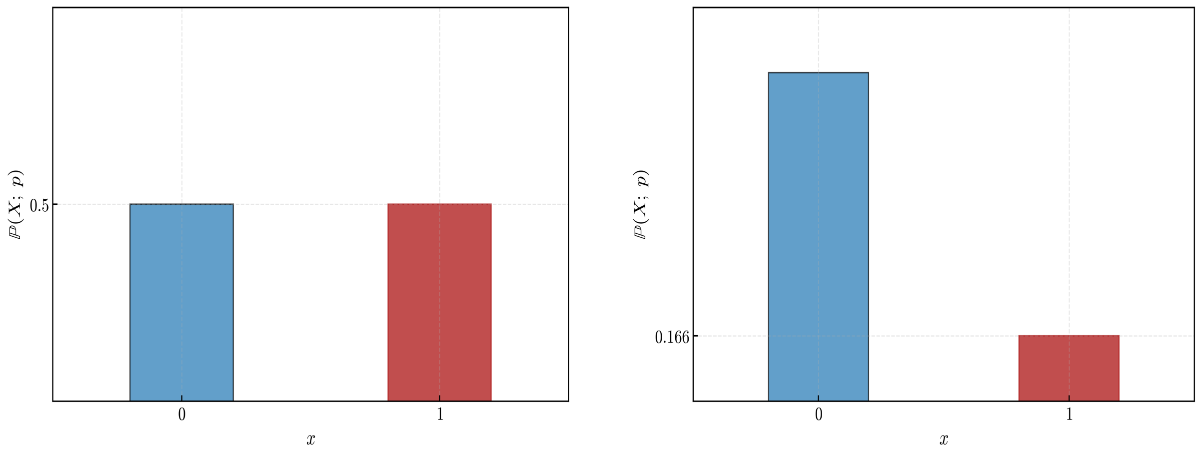
A Bernoulli random variable x takes only two values, usually 1 (success) and 0 (failure). Its probability mass function is

$$\mathbb{P}(x;\, p) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0, \end{cases} \qquad 0 \leq p \leq 1. \tag{2.2}$$

Bernoulli trials are used whenever an experiment has exactly two possible outcomes. Typical examples include the heads and tails of the coin, success or failure of a medical treatment, or whether a user clicks on a digital advertisement. In all these cases, the outcome is binary, even if the underlying process is complex.

Example: A single coin toss of a fair coin can be modeled as a Bernoulli trial, with $x = 1$ representing heads and $x = 0$ tails. For a fair coin, symmetry suggests $p = 1/2$.

Example: A single coin toss of a biased coin can be modeled as a Bernoulli trial, with $x = 1$ representing heads and $x = 0$ tails, but now the probability of successwould be, for instance $p = 2/3$.



(a) Bernoulli trial on a fair coin.

(b) Bernoulli trial on a biased coin.

Figure 2.1: Bernoulli probability mass distribution.

2.2.2 Discrete uniform distribution

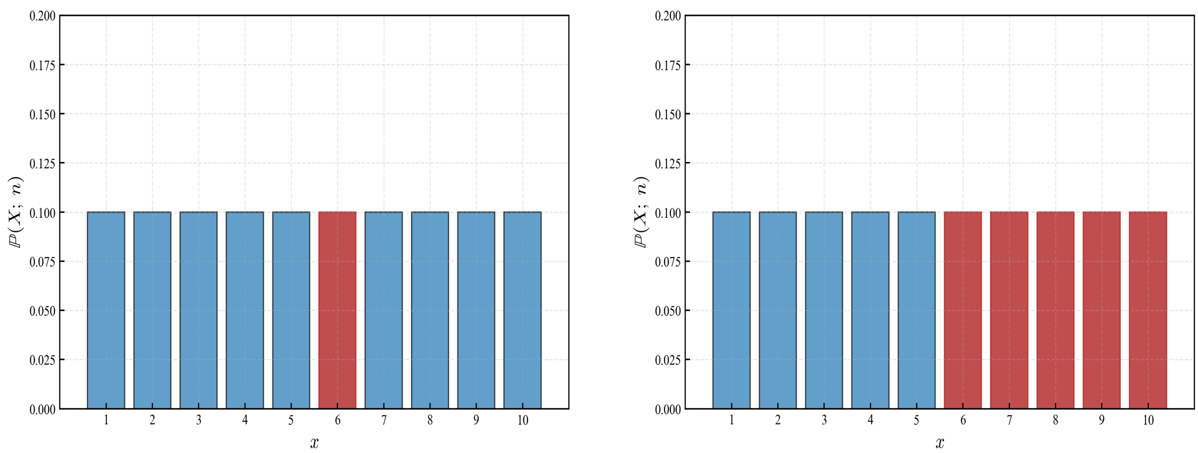
The discrete uniform distribution has its roots in classical symmetry arguments used in early probability theory. It formalizes the idea that, in the absence of distinguishing information, all outcomes should be treated equally. This idea reflects a principle already present in early probability theory: when no outcome can be distinguished from another based on available information, they should be treated symmetrically. Laplace formalized this reasoning as "the principle of insufficient reason".

If a random variable x can take n distinct values $\{x_1, \ldots, x_n\}$, the discrete uniform distribution assigns

$$\mathbb{P}(x;\; n) = \frac{1}{n}, \qquad i = 1\,, \ldots, n\,. \tag{2.3}$$

Discrete uniform distributions appear whenever outcomes are assumed to be equally likely. Examples include faces of dice, card draws from a well-shuffled deck, or randomized experimental assignments where each category is given equal probability.

Example: Rolling a fair six-sided die can be modeled as a discrete uniform distribution on $\{1, 2, 3, 4, 5, 6\}$, where each face has probability $1/6$.



(a) Probability distribution.

(b) Cumulative probability.

Figure 2.2: Discrete uniform probability mass distribution.

2.2.3 Binomial distribution

The binomial distribution was systematically studied by Abraham de Moivre in the early 18^{th} century. His analysis of repeated Bernoulli trials led not only to the binomial formula but also to the first appearance of the normal approximation. De Moivre introduced the binomial distribution while studying games of chance, but its importance quickly extended far beyond gambling. By considering repeated trials under identical conditions, the binomial distribution became a central model for understanding variability in counting processes.

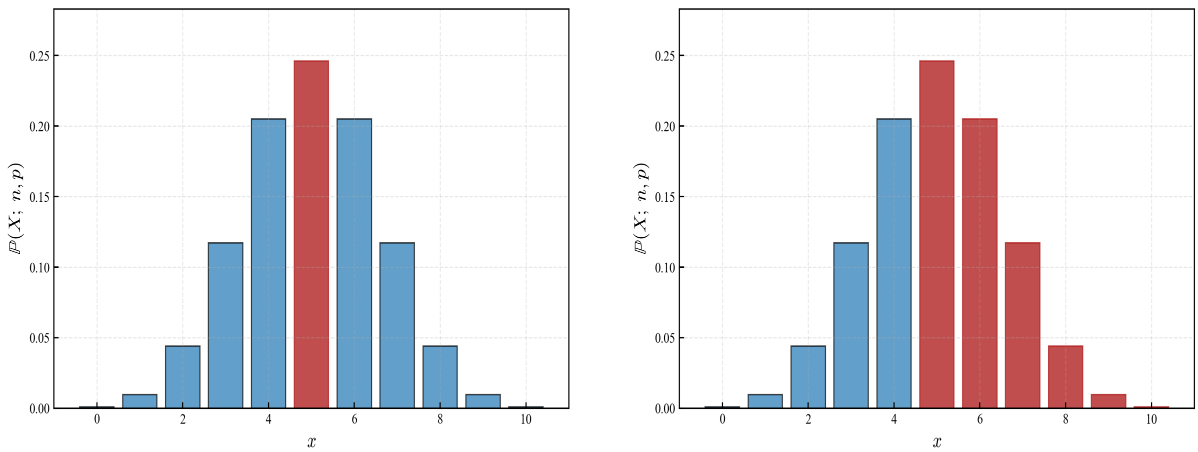
The binomial distribution models the number of successes in n independent Bernoulli trials with success probability p. Its probability mass function, given a total number of attempts n, and the individual probability p of each success, is

$$\mathbb{P}(x;\ n,\ p) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n\,. \tag{2.4}$$

Binomial models naturally arise when we count how many times a certain event occurs in a fixed number of attempts. Examples include the number of defective items in a batch, the number of patients responding to a treatment, or the number of voters favoring a candidate in a survey.

Example: The number of heads obtained when tossing a fair coin 10 times follows a binomial distribution with $n = 10$ and $p = 1/2$.

Example: The number of 4s obtained when rolling a fair dice 10 times follows a binomial distribution with $n = 10$ and $p = 1/6$.



(a) Probability distribution. (b) Cumulative probability.

Figure 2.3: Binomial probability mass distribution.

2.2.4 Poisson distribution

The Poisson distribution was introduced by Siméon Denis Poisson in 1837 while studying rare events in judicial statistics. It arises as a limiting case of the binomial distribution when events are rare but opportunities are numerous. Poisson originally introduced this distribution to study rare events, such as wrongful convictions in court cases. Its mathematical simplicity and clear interpretation soon made it a fundamental model for random counts occurring over time or space.

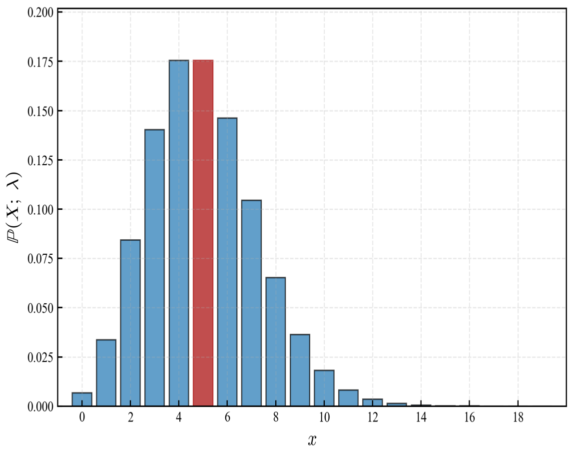
The Poisson distribution models the number of events occurring in a fixed interval of time or space. Its probability mass function, known the observed historical average λ, also known as "rate of occurrence", is

$$\mathbb{P}(x;\ \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \dots, \tag{2.5}$$

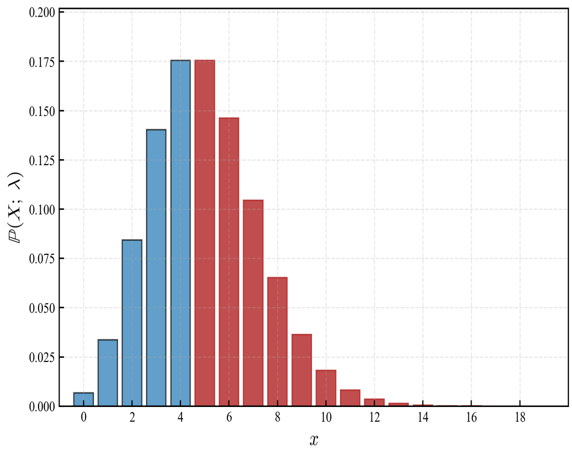
Poisson distributions are commonly used to model events that occur independently and sporadically. Typical examples include the number of phone calls received by a call center, the number of typing errors on a page, or the number of decay events detected by a sensor during a fixed time interval.

Example: The number of emails received in one hour, when messages arrive independently at an average rate of $\lambda = 5$ per hour, can be modeled using a Poisson distribution.

Example: The number of cancer patients observed in a hospital over a week, when the average patients happen at a rate rate of $\lambda = 7$ per week, can also be modeled using a Poisson distribution.



(a) Probability distribution.



(b) Cumulative probability.

Figure 2.4: Poisson probability mass distribution.

Cumulative probabilities:

So far, we have described probability distributions by assigning probabilities to individual outcomes. While this is natural for discrete models, it is often more informative to consider the probability that a random variable takes a value *up to*, greater or smaller than a given threshold. This leads to the notion of *cumulative probability*.

As a note, it is common notation to use uppercase letters (such as X) to denote random variables, and lowercase letters (such as x) to denote their possible values. For a discrete random variable X, the *cumulative distribution function* (CDF) is defined as

$$F(x) = \mathbb{P}(X \leq x), \tag{2.6}$$

that is, the probability that the outcome of the experiment is less than or equal to x.

If the possible values of X are $\{x_1, x_2, \dots\}$, the cumulative probability at a point x is obtained by summing the probabilities of all outcomes less than or equal to x:

$$F(x) = \sum_{x_i \leq x} \mathbb{P}(X = x_i). \tag{2.7}$$

The cumulative distribution function provides a global view of the distribution. While the probability mass function tells us how probability is assigned locally to each outcome, the CDF tells us how probability accumulates as we move along the real line.

Several important properties follow immediately from the definition:

- $F(x)$ is non-decreasing.
- $0 \leq F(x) \leq 1$ for all x.
- $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$.

Example. For a fair six-sided die, the probability that the outcome is *at most* 4 is

$$\mathbb{P}(X \leq 4) = \mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(4) = \frac{4}{6}.$$

The corresponding cumulative distribution function increases in steps of size 1/6 at each integer outcome.

Cumulative probabilities play a central role in probability theory because they naturally generalize to the continuous case, where individual outcomes no longer carry positive probability.

2.3 Continuous events

By *continuous* events, in contrast to the discrete case, we mean that the set of possible outcomes is uncountably infinite, typically forming an interval of real numbers. In such cases, individual outcomes have zero probability, and uncertainty is described using probability *densities*. Probabilities are obtained by integrating the density over ranges of values.

When moving from discrete to continuous outcomes, the frequentist intuition that works well for counting events begins to break down. In a discrete setting, probabilities can be interpreted as long-run relative frequencies of individual outcomes. For example, the probability of rolling a 3 with a fair die can be understood as the fraction of times the outcome 3 appears when the die is rolled repeatedly. Each outcome has a positive probability, and frequencies converge to these values as the number of trials grows.

In a continuous setting, this interpretation can no longer be applied directly. If outcomes lie on a continuous interval, such as all real numbers between 0 and 1, the probability of observing any exact value is zero. No matter how many times the experiment is repeated, the relative frequency of obtaining exactly 0.37 will be zero. Same would happen with the probability of measuring a specific temperature in a room, let's say 25.0 degrees. Since there is an infinite, continuous amount of values from 24.999... to 25.000, and from 25.000 to 25.001, the probability of measuring a single value would also be mathematically zero. This does not mean that the outcome is impossible, but rather that probability must now be assigned to *ranges* of values rather than to individual points. We just need a new mathematical object that represents this information.

The concept of a probability *density* is introduced precisely to resolve this issue: densities describe how probability is distributed locally, while actual probabilities are obtained by integrating the density over intervals. In this way, the frequentist idea of long-run relative frequency is preserved, but it applies to intervals of outcomes rather than to single values.

Mathematically, a continuous probability distribution is described by a density function $f(x)$ such that

$$f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\, dx = 1. \tag{2.8}$$

The probability that a random variable lies in an interval $[a, b]$ is then given by the area under the density curve between a and b.

2.3.1 Gaussian distribution

The Gaussian distribution emerged from the work of Abraham de Moivre in the early 18^{th} century and was later developed systematically by Gauss and Laplace. Its physical interpretation was provided by Carl Friedrich Gauss in *"Theoria Motus Corporum Coelestium"* (1809) [15], in the context of measuring errors of astronomical observations, where repeated measurements of the same quantity produced small deviations around a central value. This interpretation linked probability theory directly to experimental science.

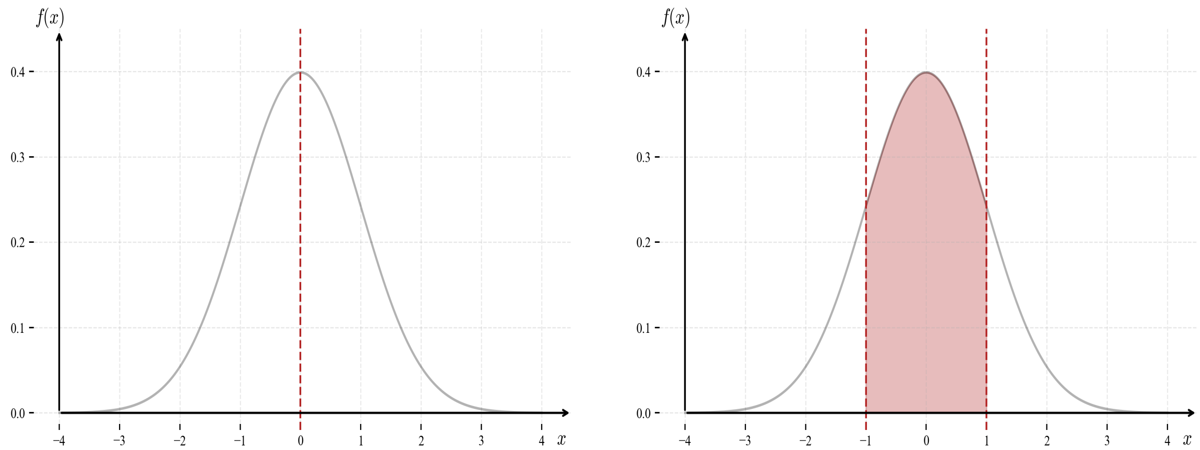
The Gaussian distribution models the accumulation of many small, independent effects. Its central role in probability theory is explained by the central limit theorem, which establishes it as a universal limiting distribution, and that we will cover in Chapter 3.

The probability density function of a Gaussian random variable x with mean μ and variance σ^2 is

$$f(x;\ \mu,\ \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}. \tag{2.9}$$

Gaussian distributions are used to model many natural and social phenomena where values cluster around an average, and it appears incredibly often. Examples include measurement errors, biological traits such as height, and aggregated effects of many small influences acting together. Why so many physical phenomena tend to appear centered around a mean value (μ), with some notion of spread (σ), and converging to a perfect Gaussian bell-shape as the number of observations increase, is still an open question which triggers various philosophical debates.

Example: Measurement errors in physical experiments are often modeled as Gaussian, with $\mu = 0$ representing no systematic bias and σ describing measurement precision.



(a) Probability distribution.

(b) Cumulative probability.

Figure 2.5: Gaussian probability density distribution.

2.3.2 Exponential distribution

The exponential distribution arose in the 19^{th} century in the study of waiting times and decay processes, closely connected to Poisson's work on random events and later developments in queueing theory and stochastic processes [46, 22]. The exponential distribution emerged naturally from the study of random event timing, particularly in physics and telecommunications. Its mathematical form reflects the assumption that events occur independently and at a constant average rate.

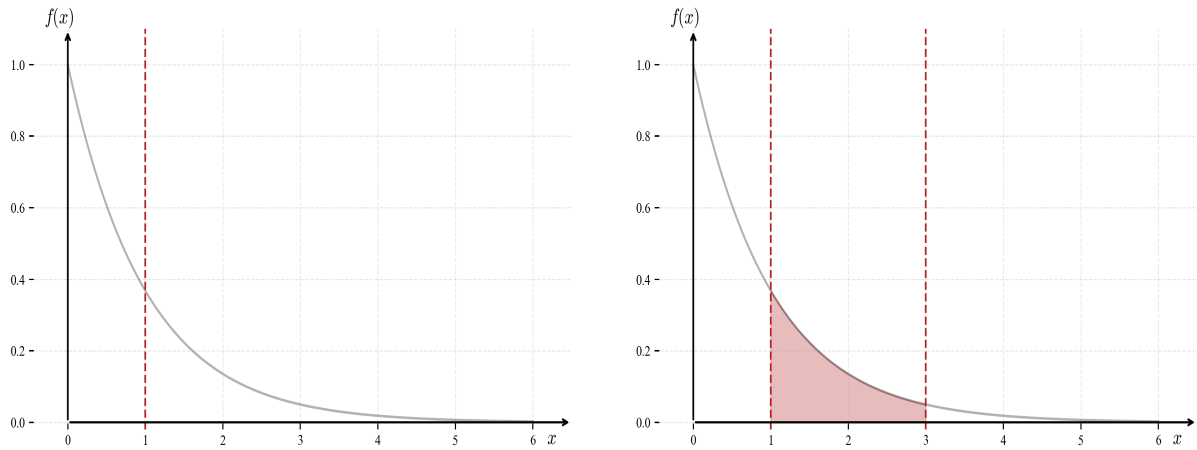
It naturally models the time until the first occurrence of a random event and is characterized by the absence of memory: the future waiting time does not depend on how much time has already elapsed.

The probability density function of an exponential random variable X with rate $\lambda > 0$ is

$$f(x;\ \lambda) = \lambda e^{-\lambda x}, \qquad x \geq 0. \tag{2.10}$$

Exponential models are appropriate for waiting-time phenomena. Examples include the time until a machine fails, the time until the next customer arrives, or the time between successive radioactive decay events.

Example. The time until the next phone call arrives at a call center, assuming calls arrive independently at a constant average rate, is often modeled using an exponential distribution.



(a) Probability distribution.

(b) Cumulative probability.

Figure 2.6: Exponential probability density distribution.

2.3.3 Continuous uniform distribution

The continuous uniform distribution, also called flat distribution, extends classical symmetry arguments already present in Laplace's *Théorie Analytique des Probabilités* (1812) [16]. It represents complete ignorance about where within a bounded interval an outcome may fall. The continuous uniform distribution formalizes the idea of complete uncertainty over a bounded range. Unlike other distributions, it does not privilege any value within the interval, making it a neutral reference model.

If a random variable x is uniformly distributed on an interval $[a, b]$, its probability density function is

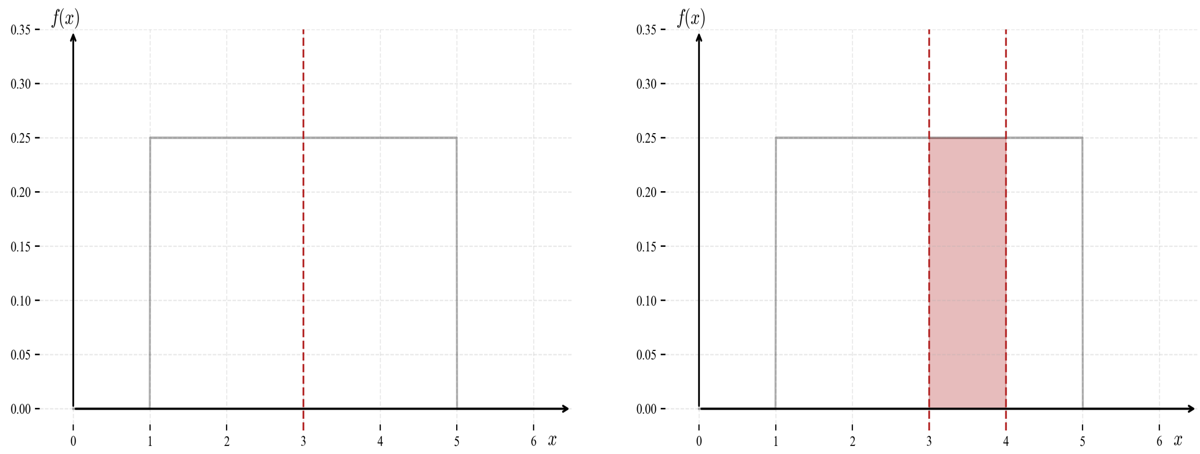
$$f(x;\ a,\ b) = \frac{1}{b-a}, \qquad a \leq x \leq b. \tag{2.11}$$

Continuous uniform distributions are often used in simulations and random sampling. They arise, for example, when generating random starting points, choosing random times within a fixed interval, or modeling unknown quantities constrained only by upper and lower bounds.

Example: If a random number generator produces values evenly between 0 and 1, the outcome can be modeled as a continuous uniform distribution on $[0, 1]$.

Cumulative probability in the continuous case:

When outcomes are continuous, individual values have zero probability and probability is described using densities rather than masses. Nevertheless, the idea of cumulative probability



(a) Probability distribution.

(b) Cumulative probability.

Figure 2.7: Continous uniform probability density distribution.

remains fundamental. As before, to get used to standard notation, we use uppercase letters (such as X) to denote random variables, and lowercase letters (such as x) to denote their possible values.

For a continuous random variable X with probability density function $f(x)$, the cumulative distribution function is defined in exactly the same way:

$$F(x) = \mathbb{P}(X \leq x). \tag{2.12}$$

The difference lies in how this probability is computed. Since probability is distributed continuously, cumulative probability is obtained by integrating the density:

$$F(x) = \int_{-\infty}^{x} f(t)\, dt. \tag{2.13}$$

In this setting, the density $f(x)$ does not represent probability itself, but rather the *rate at which probability accumulates*. The probability that X lies in an interval $[a, b]$ is given by

$$\mathbb{P}(a \leq X \leq b) = \int_{a}^{b} f(t)\, dt = F(b) - F(a). \tag{2.14}$$

Thus, while probability mass functions assign probabilities to individual points, probability density functions describe how probability is spread across the real line. In both discrete and continuous cases, the cumulative distribution function provides a unified description of uncertainty and serves as the primary object connecting probability to integration.

2.4 Expected values

Probability distributions describe uncertainty, but to summarize and compare them we often want a small number of representative quantities. The most important of these are the *moments* of a random variable. Moments capture different aspects of a distribution such as its location, spread, and shape.

Throughout this section, we will again denote random variables by uppercase letters, such as X. In the discrete case, probabilities are assigned to individual values via $\mathbb{P}(X = x_i)$, while in the continuous case probabilities are assigned to intervals and described through a probability density function.

Mean as expected value:

The most fundamental moment is the *expected value*, also called the mean. Naming here may seem misleading, as this mean is not to be confused with the sample mean we discussed in previous chapter. One thing is the mean and variance *as expected values of random variables*, and a different one is as a number that summarizes a sample, which we referred in last chapter as *sample mean.* Informally, the expected value represents the long-run average outcome of a random experiment repeated many times. It answers the question: *where is the distribution centered?*

For a discrete random variable X taking values $\{x_i\}$ with probabilities $\mathbb{P}(X = x_i)$, the expected value is defined as

$$\mathbb{E}[X] = \sum_i x_i \, \mathbb{P}(X = x_i). \tag{2.15}$$

For a continuous random variable with probability density function $f(x)$, the expected value is defined analogously by replacing the sum with an integral:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \, f(x) \, dx. \tag{2.16}$$

The expected value is a *first-moment* quantity: it captures the location of a distribution but provides no information about its variability. A key property of expectation is linearity. For any random variables X_i,

$$\mathbb{E}\left[\sum_i X_i\right] = \sum_i \mathbb{E}[X_i], \tag{2.17}$$

regardless of whether the variables are independent.

Variance as expected value:

The second moment of central importance is the *variance*, which measures how spread out the distribution is around its mean. Variance is defined as the expected squared deviation from the mean:

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] \, . \tag{2.18}$$

An equivalent and often more convenient expression for the variance is

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2. \tag{2.19}$$

For a discrete random variable, this corresponds to

$$\mathrm{Var}(X) = \sum_i (x_i - \mu)^2 \, \mathbb{P}(X = x_i), \qquad \mu = \mathbb{E}[X], \tag{2.20}$$

while for a continuous random variable it is given by

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \, f(x) \, dx. \tag{2.21}$$

More generally, the k-th *moment* of a random variable describes higher-order features of its distribution:

- The *first moment*, referred to as the *mean*, describes central location.

- The *second moment*, referred to as the *variance*, describes spread.
- The *third moment* is related to *skewness*, measuring asymmetry.
- The *fourth moment* is related to *kurtosis*, measuring tail heaviness.

In practice, mean and variance already provide a powerful summary of most distributions. Classical statistical inference—such as confidence intervals, hypothesis tests, and error propagation—relies heavily on estimators of these two quantities and on their sampling distributions.

2.5 Mean and variance of common distributions

We conclude this chapter by collecting the expected value and variance of the probability distributions introduced earlier. These results provide concrete examples of the abstract definitions given in the previous section and will be used repeatedly in later chapters.

Bernoulli distribution:

Let $X \sim \mathrm{Bern}(p)$, with $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. Then

$$\mathbb{E}[X] = p, \tag{2.22}$$

$$\mathrm{Var}(X) = p(1 - p). \tag{2.23}$$

The mean equals the success probability, while the variance is largest when $p = 1/2$.

Binomial distribution:

Let $X \sim \mathrm{Bin}(n, p)$ represent the number of successes in n independent Bernoulli trials. Then

$$\mathbb{E}[X] = np, \tag{2.24}$$

$$\mathrm{Var}(X) = np(1 - p). \tag{2.25}$$

Both the mean and variance scale linearly with the number of trials.

Poisson distribution:

Let $X \sim \mathrm{Pois}(\lambda)$, where $\lambda > 0$ is the average rate of occurrence. Then

$$\mathbb{E}[X] = \lambda, \tag{2.26}$$

$$\mathrm{Var}(X) = \lambda. \tag{2.27}$$

A defining feature of the Poisson distribution is that its mean and variance coincide.

Discrete uniform distribution:

Let X be uniformly distributed on the set $\{1, 2, \ldots, n\}$. Then

$$\mathbb{E}[X] = \frac{n + 1}{2}, \tag{2.28}$$

$$\mathrm{Var}(X) = \frac{n^2 - 1}{12}. \tag{2.29}$$

The mean lies at the center of the interval, while the variance depends only on its width.

Gaussian distribution:

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then

$$\mathbb{E}[X] = \mu, \tag{2.30}$$

$$\mathrm{Var}(X) = \sigma^2. \tag{2.31}$$

The parameters μ and σ^2 directly control the location and spread of the distribution.

Exponential distribution:

Let $X \sim \mathrm{Exp}(\lambda)$, with $\lambda > 0$. Then

$$\mathbb{E}[X] = \frac{1}{\lambda}, \tag{2.32}$$

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}. \tag{2.33}$$

Larger values of λ correspond to shorter expected waiting times and reduced variability.

Continuous uniform distribution:

Let $X \sim \mathrm{Unif}(a, b)$. Then

$$\mathbb{E}[X] = \frac{a+b}{2}, \tag{2.34}$$

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}. \tag{2.35}$$

As in the discrete case, the mean lies at the midpoint of the interval and the variance depends only on its length.

These examples illustrate how mean and variance summarize probability distributions in concrete terms. In later chapters, we will study how these quantities are estimated from data and how their sampling variability affects statistical inference.

Chapter 3

Estimation, variability and confidence

The development of mathematics is a continuous process of abstraction.

— Emmy Noether

In the previous chapters, we described data using summary statistics and introduced probability models to represent uncertainty. In this chapter we connect these two perspectives. Estimation is the process through which data is used to learn about unknown features of a population, while probability provides a language to quantify how reliable such learning is, and focuses on making predictions [3, 4].

As we discussed already, a crucial distinction must be made between *prediction* and *inference.* Prediction focuses on forecasting future observations, whereas inference aims to draw conclusions about underlying parameters or mechanisms that generate the data [5]. Hence, estimation lies at the heart of inference: with some general results such as *the Law of Large Numbers* and *the Central Limit Theorem*, together with the idea of *estimators* and *bias*, we will transform random samples into numerical statements about unknown quantities.

Because data is inherently variable, different samples lead to different estimates. Understanding how estimates behave across repeated samples is therefore essential. Rather than focusing on a single observed value, we study the *distribution of the estimator* across all possible samples. This shift—from a fixed number to a random quantity—is central to statistical thinking [3, 4].

Two fundamental results from probability theory provide the foundation for this analysis. The first is *the Law of Large Numbers*, which states, informally, that averages computed from large samples tend to stabilize around the true value. First studied systematically by Jacob Bernoulli, this result formalizes the idea that more data improves accuracy [13].

The second is *the Central Limit Theorem*, whose roots can be traced to Laplace [16]. It explains why many estimators exhibit approximately Gaussian variability when the sample size is large, providing the basis for confidence intervals and hypothesis testing.

To compare estimators, we introduce two key concepts: *bias* and *variance.* The bias measures systematic deviation from the true value, while the variance captures fluctuations across samples. These two aspects are often in tension, and understanding their trade-off is central to statistical practice.

Throughout this chapter, we will also prepare the topics that will be central for hypothesis testing and modelling, rooted on the formalization of estimators as random variables, their behavior through the Law of Large Numbers and the Central Limit Theorem, and the use bias and variance to evaluate their performance. The emphasis will be on intuition supported by probabilistic reasoning, providing tools to assess what can be reliably learned from data.

3.1 The Law of Large Numbers

The Law of Large Numbers was first proved by Jacob Bernoulli in his posthumously published *Ars Conjectandi* (1713). His aim was to explain how stable patterns emerge from random phenomena, providing a rigorous foundation for interpreting probability as long-run relative frequency [13, 14].

Historically, this result played a central role in the development of probability as a scientific discipline. In the 18^{th} and 19^{th} centuries, it supported the view—later formalized by thinkers such as Laplace—that randomness at the individual level can coexist with regularity at the collective level. This idea influenced not only mathematics but also fields such as astronomy, physics, and the social sciences, where large populations exhibit stable statistical laws despite individual unpredictability.

The theorem formalizes a simple but powerful idea: while individual observations may be highly variable, their average becomes increasingly stable as more data is collected. In this sense, randomness averages out.

Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with finite expectation $\mu = \mathbb{E}[X_1]$. The sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{3.1}$$

satisfies that $\bar{X}_n$ converges in probability to μ

$$\bar{X}_n \xrightarrow{P} \mu \quad \text{as } n \to \infty, \tag{3.2}$$

This version is known as the *Weak Law of Large Numbers.* As a note, it is remember again the notation we to used in the previous chapter, where uppercase letters (such as X) denote random variables, and lowercase letters (such as x) to denote their possible values.

A stronger statement is also possible. Under the same conditions, one has

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu, \tag{3.3}$$

meaning that the sample average converges to μ with probability one. This is called the *Strong Law of Large Numbers.* Informally, while the weak law guarantees that large deviations become unlikely, the strong law ensures that the sequence of averages almost surely settles at the true value.

Example: Consider repeated tosses of a fair coin, where $X_i = 1$ for heads and $X_i = 0$ for tails. Then $\mu = 1/2$, and

$$\bar{X}_n = \frac{\text{number of heads in } n \text{ tosses}}{n}.$$

Although early results may fluctuate significantly, the proportion of heads stabilizes near $1/2$ as the sample size n increases.

Example: Suppose we repeatedly measure the length of an object with small random errors. Each measurement may differ slightly, but their average provides a more reliable estimate of the true length. Increasing the number of measurements reduces the impact of random fluctuations.

The Law of Large Numbers justifies the basic principle, for which we share a common intuition, of statistical estimation: averages computed from sufficiently large samples (of large n) provide reliable information about underlying quantities.

3.2 The Central Limit Theorem

Consider the following simple situation. Suppose a student records their daily study time (in hours) over many days. On some days they study 1 hour, on others 3 or 4 hours, and occasionally not at all. The data is clearly uneven and not symmetric. Now imagine taking the average study time over 30 days, and repeating this process many times for different months. Although the individual daily values are irregular, the distribution of these 30-day averages will tend to form a roughly bell-shaped pattern centered around the true average study time. This phenomenon—where averages of many observations become approximately normally distributed—is the essence of the Central Limit Theorem.

The origins of the Central Limit Theorem trace back to Abraham de Moivre, who observed that binomial distributions can be approximated by a normal curve for large sample sizes. Laplace later extended this idea, showing that the normal distribution arises naturally from the aggregation of many independent random effects [16, 26].

From a broader historical perspective, the Central Limit Theorem reflects a shift toward understanding randomness through structure rather than irregularity. During the nineteenth century, the normal distribution came to be seen not merely as a mathematical curiosity, but as a universal pattern underlying diverse phenomena—from measurement errors to social statistics. This viewpoint was reinforced by developments in astronomy and physics, where repeated observations revealed consistent bell-shaped variability.

While the Law of Large Numbers explains where averages converge, it does not describe how they fluctuate around their limiting value. The Central Limit Theorem addresses this question by describing the shape of these fluctuations.

Let $X_1, \ldots, X_n$ be independent and identically distributed random variables with mean μ and finite variance σ^2. Then

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } n \to \infty, \tag{3.4}$$

that is, the standardized sample mean converges in distribution to a standard normal random variable.

A useful informal interpretation is the following: for large n,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

This approximation becomes increasingly accurate as the sample size grows and forms the basis for many statistical methods.

There are different levels of generality in the Central Limit Theorem. The classical version assumes independence and identical distribution, while more advanced forms allow for certain types of dependence or variation between observations, provided no single term dominates the sum.

Example (Averaging Measurements): Even if individual observations are skewed or irregular, the average of many such observations tends to have an approximately bell-shaped distribution.

Example (Binomial Approximation): Let X be the number of successes in n independent trials with success probability p. For large n, the distribution of X is well approximated by a normal distribution with mean np and variance $np(1-p)$.

The Central Limit Theorem explains the widespread appearance of the normal distribution in statistics. Together with the Law of Large Numbers, it provides both a guarantee of accuracy (convergence) and a description of uncertainty (fluctuations), forming a cornerstone of statistical inference.

3.3 Bias, variance and Mean Squared Error

The systematic study of estimators and their properties developed alongside the rise of mathematical statistics in the late 19th and early 20th centuries. Early statisticians recognized that good estimation procedures must balance systematic error and random variability. This insight led to the decomposition of mean squared error into variance and squared bias, a framework that remains central in both classical statistics and modern machine learning, and that we will revisit in Chapter 5 [26, 47, 48].

Before introducing the formal definitions, it is worth pausing on what an estimator actually is. Suppose we want to know the average height of all adults in a country—again, a quantity we cannot measure directly. We instead collect a random sample and compute the sample mean, as we saw in Chapter 1. That computed number is our *estimator*: a recipe that turns data into a guess about an unknown quantity. But because the sample is random, the estimator is itself a random variable. If we repeated the study with a different sample, we would get a slightly different number. The natural question is: how good is our recipe, on average?

Two distinct things can go wrong. The estimator might be *systematically off*—always a little too high, or always a little too low. This is *bias*. Or it might be *erratic*—sometimes far too high, sometimes far too low, even if it is correct on average. This is *variance*. A good estimator should ideally be free of both, but as we will see, the two are often in tension.

Bias and variance

If $\hat{\theta}$ is an estimator of an unknown parameter θ ($\bar{x}$ for μ, s for σ), we define its *bias* as

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta \; , \tag{3.5}$$

the difference between what the estimator returns on average and the true value. If $\text{Bias}(\hat{\theta}) = 0$, the estimator is called *unbiased*: it neither systematically over- nor underestimates. In the same way, the *variance* of the estimator is

$$\text{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right] \; , \tag{3.6}$$

which measures how much $\hat{\theta}$ fluctuates around its own average across different samples.

These two quantities can be combined into a single number, the *mean squared error* (MSE), defined as the average squared distance between the estimator and the true value:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] \; . \tag{3.7}$$

The connection to bias and variance follows directly from the definition of expectation. Writing $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta)$ and expanding the square, the cross-term vanishes and one obtains

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2 \; . \tag{3.8}$$

Total error is therefore the sum of spread and systematic offset, each squared. For an unbiased estimator, the second term vanishes and MSE reduces to variance alone.

For the purposes of this chapter, MSE serves mainly as a compact summary of estimation quality. Its deeper role—and the tension it reveals between bias and variance when choosing among competing models—will be developed in Chapter 5, once we have the language of statistical modelling to make that trade-off concrete.

An analogy: Imagine throwing darts at a target. An unbiased estimator with high variance is like a player whose throws cluster around the bullseye on average, but land all over the board on any given throw. A biased estimator with low variance is like a player whose throws are tightly grouped, but systematically to the left of center. Good statistical practice aims for tight grouping *and* correct centering simultaneously.

The sample mean

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, each with mean μ and variance σ^2. The most natural estimator of the population mean μ is the *sample mean*:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \ . \tag{3.9}$$

Is the sample mean biased? We compute its expectation directly:

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \frac{1}{n}\cdot n\mu = \mu \ .$$

So $\mathrm{Bias}(\bar{X}) = \mu - \mu = 0$. The sample mean is an unbiased estimator of the population mean.

What is its variance? Since the X_i are independent,

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n} \ .$$

Two things are worth noting. First, the variance decreases as n grows: larger samples yield more stable estimates, exactly as the Law of Large Numbers predicts. Second, the square root $\sigma/\sqrt{n}$ is called the *standard error of the mean*—it measures the typical size of the fluctuations of $\bar{X}$ around μ, and it is this quantity that appears in the confidence interval formula of the next section.

Example: Suppose we measure the weight of a cereal box five times and obtain, in grams: $502, 498, 501, 499, 500$. The sample mean is

$$\bar{X} = \frac{502 + 498 + 501 + 499 + 500}{5} = \frac{2500}{5} = 500 \ .$$

If the true mean weight is $\mu = 500\,\mathrm{g}$, this particular sample happens to be exactly on target. Of course, a different set of five measurements would give a slightly different answer. The variance of the estimator, $\sigma^2/5$, tells us how large those fluctuations typically are.

The sample variance and Bessel's correction

A natural estimator of the population variance σ^2 is constructed by measuring how spread out the observations are around their mean. There are two closely related candidates. The first is the *naive variance estimator*:

$$\tilde{s}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \ . \tag{3.10}$$

The second, which differs only in the denominator, is the *sample variance*:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \ . \tag{3.11}$$

At first glance, the choice of $n-1$ rather than n looks like a typographical error. It is not. The factor $n-1$ is known as *Bessel's correction*, and it exists to correct a systematic bias that arises from a subtle but important source.

Why does the naive estimator underestimate? When we compute $(X_i - \bar{X})^2$, we measure deviations from the *sample mean* $\bar{X}$, not from the *true mean* μ. The sample mean is chosen to minimize the sum of squared deviations from itself—it is, by construction, the value that makes the observed data look as concentrated as possible. This means the deviations $(X_i - \bar{X})$ are systematically *smaller* than the true deviations $(X_i - \mu)$ would be. The sample mean is pulled toward the data in a way the true mean is not. As a consequence, the naive estimator $\tilde{s}^2$ underestimates the true spread.

The calculation. Let us verify this precisely. We wish to compute $\mathbb{E}[\tilde{s}^2]$. A useful algebraic identity is:

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2 \ .$$

Taking expectations of both sides, and recalling that $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$ and $\mathbb{E}[(\bar{X} - \mu)^2] = \sigma^2/n$:

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = n\sigma^2 - \sigma^2 = (n-1)\sigma^2 \ .$$

Therefore

$$\mathbb{E}[\tilde{s}^2] = \frac{1}{n}\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = \frac{(n-1)\sigma^2}{n} = \sigma^2 - \frac{\sigma^2}{n} \ ,$$

which is strictly less than σ^2. The naive estimator is *biased downward* by a factor of $(n-1)/n$.

The fix is immediate: divide by $n-1$ instead of n. Then

$$\mathbb{E}[s^2] = \frac{1}{n-1}\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = \frac{(n-1)\sigma^2}{n-1} = \sigma^2 \ ,$$

and s^2 is an unbiased estimator of σ^2.

Intuition for the degrees of freedom. Another way to understand the factor $n-1$ is through the concept of *degrees of freedom*. We have n observations, but we have already used them to estimate one quantity—the mean $\bar{X}$—before computing the deviations. Once the mean is fixed, only $n-1$ of the n deviations $(X_i - \bar{X})$ can vary freely: the last one is determined by the constraint that they sum to zero. We have, in a sense, *spent* one degree of freedom on estimating the mean, and only $n-1$ remain for estimating the spread.

Example: Suppose we measure the same five cereal boxes and record: $502, 498, 501, 499, 500$. The sample mean is $\bar{X} = 500$. The deviations from the mean are:

$$2, \quad -2, \quad 1, \quad -1, \quad 0 \ .$$

Note that these sum to zero, as they must. The sum of squared deviations is

$$2^2 + (-2)^2 + 1^2 + (-1)^2 + 0^2 = 4 + 4 + 1 + 1 + 0 = 10 \ .$$

The naive estimate divides by $n = 5$: $\tilde{s}^2 = 10/5 = 2.0\,\mathrm{g}^2$. The corrected sample variance divides by $n - 1 = 4$: $s^2 = 10/4 = 2.5\,\mathrm{g}^2$.

The corrected value is larger, reflecting the fact that the naive estimator systematically underestimates the true variability. For large samples, the difference between n and $n-1$ is negligible. For small samples—such as $n = 5$—the correction matters.

In many problems, we observe not one but two measurements on each individual: height and weight, temperature and pressure, exam score and hours studied. The *covariance* between two variables X and Y measures the direction and strength of their linear relationship. If tall people tend to weigh more, the covariance between height and weight is positive. If hot days tend to have lower humidity, the covariance is negative.

The sample covariance

The population covariance is defined as

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] , \tag{3.12}$$

where $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Note that $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$, so variance is a special case.

Given n paired observations $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$, the natural estimator of $\mathrm{Cov}(X, Y)$ is the *sample covariance*:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y}) . \tag{3.13}$$

The same Bessel correction applies: dividing by $n-1$ rather than n makes the estimator unbiased, for exactly the same reason as before. Both the sample mean $\bar{X}$ and the sample mean $\bar{Y}$ have been estimated from the data before computing the deviations, leaving $n - 1$ free degrees of freedom.

Example: Suppose we record the number of hours five students studied and their exam scores:

Student	Hours studied (X)	Exam score (Y)
1	2	55
2	4	65
3	6	70
4	8	80
5	10	85

The sample means are $\bar{X} = 6$ and $\bar{Y} = 71$. The deviations and their products are:

Student	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	-4	-16	64
2	-2	-6	12
3	0	-1	0
4	2	9	18
5	4	14	56
			$\sum = 150$

The sample covariance is

$$s_{XY} = \frac{150}{n-1} = \frac{150}{4} = 37.5 .$$

The positive value confirms what the data suggest: students who study more tend to score higher. Note that the units of s_{XY} are hours $\times$ points, which can be hard to interpret directly. The *sample correlation coefficient*, obtained by dividing the covariance by the product of the two sample standard deviations, rescales this to a dimensionless number between -1 and 1:

$$r_{XY} = \frac{s_{XY}}{s_X \cdot s_Y} , \tag{3.14}$$

where s_X and s_Y are the sample standard deviations of X and Y respectively.

A note on interpretation. Covariance and correlation measure linear association. Two variables can have zero covariance and yet be strongly related in a nonlinear way—a point to bear in mind when interpreting real data.

3.4 Confidence intervals and critical regions

Confidence intervals emerged as a practical response to the limitations of point estimation. Rather than asking for a single best value, statisticians sought procedures that quantify uncertainty in a repeatable and operational way. The concept was formalized in the early twentieth century as part of the frequentist framework, emphasizing long-run coverage properties under repeated sampling. This interpretation remains foundational in applied statistics and experimental science [3, 4].

Point estimates summarize data with a single number, but they do not convey uncertainty. A sample mean of 170 cm tells us our best guess for the population mean, but says nothing about how far off that guess might be. Confidence intervals address this limitation by providing a range of plausible values for an unknown parameter, together with a precise statement about how often such ranges succeed in capturing the truth.

For a population with mean μ and known variance σ^2, an approximate $100(1-\alpha)\%$ confidence interval for μ is given by

$$\bar{X}_n \pm z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}\ , \tag{3.15}$$

where $z_{\alpha/2}$ is the quantile of the standard normal distribution such that $\mathbb{P}(|Z| \leq z_{\alpha/2}) = 1-\alpha$.

To understand this formula, it helps to read it piece by piece. The term $\bar{X}_n$ is the sample mean, our best point estimate of μ. The quantity $\sigma/\sqrt{n}$ is the *standard error*: it measures how much the sample mean typically varies from sample to sample, and shrinks as n grows. Finally, $z_{\alpha/2}$ determines how wide the interval must be to achieve the desired coverage—for 95% confidence, $z_{\alpha/2} \approx 1.96$. The formula says: take the sample mean, and add and subtract a margin of error that accounts for both the natural variability of the data and the sample size.

Interpretation. Before working through the numbers, it is essential to be clear about what a confidence interval does and does not mean—because the natural reading is wrong. A 95% confidence interval does *not* mean that there is a 95% probability that the true parameter μ lies inside the interval we computed. The true μ is a fixed (if unknown) number; it either lies in the interval or it does not. What the 95% refers to is the *procedure*: if we repeated our experiment many times and constructed a confidence interval each time, approximately 95% of those intervals would contain the true value. The randomness is in the interval, not in the parameter.

Example. Suppose we collect a sample of $n = 4$ observations: $10, 12, 9, 11$. The sample mean is

$$\bar{X}_n = \frac{10 + 12 + 9 + 11}{4} = 10.5\ .$$

Assume the population standard deviation is known to be $\sigma = 2$. The standard error is $\sigma/\sqrt{n} = 2/2 = 1$. For a 95% confidence level, $z_{\alpha/2} \approx 1.96$, giving the interval

$$10.5 \pm 1.96 \times 1 = (8.54,\ 12.46)\ .$$

Based on this small sample, plausible values for μ range from roughly 8.5 to 12.5. The width of the interval reflects both the variability of the data and the limited sample size.

The σ-rules

Because of the Central Limit Theorem, sample means are approximately normally distributed for large samples. This means the coverage probabilities of intervals of the form $\bar{X}_n \pm k\,(\sigma/\sqrt{n})$ can be read directly off the standard normal distribution. The three most commonly used values are:

One standard deviation (1σ).

$$\mathbb{P}(|Z| \leq 1) \approx 0.683 ,$$

so a 1σ interval covers approximately 68% of the distribution.

Two standard deviations (2σ).

$$\mathbb{P}(|Z| \leq 2) \approx 0.954 ,$$

which is approximately 95%, and explains why $z_{\alpha/2} \approx 1.96$ appears so often in practice.

Three standard deviations (3σ).

$$\mathbb{P}(|Z| \leq 3) \approx 0.997 ,$$

so approximately 99.7% of the probability mass lies within three standard deviations of the mean.

Example. Suppose $\bar{X}_n = 170$, $\sigma = 10$, and $n = 100$, so the standard error is $\sigma/\sqrt{n} = 1$. The three intervals are:

$$170 \pm 1 \quad (68\%) , \qquad 170 \pm 2 \quad (95\%) , \qquad 170 \pm 3 \quad (99.7\%) .$$

Wider intervals are more likely to contain the true mean, but are also less informative. The choice of confidence level is a deliberate trade-off between precision and reliability.

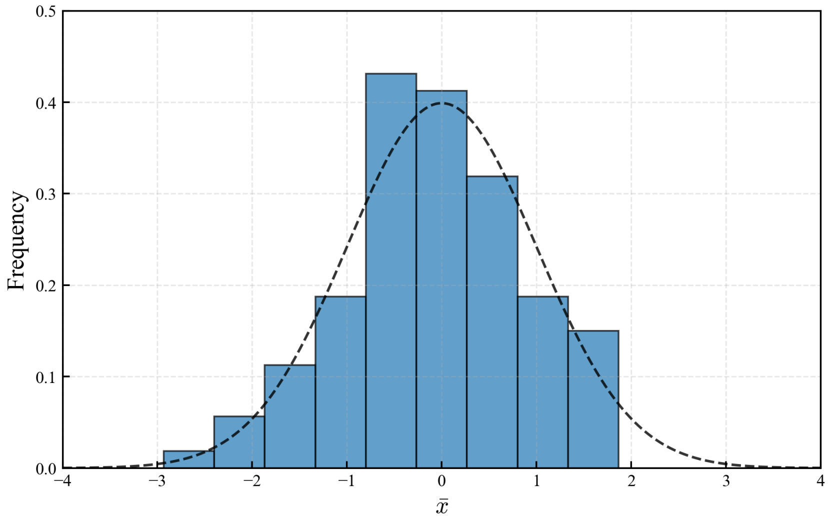
From 3σ to 5σ: evidence and discovery

In some scientific fields, particularly physics, much stronger evidence is required before announcing a discovery. The reason is partly philosophical and partly practical: when searching for a rare signal among many possible measurements, a result that would occur by chance one time in a hundred—a 2σ fluctuation—is simply not surprising enough to be convincing.

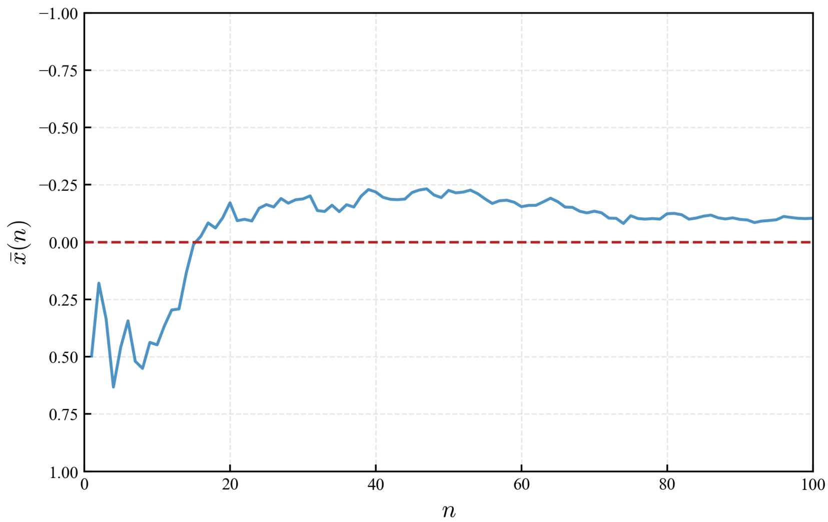
The convention in particle physics is that a 3σ result ($p \approx 0.003$) constitutes *evidence* for a new phenomenon, while a 5σ result ($p \approx 3 \times 10^{-7}$, or roughly one chance in three million) is required to claim a *discovery*. This standard was famously applied in July 2012, when physicists at CERN announced the discovery of the Higgs boson: both the ATLAS and CMS experiments independently reported signals exceeding the 5σ threshold, ruling out a statistical fluctuation to an exceptionally high degree of confidence. The strictness of this criterion reflects the asymmetry between the cost of a false claim—announcing a particle that does not exist—and the cost of waiting longer for more data.

Confidence intervals and hypothesis tests

Confidence intervals and hypothesis tests are two sides of the same coin. A $100(1-\alpha)\%$ confidence interval for μ consists precisely of all the values μ_0 that would *not* be rejected by a two-sided hypothesis test at significance level α. Equivalently, a value μ_0 lies outside the confidence interval if and only if the data is sufficiently far from μ_0 to reject it. This duality means that constructing a confidence interval and performing a hypothesis test are, in a formal sense, the same procedure viewed from different angles—one reports a range of compatible values, the other delivers a binary decision about a specific value. We will return to this connection in the next chapter, where hypothesis testing is developed in full.

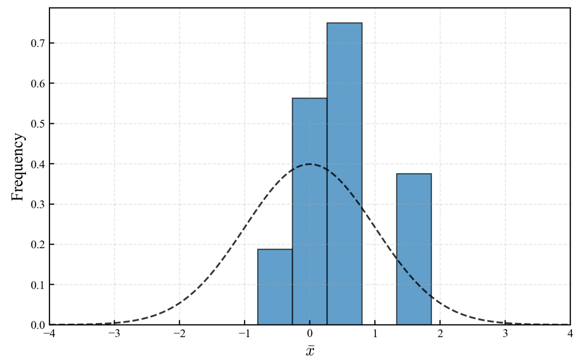


(a) Histogram of the sample mean $\bar{x}$ with a Gaussian fit.

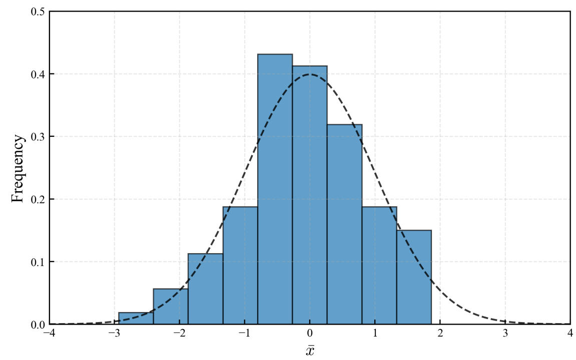


(b) Running sample mean $\bar{x}(n)$ converging to the expected value μ, as the sample size increases.

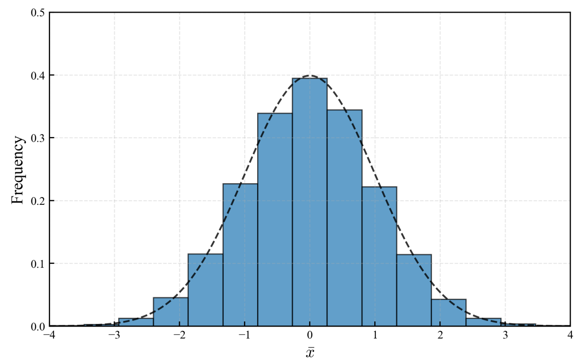
Figure 3.1: The Law of Large Numbers: For a set of independent and identically distributed (iid) random variables, the sample mean $\bar{x}$ asymptotically tends to the expected value μ, as the sample size n increases.



(a) Histogram of sample mean for a sample size $n = 10$ and Gaussian fit (dashed).

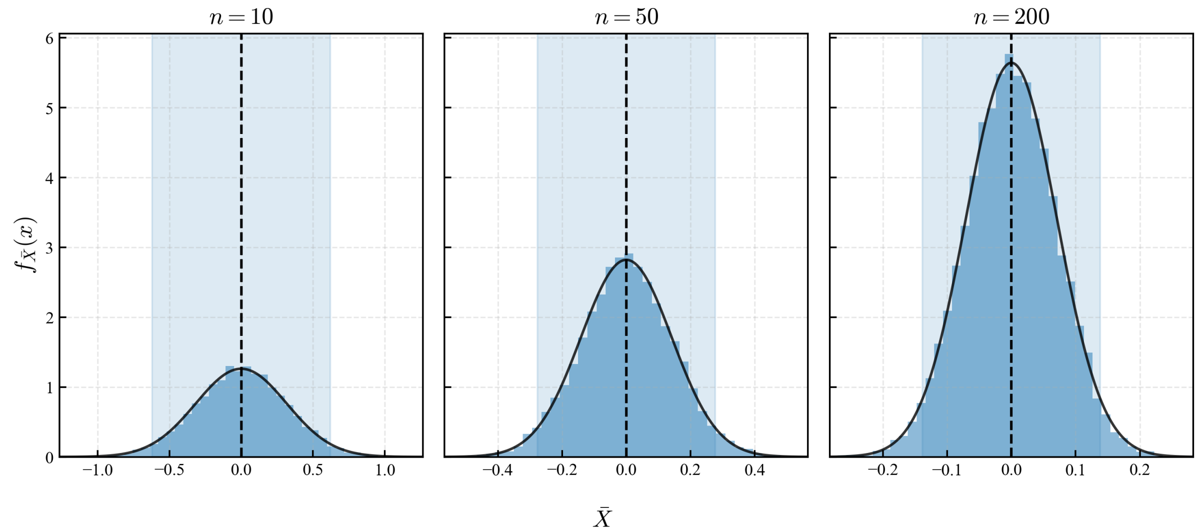


(b) Histogram of sample mean for a sample size $n = 50$ and Gaussian fit (dashed).

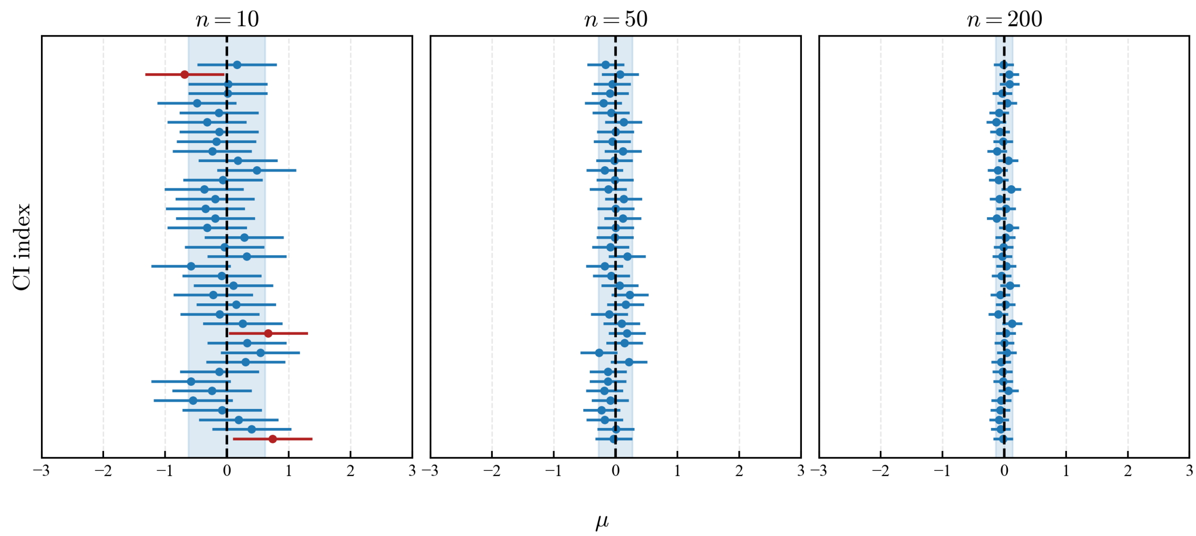


(c) Histogram of sample mean for a sample size $n = 100$ and Gaussian fit (dashed).

Figure 3.2: The Central Limit Theorem: For a set of independent and identically distributed (iid) random variables, the sample mean $\bar{x}$ asymptotically tends to a Gaussian distribution, as the sample size n increases.



(a) Sample mean distribution, illustrating convergence to a perfect Gaussian curve as the sample size n increases, as expected under the Law of Large Numbers and the Central Limit Theorem.



(b) Confidence intervals: As the sample size n increases, a narrower region from the estimator (e.g. sample mean, sample variance) is more likely to contain the true population parameter.

Figure 3.3: Confidence intervals and critical regions

Chapter 4

Introduction to hypothesis testing

The object of statistical science is the reduction of data to relevant information.

— Ronald A. Fisher

The term *hypothesis testing* lies on top of the two pillars we have mentioned in previous chapters. From statistical analysis, as discussed in Chapter 1, we will use sampling, estimators, and graphical summaries to describe data. From probability theory, as discussed in Chapter 2, we will rely on probability distributions and expected values to model random variation and uncertainty. Throug this chapter, we will mostly work in a *parametric* setting, where data is assumed to follow a simple, smooth, and well-behaved distribution—most often the Gaussian distribution. Under these assumptions, and supported by the Law of Large Numbers and the Central Limit Theorem, estimators such as the sample mean and variance provide reliable information about the true population parameters. Confidence intervals and critical regions, introduced in Chapter 3, formalize this idea of reliability.

Within this framework, we introduce the notion of *test statistic*, also referred to as *statistic*, or *statistic test*: a numerical quantity computed from data, designed to measure how close our observations are to what we would expect under a given hypothesis. Statistical tests are built by comparing such quantities—such *statistics*—to their behavior under an assumed model. In modern hypothesis testing, this comparison is often summarized through the *p-value*, which quantifies how unusual the observed data would be if the null hypothesis were true. We will discuss such topics in detail in the throughout the chapter.

Let us begin with a brief historical note, often skipped. The ideas behind hypothesis testing did not emerge all at once. While the general notion of testing expectations against observations is very old, the mathematical tools required for modern statistical testing are relatively recent. As we have seen, systematic use of estimators and probability models developed gradually during the 19^{th} and early 20^{th} centuries, and the axiomatic foundations of probability were only formalized in 1933 with Kolmogorov's work [23].

The first statistical tests were developed in the early 20^{th} century by researchers such as Pearson and Fisher, among many others, and were originally designed to measure discrepancies between data and theoretical expectations, not to support automatic accept–reject decisions. The formal structure of hypothesis testing, including the very idea of null and alternative hypotheses, error rates, and decision rules, was later introduced by Jerzy Neyman and Egon Pearson, son of Karl Pearson. Because these ideas developed in stages, their interpretation requires some care. Throughout this chapter, we will emphasize both the practical use of statistical tests and the assumptions on which they are based.

It is worth noting, already at this stage, that the interpretation of statistical tests is not purely technical, but also deeply philosophical. Fisher's original conception of significance testing treated the p-value as a flexible measure of evidence against a null hypothesis, while the Neyman-Pearson framework emphasized long-run error control and decision rules [49]. These distinct viewpoints connect more broadly to 20$^{\text{th}}$ century debates about scientific inference, prediction, and falsification, notably in the work of mathematicians and philosophers such as Reichenbach, Popper, and, more recently, de Finetti and Mayo.

All these philosophical issues provide important context for understanding both the strengths and the limitations of modern hypothesis testing, and will be discussed at the end of the chapter, to motivate and introduce the very idea of Bayesian probability.

4.1 Prediction vs inference revisited

As we saw in previous chapters, when formulating hypotheses about natural phenomena, it is important to recall the distinction between population and sampling. On the one hand, we consider an idealized and generally inaccessible population, characterized by true but unknown quantities such as the mean and variance (μ, σ^2). Mathematical prediction, in this context, refers to the computation of expected values—also known as statistical moments—defined for a random variable together with its probability distribution. For a review on random variables and probability distributions, please go back to Chapter 2.

Given a random variable, expected values (or moments) can be computed directly from its probability distribution.

Population mean for a discrete random variable x and possible outcomes $\{x_i\}$:

$$\mu = \mathbb{E}[x] = \sum_i x_i \, \mathbb{P}(x = x_i) \tag{4.1}$$

Population variance for a discrete random variable x and possible outcomes $\{x_i\}$:

$$\sigma^2 = \mathbb{E}\big[(x-\mu)^2\big] = \sum_i (x_i - \mu)^2 \, \mathbb{P}(x = x_i) \tag{4.2}$$

Population mean for a continuous random variable x with density $f(x)$:

$$\mu = \mathbb{E}[x] = \int_{-\infty}^{\infty} x \, f(x) \, dx \tag{4.3}$$

Population variance for a continuous random variable x with density $f(x)$:

$$\sigma^2 = \mathbb{E}\big[(x-\mu)^2\big] = \int_{-\infty}^{\infty} (x-\mu)^2 \, f(x) \, dx \tag{4.4}$$

In any of these cases, we compute cumulative probabilities as the sum of all cases above / below a certain value, or in a given range. In the case of continuous random variables, following density distributions, this sum is computed as the definite integral of the distribution.

Hypotheses are therefore always formulated in terms of mathematical predictions about the *population-level parameters.* For example, under Newtonian mechanics, a hypothesis may consist of Newton's second law, from which predictions can be made about the position of a falling object as a function of time. In biology, a hypothesis may concern the effect of a gene on a disease or on stress response, formulated in terms of expected expression levels or count. In epidemiology or medical sciences, about the relationship between smoking prevalence and lung cancer risk. In all such cases, hypotheses concern population properties, while access to them is obtained only through finite samples of observations, collectively referred to as *data.* Out of that data, we compute estimators sucn as the sample mean and sample variance ($\bar{x}$, s^2). For a review on describing populations and statistical estimators, please go back to Chapter 1.

Observed sample mean for a sample $\mathcal{X} = \{x_1, x_2, \ldots, x_n\}$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{4.5}$$

Observed sample variance for the same sample

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \tag{4.6}$$

Hypothesis testing formalizes this comparison between population-level predictions and sample-based estimates by quantifying how compatible the observed data are with a hypothesized model.

4.2 General approach to hypothesis testing

When dealing with hypotheses, predictions, experiments, and data, there exist many approaches and formulations, as many as instruments, scales, and fields of study. These approaches change from one field to another, and they also change over time. Our very ideas of hypothesis, prediction, measurement, and scientific law have evolved historically. Nowadays, when people refer to *hypothesis testing*, they usually mean a very specific approach: an almost algorithmic set of rules applied broadly to inference and data analysis problems. In this chapter, we will define such approach as the *modern* or *general* approach to hypothesis testing. It assumes basic notions of probability theory, randomness, and probability distributions, together with statistical concepts such as estimators and sample-based descriptions. The core ideas of statistical tests, p-values, and significance that we discuss here are relatively recent, tracing back to the work of Pearson, Fisher, and Neyman in the early 20$^{\text{th}}$ century [49].

The general approach to hypothesis testing can be summarized in the following steps:

- **Formulate hypotheses.** One specifies a *null hypothesis* H_0, representing the expected or reference case, and an *alternative hypothesis* H_1, representing a departure from H_0 that would be considered scientifically relevant or surprising.
- **Experiment, measurement, observation.** Any process—regardless of instrumentation or field of study—that produces measurements or observations, resulting in one or more samples of data.
- **Compute a statistic or test statistic.** From the observed data, one computes an informative quantity, which may be a simple estimator such as the sample mean or variance, or a more elaborate statistic designed to quantify how far the observed data deviate from what is expected under H_0.
- **Compute a p-value.** The p-value is the probability that, assuming the null hypothesis and its associated population parameters are true, one would obtain a value of the statistic at least as extreme as the one observed.
- **Interpret the result.** The p-value is compared to a chosen significance level, leading to a decision or conclusion, commonly phrased as rejecting or not rejecting the null hypothesis. This is already a subtle topic with open philosophical questions, as we will see.

A few remarks are in order regarding this general roadmap. A statistic may be as simple as an estimator, such as the sample mean, or a more abstract quantity derived from the data. Fisher originally introduced the p-value as a continuous measure of evidence against the null hypothesis, rather than as a strict decision rule. The widespread practice of fixed significance thresholds and accept–reject decisions reflects a later synthesis of Fisher's ideas with the Neyman-Pearson framework. As a result, the modern approach to hypothesis testing combines elements of both traditions, a point we will return to later in this chapter. *Warning.* A p-value *does not* measure the probability that a hypothesis is true or false, but rather how compatible the observed data are with the null hypothesis under the assumed model.

4.3 Statistical tests: common examples

4.3.1 One-sample t-test: compare sample mean with hypothesized value

The one-sample t-test is arguably the simplest example of a statistical test we will discuss. It was developed in 1908 by William S. Gosset, a statistician working at the Guinness brewery in Dublin, who was concerned with accurately estimating the variability of the sample mean when the population variance is unknown, particularly for small sample sizes. Due to restrictions imposed by his employer, Gosset published his work in the *Biometrika* journal under the pseudonym *Student*. For this reason, the test is still widely known as the *Student's t-test* [50].

The test begins by formulating a null hypothesis about the true population mean *prior to any data collection*. Typically, the null hypothesis is written as

$$H_0 : \mu = \mu_0,$$

where μ denotes the true (and generally inaccessible) population mean, and μ_0 is a hypothesized reference value. As emphasized in Chapter 2, such population-level quantities can be predicted mathematically through expected values, or estimated empirically from observed data.

We then collect a sample of observations

$$\mathcal{X} = \{x_1, x_2, \dots, x_n\},$$

from which we compute the sample mean $\bar{x}$ as an estimator of μ, and the sample standard deviation s as an estimator of the unknown population standard deviation σ.

Combining these elements, we define the one-sample t-statistic as

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}\ . \tag{4.7}$$

This statistic quantifies how far the observed sample mean deviates from the hypothesized value, relative to the estimated variability of the data. In particular, as $\bar{x} \to \mu_0$, we have $t \to 0$.

Because $\bar{x}$ and s depend on random samples, the statistic t itself is a *real-valued random variable*, and it will follow *some* distribution. Under the null hypothesis H_0, its probability distribution is given by the *Student's t-distribution*,

$$f(t;\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \tag{4.8}$$

where $\nu = n - 1$ denotes the number of *degrees of freedom*. It can be demonstrated that as $\nu \to \infty$, the t-distribution converges to the standard normal distribution.

Once the distribution of the test statistic is known, we compute the p-value. Following Fisher's original definition, the p-value is the probability of obtaining a value of the statistic at least as extreme as the one observed, assuming that the null hypothesis H_0 is true.

For a one-sided test, this probability is given by

$$p_{\text{one-sided}} = \mathbb{P}(t \geq t_{\text{obs}}) = \int_{t_{\text{obs}}}^{\infty} f(t;\nu)\,dt,$$

while for a two-sided test,

$$p_{\text{two-sided}} = \mathbb{P}(|t| \geq |t_{\text{obs}}|) = 2\int_{|t_{\text{obs}}|}^{\infty} f(t;\nu)\,dt.$$

Example. For a sample of size $n = 10$, with observed mean $\bar{x} = 5.2$, sample standard deviation $s = 1.0$, and hypothesized value $\mu_0 = 5$, the observed statistic is

$$t_{\text{obs}} = \frac{5.2 - 5}{1/\sqrt{10}} \approx 0.63.$$

With $\nu = 9$ degrees of freedom, the corresponding two-sided p-value is approximately $p \approx 0.54$.

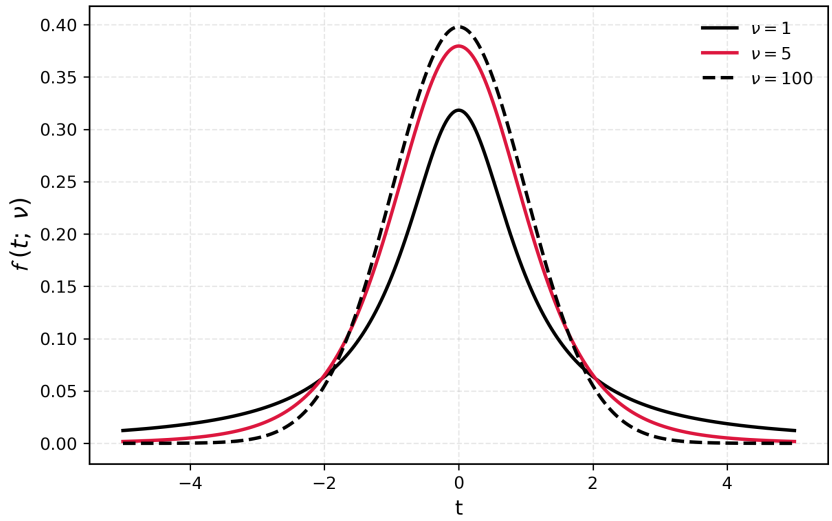


Figure 4.1: The Student's t-distribution for different values of the degrees of freedom ν. As they increase ($\nu \to \infty$), the distribution asymptotically approaches a Gaussian (Normal) curve.

4.3.2 Two-sample t-test: compare sample means of two independent groups

The two-sample t-test extends the ideas of the one-sample case to the comparison of two independent samples,

$$\mathcal{X}_1 = \{x_{1,1}, \ldots, x_{1,n_1}\}, \qquad \mathcal{X}_2 = \{x_{2,1}, \ldots, x_{2,n_2}\}.$$

The null hypothesis now concerns the equality of the two population means,

$$H_0 : \mu_1 = \mu_2,$$

where μ_1 and μ_2 denote the true means of the populations from which the samples are drawn.

From the observed data, we compute the sample means $\bar{x}_1$, $\bar{x}_2$ and sample variances s_1^2, s_2^2, which serve as estimators of the corresponding population quantities.

A commonly used test statistic for this problem is given by

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}\,. \tag{4.9}$$

This statistic measures the difference between the two sample means relative to the estimated standard error of that difference. As $\bar{x}_1 \to \bar{x}_2$, we again have $t \to 0$.

This formulation corresponds to *Welch's two-sample t-test*, which does not assume equal population variances. Under the null hypothesis, the statistic approximately follows a Student's t-distribution, with degrees of freedom given by the Welch–Satterthwaite approximation. As in the one-sample case, the p-value is obtained by integrating the appropriate tail(s) of the t-distribution [51].

An alternative formulation is the two-sample t-test with *pooled variance*, which additionally assumes that the two populations have the same variance, $\sigma_1^2 = \sigma_2^2$. Under this assumption, the sample variances s_1^2 and s_2^2 are combined into a single pooled estimator of the common variance, leading to a simplified standard error and a Student's t-distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom. While this test is exact under normality and equal variances, it is more sensitive to heteroscedasticity than Welch's test and is therefore less robust when the equality of variances is in doubt.

4.3.3 Fisher's F-test: compare variances of two independent groups

Fisher's variance-ratio test, commonly known as the F-test, was introduced in the 1920s and formally developed in Fisher's foundational works *"Statistical Methods for Research Workers"* (1925) and *"The Design of Experiments"* (1935). Fisher's contributions to statistics, experimental design, and mathematical modelling are widely regarded as, by far, among the most influential of the 20$^{\text{th}}$ century [52, 53, 54].

The F-test begins by formulating a null hypothesis about the equality of population variances *prior to any data collection*. In its simplest form, the null hypothesis is

$$H_0 : \sigma_1^2 = \sigma_2^2,$$

where σ_1^2 and σ_2^2 denote the true (and generally inaccessible) variances of the two populations under study. As in previous examples, these population-level quantities may be hypothesized based on prior knowledge or theoretical considerations, as discussed in Chapter 2.

We then collect two independent samples

$$\mathcal{X}_1 = \{x_{1,1}, \ldots, x_{1,n_1}\}, \qquad \mathcal{X}_2 = \{x_{2,1}, \ldots, x_{2,n_2}\},$$

from which we compute the sample variances s_1^2 and s_2^2, serving as estimators of σ_1^2 and σ_2^2, respectively. The Fisher F-statistic is defined as the ratio

$$F = \frac{s_1^2}{s_2^2} \,. \tag{4.10}$$

By construction, when the sample variances are similar, the statistic remains close to one, and in the ideal case $s_1^2 \to s_2^2$, we have $F \to 1$. As in earlier examples, the test statistic is designed so that, under the null hypothesis, it takes a simple and characteristic value.

Unlike the t-test, where the null hypothesis appears explicitly as a parameter in the definition of the statistic, the F-statistic is defined purely as a ratio of estimators. Historically, this reflects Fisher's focus on sampling distributions rather than on explicit parameter-based hypotheses. The null hypothesis is encoded implicitly through the condition under which the ratio of sample variances follows a specific probability distribution.

Because the statistic F is computed from random samples, it is itself a real-valued random variable. Under the null hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$, its distribution is given by Fisher's *F-distribution*,

$$f(F;\nu_1,\nu_2) = \frac{1}{B\left(\frac{\nu_1}{2},\frac{\nu_2}{2}\right)} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} F^{\nu_1/2-1} \left(1 + \frac{\nu_1}{\nu_2}F\right)^{-(\nu_1+\nu_2)/2}, \tag{4.11}$$

where $\nu_1 = n_1 - 1$ and $\nu_2 = n_2 - 1$ denote the degrees of freedom. Some texts denote the distribution by F_{ν_1,ν_2}; here we reserve F for the statistic itself and $f(F;\nu_1,\nu_2)$ for its probability density. As both degrees of freedom increase, the distribution becomes increasingly concentrated around $F = 1$, and the logarithm of F is approximately normally distributed.

Once the sampling distribution is specified, the p-value is computed by integrating the appropriate tail of the F-distribution. For a one-sided test,

$$p_{\text{one-sided}} = \mathbb{P}(F \geq F_{\text{obs}}) = \int_{F_{\text{obs}}}^{\infty} f(F;\nu_1,\nu_2)\,dF.$$

Because the F-distribution is asymmetric, two-sided p-values are defined by doubling the smaller tail probability,

$$p_{\text{two-sided}} = 2\min\left\{\int_0^{F_{\text{obs}}} f(F;\nu_1,\nu_2)\,dF,\ \int_{F_{\text{obs}}}^{\infty} f(F;\nu_1,\nu_2)\,dF\right\}.$$

Example. If $n_1 = n_2 = 10$, with sample variances $s_1^2 = 4$ and $s_2^2 = 2$, then

$$F_{\text{obs}} = 2, \qquad \nu_1 = \nu_2 = 9,$$

yielding a one-sided p-value of approximately $p \approx 0.12$.

Remark (Fisher's exact test).

It is worth noting that not all of Fisher's contributions to hypothesis testing rely on continuous distributions or asymptotic arguments. In situations involving small samples and categorical data—most notably 2×2 contingency tables—Fisher introduced what is now known as *Fisher's exact test*. Rather than comparing estimators such as means or variances, this test is based on the exact sampling distribution of cell counts under a fixed-margins assumption, which follows a hypergeometric distribution. As in the variance-ratio test, the null hypothesis is encoded through the conditions under which the observed statistic has a known distribution. We will return to this test later when discussing inference for categorical data and contingency tables.

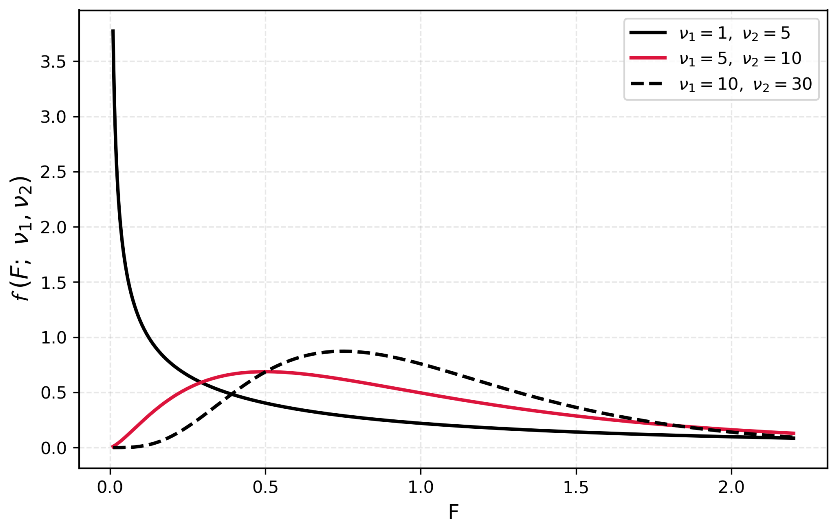


Figure 4.2: The Fisher's F-distribution for different values of the degrees of freedom ν_1, ν_2. As they increase ($\nu_1, \nu_2 \to \infty$), the distribution asymptotically approaches a Gaussian (Normal) curve.

4.3.4 Fisher's ANOVA: compare variability across multiple groups

Fisher's analysis of variance (ANOVA) generalizes the variance-ratio F-test to more than two groups. Developed in the 1920s in the context of experimental design, ANOVA addresses the question of whether observed differences between several group means can plausibly be attributed to random variation alone [53].

In essence, ANOVA compares the variability between group means to the variability within groups.

Consider k independent samples

$$\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_k,$$

with sizes $n_1, n_2, \ldots, n_k$, sample means $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_k$, and overall mean $\bar{x}$. The null hypothesis is

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k,$$

stating that all groups share the same true population mean.

The ANOVA F-statistic is defined as

$$F = \frac{s^2_{\text{between}}}{s^2_{\text{within}}}, \tag{4.12}$$

where s^2_{between} and s^2_{within} represent the variation between samples, and within samples, and are computed as

$$s^2_{\text{between}} = \frac{1}{k-1} \sum_{i=1}^{k} n_i (\bar{x}_i - \bar{x})^2,$$

$$s^2_{\text{within}} = \frac{1}{N-k} \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2,$$

and $N = \sum_{i=1}^{k} n_i$. When the null hypothesis is true and group means are similar, the ratio remains close to one.

Under H_0, the ANOVA F-statistic follows a Fisher F-distribution with degrees of freedom $\nu_1 = k - 1$ and $\nu_2 = N - k$. The corresponding p-value is obtained by integrating the right tail of this distribution, exactly as in the two-sample F-test.

4.3.5 Pearson's χ^2 test: goodness-of-fit and tests of independence

The chi-square test was introduced by Karl Pearson in 1900 as part of his work on goodness-of-fit and contingency tables [55]. Pearson's original formulation was oriented toward measuring discrepancy between observed and expected frequencies, rather than toward formal decision-based hypothesis testing. Given observed counts $\{O_1, O_2, \ldots, O_k\}$ and expected counts $\{E_1, E_2, \ldots, E_k\}$, the χ^2-statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \tag{4.13}$$

was conceived as a numerical measure of lack of agreement between data and model, with larger values indicating poorer fit.[1]

As in the previous examples, the χ^2-statistic is computed from random data, and repeating the experiment would generally lead to different observed counts and hence a different value

[1] Pearson's original formulation predates the modern null/alternative hypothesis framework and the notion of Type I and Type II errors.

of χ^2. This means that χ^2 *is itself a real-valued random variable.* Under suitable conditions and sufficiently large samples, it follows a probability distribution known as the Pearson χ^2-*distribution*,

$$f(\chi^2; \nu) = \frac{(\chi^2)^{\nu/2-1}\, e^{-\chi^2/2}}{2^{\nu/2}\, \Gamma(\nu/2)}, \tag{4.14}$$

defined for $\chi^2 > 0$.

Unlike the t-test, where the degrees of freedom are always $n-1$, the degrees of freedom ν of the χ^2-test depend on the structure of the data and on how many parameters are estimated under the null hypothesis. In particular:

- For a goodness-of-fit test with k categories,

$$\nu = k - 1 - p,$$

  where p is the number of parameters estimated from the data.

- For a test of independence in an $r \times c$ contingency table,

$$\nu = (r-1)(c-1).$$

In the modern framework, the χ^2 test is formulated with an explicit null hypothesis and a p-value. One specifies a null hypothesis H_0 describing the expected distribution or independence structure, derives the asymptotic distribution $\chi^2 \sim \chi^2_\nu$ under H_0, and computes the p-value as the right-tail probability

$$p = \mathbb{P}(\chi^2 \geq \chi^2_{\text{obs}}) = \int_{\chi^2_{\text{obs}}}^{\infty} f(\chi^2; \nu)\, d\chi^2.$$

A deeper theoretical connection emerges through likelihood-based testing. Pearson's χ^2-statistic is asymptotically equivalent to the likelihood-ratio statistic $-2\log\Lambda$, a result formalized by Wilks in the 1930s [56]. Together with Wald and score tests, this establishes a unifying asymptotic framework for hypothesis testing in parametric models.

Two remarks are in order. First, the validity of the Pearson χ^2-test relies on large-sample approximations: expected counts should not be too small, and the approximation improves as sample size increases. When these conditions fail—most notably in small samples or sparse contingency tables—exact procedures such as Fisher's exact test provide a more reliable alternative. Second, the appearance of the chi-square distribution in this context is not accidental. Together with the Wald and score (Lagrange multiplier) tests, the likelihood-ratio test forms a triad of asymptotically equivalent procedures. Although these tests differ in construction, all converge to the same χ^2 distribution under the null hypothesis in large samples, providing a unifying framework for parametric inference.

A remark is in order regarding sidedness. Unlike tests such as the t-test, where deviations in either direction from the null hypothesis are meaningful, the χ^2-statistic is nonnegative by construction and measures the overall discrepancy between observed and expected frequencies. Small values of χ^2 indicate unusually good agreement with the model, whereas large values indicate poor fit. Consequently, evidence against the null hypothesis arises only through large values of the statistic, and hypothesis testing with the χ^2-distribution is inherently one-sided. While it is mathematically possible to define a two-sided p-value by symmetrizing tail probabilities, such constructions are nonstandard and rarely meaningful in practice, as unusually small values of χ^2 do not correspond to a distinct or interpretable alternative hypothesis.

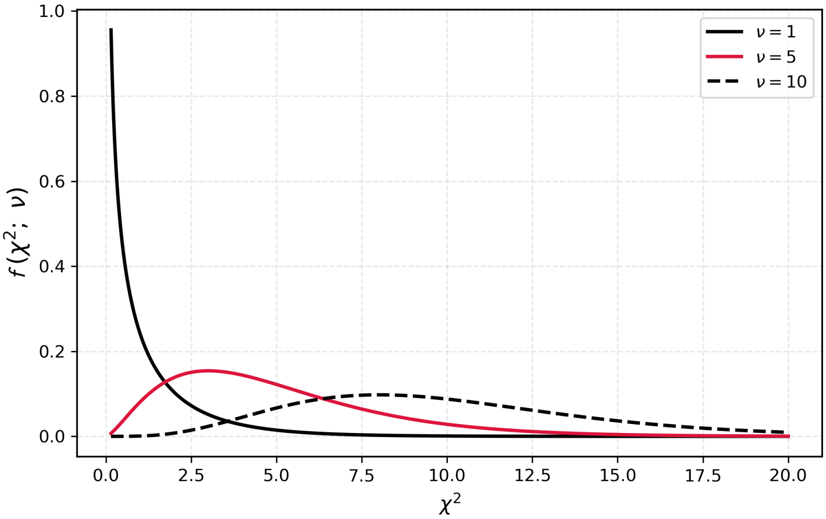


Figure 4.3: The Pearson's χ^2-distribution for different values of the degrees of freedom ν. As they increase ($\nu \to \infty$), the distribution asymptotically approaches a Gaussian (Normal) curve.

4.3.6 The Wald test: asymptotic behavior

The Wald test was developed by Abraham Wald in the 1930s as part of a general asymptotic theory of hypothesis testing for parametric models with large samples [57]. Its purpose is to test hypotheses about finite-dimensional parameters using large-sample approximations. Unlike classical tests tailored to specific settings, the Wald test provides a general framework applicable to linear models, generalized linear models, and maximum likelihood estimation.

Conceptually, the Wald test generalizes classical parametric tests such as the t- and F-tests by formulating hypotheses directly in terms of model parameters, and by relying on the asymptotic normality of estimators. In this sense, it unifies earlier procedures within a common large-sample theory.

Let $\hat{\theta}$ be an estimator of a parameter θ. The Wald statistic is defined as

$$W = (\hat{\theta} - \theta_0)^\top \widehat{\mathrm{Var}}(\hat{\theta})^{-1} (\hat{\theta} - \theta_0).$$

Under the null hypothesis and suitable regularity conditions,

$$W \xrightarrow{d} \chi^2_p,$$

where p is the number of tested constraints. Corresponding p-values are obtained from the upper tail of the chi-square distribution.

Example (one-dimensional Wald test). Consider the case $p = 1$, where θ is a scalar parameter and $\hat{\theta}$ is an estimator that is asymptotically normal,

$$\hat{\theta} \mathrel{\dot\sim} \mathcal{N}\left(\theta, \sigma^2\right).$$

Suppose we wish to test the null hypothesis $H_0 : \theta = \theta_0$. The Wald statistic then reduces to

$$W = \frac{(\hat{\theta} - \theta_0)^2}{\widehat{\sigma}^2},$$

which is simply the square of a standardized deviation. Under the null hypothesis, W converges in distribution to a χ^2_1 random variable. Equivalently, the signed statistic

$$Z = \frac{\hat{\theta} - \theta_0}{\widehat{\sigma}}$$

is asymptotically standard normal. In this one-dimensional setting, the Wald test therefore coincides with the familiar large-sample z-test, illustrating how classical tests arise as special cases of the general Wald framework.

This one-dimensional Wald formulation is used extensively in applied settings. For example, the DESeq2 software for differential gene expression analysis models log fold-changes using generalized linear models and tests individual coefficients via Wald statistics, relying on large-sample normal approximations to obtain χ^2_1 or equivalently normal-based p-values [58].

4.4 Parametric and non-parametric tests

Statistical tests are often described as *parametric* or *non-parametric*, a distinction that reflects both historical development and underlying philosophical views about statistical modelling. Parametric tests were developed first, at the beginning of the 20^{th} century, in a context where probability models were seen as idealized descriptions of data. These tests assume that observations come from a distribution which is simple smooth, and described by a small number of parameters, such as the mean and variance. Statistical inference then focuses directly on these parameters.

In practice, parametric tests are frequently associated with the Gaussian, or Normal distribution. This historical association arises because many of the foundational procedures of classical statistics—such as the t-test, the F-test, and analysis of variance—are exact under normality. As a result, the parametric versus non-parametric distinction is often informally described as "Gaussian versus non-Gaussian." This simplification is useful pedagogically, but it should be remembered that parametric models also include many non-Gaussian distributions, such as the binomial or Poisson.

Non-parametric tests emerged later, largely in the 1940s and 1950s, motivated by the recognition that real data often deviate from idealized models. Rather than assuming a specific distributional form, these methods aim to remain valid under broad and unspecified distributions. They typically rely on ranks, signs, or empirical distributions, and therefore make fewer assumptions about the shape of the data.

From a historical perspective, the development of statistical testing can be roughly organized as a timeline. Early work by Pearson and Fisher between 1900 and 1930 established the foundations of parametric inference, including the χ^2-test, the t-test, and analysis of variance. In the mid-20^{th} century, researchers such as Wilcoxon, Mann, and Whitney introduced distribution-free methods to address practical limitations of these classical procedures. Later developments, including asymptotic theory and robust statistics, provided a unifying framework that connects parametric and non-parametric approaches.

To organize the tests presented in this section, it is helpful to focus on their *goal* rather than on their label. Some tests are designed to compare central tendencies between samples, others to assess variability, and others to evaluate the overall agreement between data and a theoretical model. Viewed this way, non-parametric tests are not simply substitutes for parametric ones, but complementary tools that reflect different assumptions, historical traditions, and inferential aims.

4.4.1 Wilcoxon signed-rank test

The Wilcoxon signed-rank test was introduced by Frank Wilcoxon in 1945 as a distribution-free alternative to the one-sample and paired-sample t-tests [59]. It was motivated by situations in which normality could not be assumed but a test of central location was still desired. The parametric analogue is the one-sample or paired t-test, which tests a hypothesis about a mean.

By contrast, the Wilcoxon signed-rank test targets symmetry of the distribution about a specified location.

The test statistic is based on the ranks of the absolute deviations $|x_i - \theta_0|$, with signs retained. Under the null hypothesis of symmetry, the statistic has a known finite-sample distribution; for moderate to large samples, it is commonly approximated by a normal distribution, from which p-values are obtained.

More explicitly, let

$$d_i = x_i - \theta_0,$$

and let R_i denote the rank of $|d_i|$ among the nonzero differences. The Wilcoxon signed-rank statistic is defined as

$$W = \sum_{i:\, d_i > 0} R_i,$$

that is, the sum of the ranks corresponding to positive deviations. Under the null hypothesis of symmetry about θ_0, the distribution of W is known exactly.

4.4.2 Mann–Whitney U test

The Mann–Whitney U test, introduced by Mann and Whitney in 1947, provides a nonparametric alternative to the two-sample t-test for independent samples [60]. It was designed to compare two populations without assuming normality or equal variances. The parametric analogue is the two-sample t-test, which compares population means. The Mann–Whitney test instead assesses whether one distribution tends to produce larger observations than the other.

The test statistic U is constructed from the ranks of the pooled samples. Under the null hypothesis that the two distributions are identical, U has a known exact distribution and, asymptotically, a normal distribution. p-values are computed either exactly or via the normal approximation.

Let $R_{1,i}$ denote the rank of the i-th observation from sample 1 in the pooled sample. The Mann–Whitney statistic is

$$U = \sum_{i=1}^{n_1} R_{1,i} - \frac{n_1(n_1+1)}{2}.$$

Equivalently, U counts the number of pairs $(x_{1,i}, x_{2,j})$ such that $x_{1,i} > x_{2,j}$.

4.4.3 Levene median-based test

Levene's test was proposed by Howard Levene in 1960 as a robust alternative to Fisher's variance-ratio test [61]. The median-based version, later emphasized by Brown and Forsythe, improves robustness against non-normality. The parametric analogue is the classical F-test for equality of variances, which is highly sensitive to departures from normality. Levene's test replaces variances by absolute deviations from group centers.

The statistic is computed by applying a one-way ANOVA to the transformed data $|x_{ij} - \tilde{x}_i|$, where $\tilde{x}_i$ is the group median. Under the null hypothesis of equal spreads, the test statistic follows approximately an F distribution, from which p-values are obtained.

Formally, define the transformed observations

$$z_{ij} = |x_{ij} - \tilde{x}_i|,$$

where $\tilde{x}_i$ is the median of group i. The Levene test statistic is then the usual ANOVA F-statistic computed from the z_{ij},

$$F = \frac{s^2_{\text{between}}(z)}{s^2_{\text{within}}(z)},$$

where $s^2_{\text{between}}(z)$ and $s^2_{\text{within}}(z)$ denote the between- and within-group variances of the transformed data.

4.4.4 Kruskal–Wallis test

The Kruskal–Wallis test was introduced in 1952 by Kruskal and Wallis as a non-parametric extension of one-way ANOVA. It was motivated by the need to compare more than two groups without assuming normality. The parametric analogue is Fisher's one-way ANOVA, which tests equality of group means. The Kruskal–Wallis test instead evaluates whether the group distributions are identical [62].

The test statistic is based on the ranks of all observations:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} n_i(\bar{R}_i - \bar{R})^2 \,.$$

Under the null hypothesis, H converges in distribution to χ^2_{k-1}. p-values are computed from the chi-square distribution.

4.4.5 The Kolmogorov–Smirnov test

The Kolmogorov–Smirnov test was developed in the 1930s by Kolmogorov and later extended by Smirnov as a general goodness-of-fit procedure [63, 64]. It compares an empirical distribution to a fully specified theoretical distribution. The parametric analogue is the chi-square goodness-of-fit test, which relies on binning and asymptotic approximations. The Kolmogorov–Smirnov test instead measures the maximum discrepancy between distribution functions.

The test statistic is

$$D = \sup_x |F_n(x) - F_0(x)| \,.$$

Under the null hypothesis, D has a known distribution independent of F_0. p-values are computed from this distribution or its asymptotic form.

4.4.6 The Shapiro–Wilk test

The Shapiro–Wilk test was introduced by Shapiro and Wilk in 1965 as a powerful goodness-of-fit test specifically designed to assess normality [65]. It was motivated by the low power of general-purpose tests when applied to normal models. The parametric analogue is not a test of means or variances, but rather the assumption of normality underlying t-tests, F-tests, and ANOVA. The Shapiro–Wilk test directly targets this assumption.

The test statistic is

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \,,$$

where the coefficients a_i depend on normal order statistics. The distribution of W under the null hypothesis is obtained via approximation or simulation, and p-values are computed accordingly.

4.5 Error types in hypothesis testing

Modern hypothesis testing emerged in the early 20$^{\text{th}}$ century as an attempt to formalize uncertainty, error, and decision making in empirical science. Three major approaches—Fisherian significance testing, Neyman–Pearson hypothesis testing, and Bayesian inference—address these issues in fundamentally different ways, while later philosophical analyses by Reichenbach and Popper clarified their distinct aims. Subsequent commentators such as Cox, Mayo, and Lehmann have emphasized both the strengths of each framework and the conceptual tensions created by their later amalgamation in textbook practice. Here we explore such different approaches to hypothesis testing, with the very idea of error in the different frameworks, aiming to clarify common misconceptions.

Ronald A. Fisher introduced significance tests in the 1920s as tools for assessing the *strength of evidence* against a null hypothesis [53, 54]. In Fisher's view, a null hypothesis H_0 is a reference model, and the p-value is defined as the probability, under H_0, of observing data at least as extreme as those obtained. Small p-values indicate discordance between data and model, but Fisher rejected fixed decision thresholds and did not formalize Type II errors or power. Type I error appears implicitly as the tail probability under H_0, not as a long-run operating characteristic. Hypothesis testing, for Fisher, is evidential rather than decisional: it informs scientific judgment but does not prescribe action.

Jerzy Neyman and Egon Pearson developed a sharply different framework in the 1930s, motivated by repeated decision making [49]. Here, hypotheses H_0 and H_1 are competing models, and tests are designed to control error rates in the long run. Type I error (α) and Type II error (β) are central primitives, and optimal tests maximize power subject to a fixed α. p-values play no essential role; instead, decisions are based on pre-specified critical regions. This approach interprets hypothesis testing as a rule for action under uncertainty rather than as a measure of evidential support.

Bayesian inference, originating in Bayes's posthumous essay [43] and developed by Laplace and later subjectivists such as de Finetti [44], rejects Type I and Type II errors as fundamental concepts. Probability is interpreted as rational degree of belief, and hypotheses themselves are assigned probabilities. Inference proceeds by updating prior beliefs via Bayes' theorem to obtain posterior probabilities or Bayes factors. Hypothesis testing becomes model comparison, and decisions—if required—are made by minimizing expected loss. The Bayesian framework thus dissolves the classical error dichotomy by reframing uncertainty epistemically rather than behaviorally.

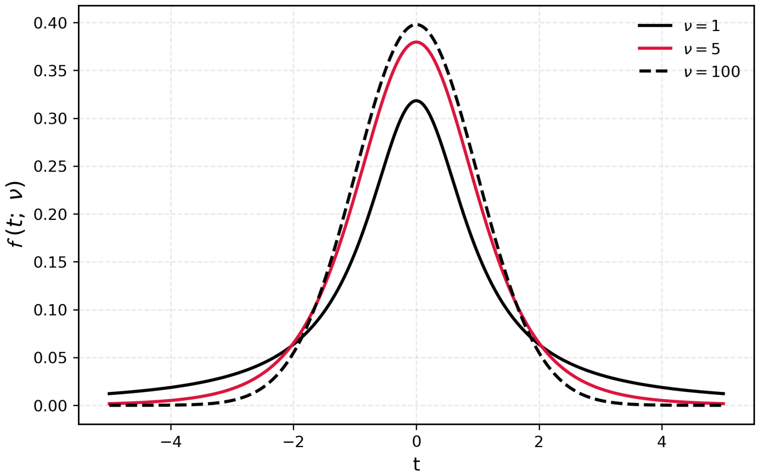
Hans Reichenbach provided the clearest philosophical articulation of the frequentist stance underlying Neyman–Pearson theory [66]. He distinguished *prediction*—statements about long-run frequencies—from *inference*—claims about truth or belief. Statistical tests, on this view, justify actions and predictions through their error properties, not through probabilistic assertions about hypotheses. This position sharply contrasts with Bayesian epistemology and clarifies why frequentist testing can function without assigning probabilities to hypotheses.

Karl Popper rejected probabilistic confirmation altogether, arguing that science advances through bold conjectures and severe attempts at falsification [67]. Statistical tests, in his view, contribute by formulating risky predictions whose failure can refute theories, not by accumulating evidence or controlling long-run errors. Popper's philosophy is incompatible with Bayesian confirmation and only partially aligned with frequentist testing, even though both emphasize error and refutation rather than belief.

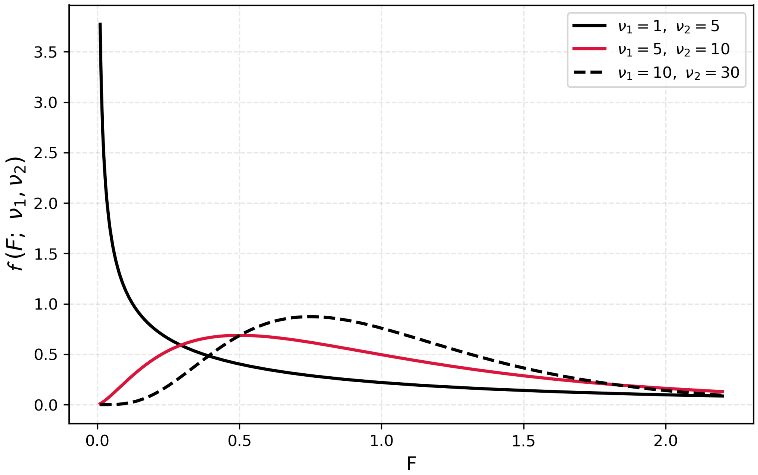
A note on contemporary studies about hypothesis testing and its interpretation. Erich Lehmann's, in his definitive work *"Testing Statistical Hypotheses"* emphasizes the formal coherence and optimality of Neyman–Pearson theory while explicitly distinguishing it from Fisher's evidential approach [68]. D. R. Cox later argues that the combination of p-values with fixed significance thresholds requires care, as both pursue distinct inferential goals [69]. More recently, Deborah Mayo further developes an error-statistical philosophy in which evidential interpretation is grounded in the severity with which hypotheses are tested [70, 71].

Together, these authors converge on a common diagnosis: the modern textbook procedure of hypothesis testing is a pragmatic but conceptually hybrid construct, blending incompatible foundations.

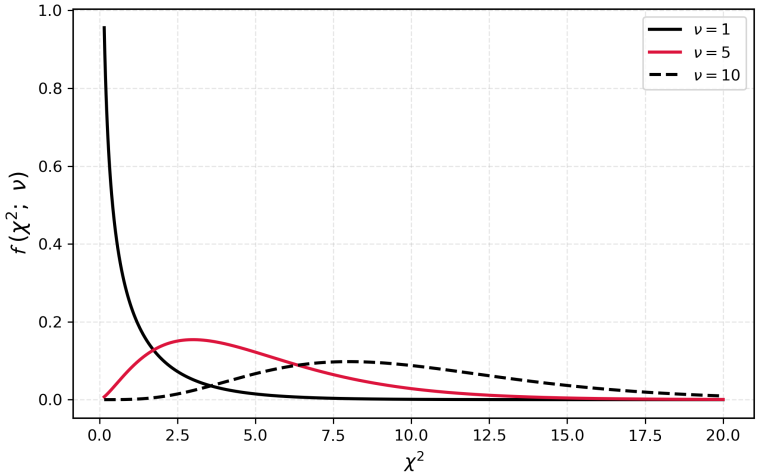
The coexistence of Fisherian evidence, Neyman–Pearson decision rules, Bayesian belief updating, Popperian falsification, and Reichenbach's predictive frequentism reflects not confusion but plurality. Each framework answers a different question—about evidence, action, belief, or prediction—and Type I and Type II errors acquire meaning only within the Neyman–Pearson decision-theoretic context. Understanding these distinctions is essential for the principled use and interpretation of hypothesis tests in modern statistics.



(a) Representation of the Student's t-distribution for different values of the degrees of freedom ν. As they increase ($\nu \to \infty$), the distribution asymptotically approaches a Gaussian(Normal) curve.

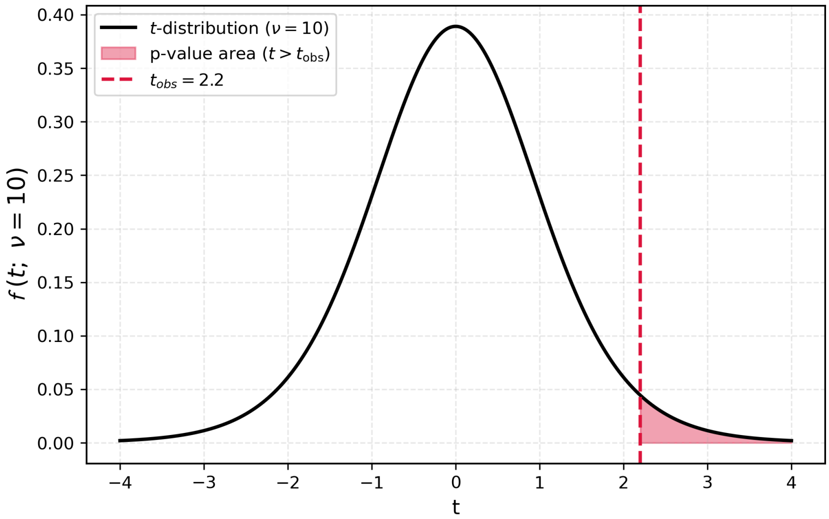


(b) Representation of the Fisher's F-distribution for different values of the degrees of freedom ν_1, ν_2. As they increase ($\nu_1, \nu_2 \to \infty$), the distribution asymptotically approaches a Gaussian (Normal) curve.

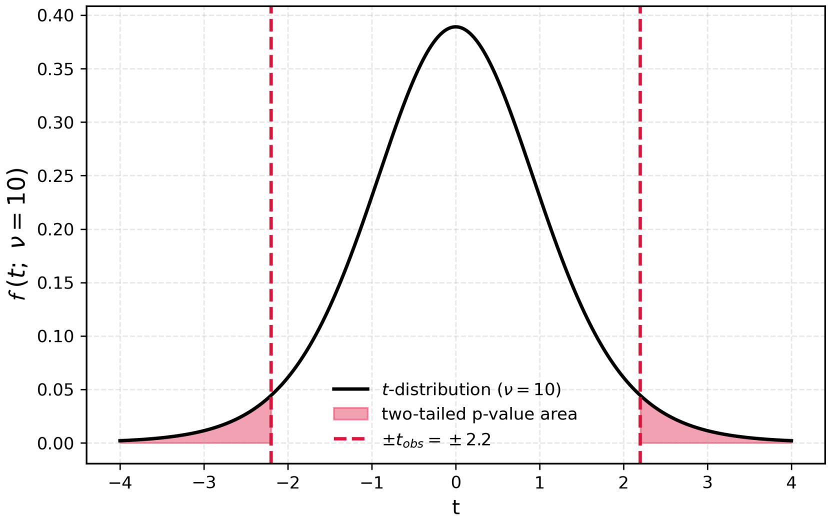


(c) Representation of the Pearson's χ^2-distribution for different value of the degrees of freedom ν. As they increase ($\nu \to \infty$), the distribution asymptotically approaches a Gaussian (Normal) curve.

Figure 4.4: Statistical distributions.

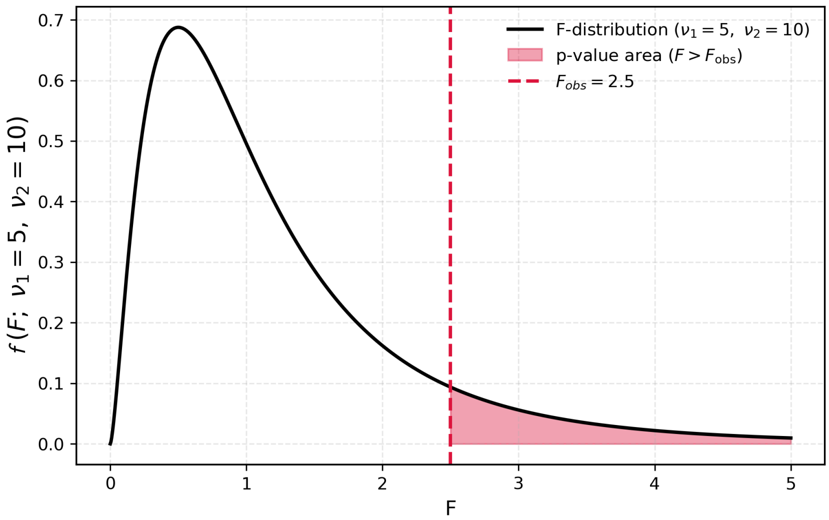


(a) Computation of the one-sided p-value, as the integral of the positive tail of the t-distribution.

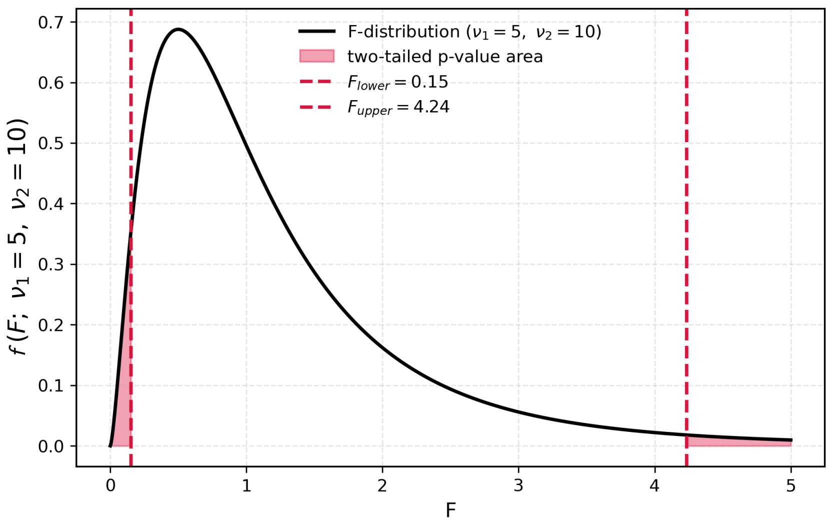


(b) Computation of the two-sided p-value, as the integral of both the positive and negative tails of the t-distribution. Given the symmetry of the t-distribution, this is exactly double the size of the one-sided p-value.

Figure 4.5: Representation of the one-sided and two-sided p-value as the cumulative probability—the integral of the tails, the area under the curve —of the Student's t-distribution.

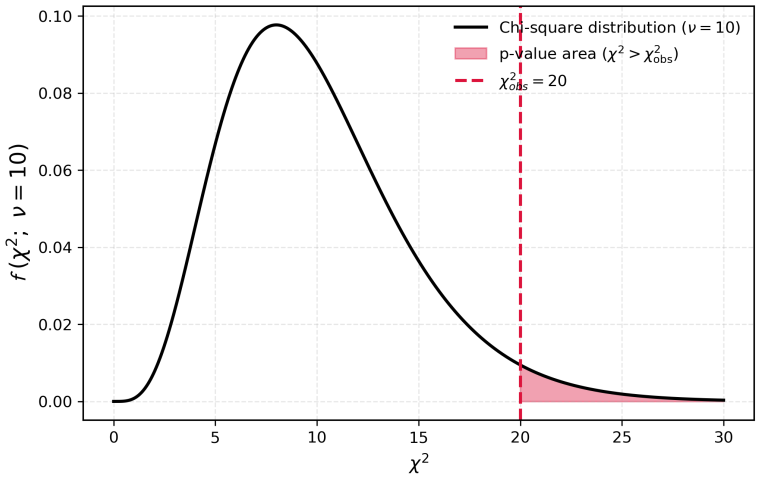


(a) Computation of the one-sided p-value, as the integral of the positive tail of the F-distribution.

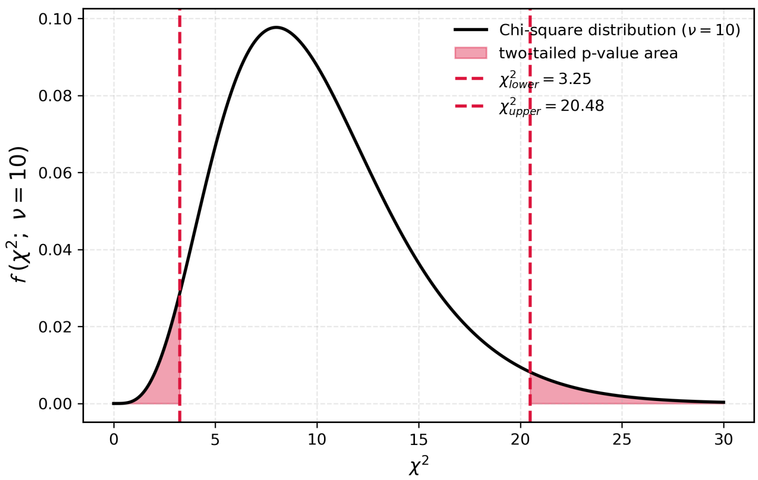


(b) Computation of the two-sided p-value, as the integral of both tails of the F-distribution. Given the asymmetry of the F-distribution, this is not double the size of the one-sided p-value, and both integrals need to be computed independently.

Figure 4.6: Representation of the one-sided and two-sided p-value as the cumulative probability—the integral of the tails, the area under the curve —of the Fisher's F-distribution.



(a) Computation of the one-sided p-value, as the integral of the positive tail of the χ^2-distribution.



(b) Computation of the two-sided p-value, as the integral of both tails of the χ^2-distribution. Given the asymmetry of the χ^2-distribution, this is not double the size of the one-sided p-value, and as we said nonstandard and rarely meaningful in practic.

Figure 4.7: Representation of the one-sided and two-sided p-value as the cumulative probability—the integral of the tails, the area under the curve —of the Pearson's χ^2-distribution.

Chapter 5

Modelling, dependency, and correlation

Numbers have an important story to tell,
if given a voice.

— Florence Nightingale

The modern scientific conception of nature rests on the idea that observable phenomena can be represented in terms of variables whose relationships follow general mathematical laws. This perspective has deep historical roots and became explicit through developments in analytic geometry and calculus, particularly through the introduction of Cartesian coordinate systems, that allowed geometric relations to be expressed algebraically. The works of Descartes and Newton, together with Kepler, Hooke, and many others, are commonly regarded as foundational in establishing this mathematical representation of physical reality [37, 72].

This shift from qualitative description to quantitative representation marked a profound transformation in scientific thought. Natural phenomena, once described in terms of causes and essences, could now be analyzed through measurable quantities and precise relations between them. Geometry and algebra, previously distinct domains, were unified into a common language in which curves, motions, and forces could be studied systematically. This mathematization of nature provided not only description, but also explanation and predictive power, foreshadowing the idea that general laws could account for a wide range of observed phenomena.

The development of calculus, rooted in the works of Newton and Leibniz, extended this framework by providing tools to describe continuous change. Relationships between quantities, as later formalized by Euler, could now be expressed as functions $y = f(x)$, whose rates of change were studied through derivatives and whose accumulated effects were analyzed through integration [73, 74]. This made it possible to model motion, growth, and variation in a unified way, capturing both instantaneous behavior and long-term evolution with a single mathematical formalism.

From this period onwards, prediction became a defining feature of scientific explanation: if the relation between variables can be expressed mathematically, then under specified assumptions one may compute future unobserved values. This predictive capacity was central to the success of classical mechanics and later to fields such as astronomy and engineering, where precise quantitative forecasts could be made from general principles.

Yet, as empirical sciences developed, it became increasingly clear that not all variability could be captured by deterministic laws alone. Measurement error, environmental influences, and intrinsic randomness introduce discrepancies between theoretical predictions and observed data. In the presence of uncertainty or stochastic variation, the attempt at prediction required what was later formalized within probabilistic modelling [66, 69]. This transition from exact laws to statistical models does not abandon the mathematical description of nature, but rather extends it to accommodate the complexity and variability inherent in real-world phenomena.

5.1 Functions and models

Since the time of Descartes [37], the idea of representing natural phenomena in terms of variables whose relationships are governed by mathematical laws has been central to science. From the deterministic laws of Newton [72] to modern statistical descriptions of uncertainty, models serve as bridges between observation and theory. As emphasized by Pearson [31], scientific laws are not direct reflections of reality, but idealized constructions that capture its regularities.

One of the central insights of this framework is that different physical mechanisms manifest themselves through distinct mathematical forms. The task of *modelling* is therefore not only to estimate parameters, but also to identify an appropriate functional structure that reflects the underlying process.

Several types of functional dependence are found quite often throughout the physical sciences and daily examples. Among the simplest examples are linear relationships, arising when the response is proportional to the input, typically near equilibrium. A classical example is Hooke's law, relating force F applied to a spring and displacement x it compresses or elongates:

$$F(x) = kx \ .$$

Quadratic relationships appear when the rate of change itself varies linearly, as in uniformly accelerated motion. The displacement of a particle x in terms of time t under constant acceleration is given by:

$$x(t) = x_0 + v_0 t + \tfrac{1}{2} a t^2 \ ,$$

where x_0 and v_0 are the initial position and velocity. Apart from polynomial dependency, periodic phenomena are described by sinusoidal functions, which model oscillatory systems such as springs, waves, and currents:

$$x(t) = A \cos(\omega t + \phi) \ .$$

Here A is the amplitude of the oscillation, ω represents the frequency, and ϕ the phase.

The importance of sinusoidal functions extends beyond simple oscillatory motion. In his study of heat conduction, Fourier showed that complex temperature distributions could be decomposed into sums of simple trigonometric functions, an insight that laid the foundations of what is now known as Fourier analysis [75]. This result demonstrated that even highly irregular phenomena, such as fire or heat waves, may be understood as superpositions of elementary components, each evolving according to simple laws. The diffusion of heat, governed by the heat equation, thus provided a striking example of how intricate physical behaviour can emerge from the combination of basic functional forms, reinforcing the central role of mathematical representation in modelling natural systems.

As a final example, exponential laws characterize processes in which the rate of change is proportional to the current state. Such behaviour arises in population dynamics, electrical circuits and radioactive decay, where the radiation counts N decrease exponentially with time:

$$N(t) = N_0 e^{-\lambda t} \ .$$

In practice, observed data rarely follow these idealized forms exactly. Measurement error, environmental variability, and incomplete knowledge introduce deviations from deterministic laws. This motivates the transition from purely functional relationships to statistical models, where such discrepancies are explicitly represented and analyzed.

5.2 Statistical modelling

Classical mathematical models describe relationships between variables as *exact* functional dependencies. In such *deterministic* settings, once the input is specified, the output is uniquely determined. This perspective underlies much of classical physics and analysis, where laws are expressed in the form $y = f(x)$. We normally refer to x as the *independent* variable, also sometimes as *explanatory*, while y is called the *dependent*, or *response* variable.

However, real data rarely follows such idealized relationships. Measurements are subject to error, systems are influenced by unobserved factors, and natural phenomena exhibit intrinsic variability. As a result, identical inputs may lead to different observed outcomes. This motivates the introduction of randomness into models.

The incorporation of uncertainty into scientific reasoning developed progressively, notably in the work of Laplace [16], who interpreted probability as a measure of uncertainty, and later in the axiomatic formulation of Kolmogorov [23], which provided a rigorous mathematical foundation. Within this framework, observable quantities are modeled as random variables X, Y and a statistical model takes the form

$$Y = f(X) + \varepsilon \; , \tag{5.1}$$

where $f(X)$ represents the systematic, deterministic component and ε is a random error term capturing deviations from the idealized relationship. This is why statistical models are referred to as *stochastic*, to distinguish them from the purely deterministic mathematical models.

The use of uppercase letters X, Y, quite common in statistics textbooks, reflects this probabilistic viewpoint: they denote random variables, that is, quantities whose values are not fixed but distributed according to some underlying probability law. Observed data correspond to particular realizations—observations—of these variables, typically denoted by lowercase x_i, y_i. This distinction, while not always emphasized in purely mathematical treatments, is central in statistics.

5.3 Linear Regression

The concept of regression originates in the work of Galton [76], who observed that extreme characteristics in parents tend to be less extreme in their offspring, a phenomenon he termed "regression toward mediocrity." This empirical insight was later formalized and extended through the development of correlation and regression analysis by Pearson [32], and placed within a broader probabilistic framework by Fisher [53].

A natural starting point, and among the simplest behaviors, is the *linear*. This leads to the simple linear regression model. In purely mathematical contexts, linear relation is written in the form

$$y(x) = ax + b + \varepsilon \; , \tag{5.2}$$

where lowercase notation emphasizes deterministic functions evaluated at specific inputs. The statistical formulation, instead, distinguishes between random variables X, Y and their realizations, and often uses the notation β_0, β_1 to denote unknown parameters to be estimated. Both representations describe the same underlying idea, but reflect different conceptual perspectives: deterministic versus probabilistic.

$$Y = \beta_0 + \beta_1 X + \varepsilon \; . \tag{5.3}$$

In this formulation, β_0 represents a baseline level or *intercept*, while β_1 quantifies the average change in Y associated with a unit change in X, also known as *slope*. The error term ε captures all sources of variation not explained by the linear relationship.

Having specified a model, one must determine how well it describes the observed data. In practice, the values predicted by a model rarely coincide exactly with observations. The differences between observed values y_i and the ones predicted by the model $\hat{y}_i$ are called *residuals*:

$$e_i = y_i - \hat{y}_i \ . \tag{5.4}$$

These residuals provide a quantitative measure of the discrepancy between the model and reality. The problem of estimation may then be formulated as one of finding parameters that make these discrepancies—the residuals—as small as possible.

For that purpose, a mathematical quantity commonly defined is the *sum of squared residuals*:

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

A model which is very accurate will produce predictions very close to the observations, hence producing a very small sum of residuals. This principle, known as the method of least squares, was first introduced by Legendre [77] in the context of astronomical calculations, and later given a probabilistic justification by Gauss [15], who showed that it arises naturally when observational errors are normally distributed. Through his work *"Theoria motus corporum coelestium"*, trying to describe planetary motions, he developed this method of least squares, building a meeting point between geometry, algebra, and probability.

The validity and interpretation of the resulting estimates rely on several assumptions:

- **Linearity:** the conditional mean is a linear function of X:

  $$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x.$$

  This means that, on average, Y changes at a constant rate as X changes. The data may be scattered, but the underlying trend follows a straight line.

- **Independence:** the error terms are independent:

  $$\varepsilon_i \text{ independent of } \varepsilon_j \quad (i \neq j).$$

  This means that the error made for one observation does not influence the error for another. In simple terms, knowing one data point does not give information about the error in another.

- **Homoscedasticity:** the variance of the errors is constant:

  $$\mathrm{Var}(\varepsilon_i \mid X) = \sigma^2.$$

  This means that the spread of the data around the regression line is roughly the same for all values of X. The variability does not increase or decrease as X changes.

- **Normality:** the errors follow a normal distribution:

  $$\varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

  This means that most errors are small, large deviations are rare, and positive and negative errors are equally likely. This assumption is mainly important when making statistical inference (e.g. confidence intervals, tests).

These assumptions do not merely simplify analysis; they determine the conditions under which the model provides reliable inference.

5.4 Covariance and Correlation

Beyond modelling the relationship between a response and a predictor, it is often of interest to quantify the degree to which two variables vary together. The earliest systematic attempts to measure such dependence arose in the late nineteenth century, notably in the work of Pearson [32], who developed the concepts of covariance and correlation as tools for studying biological and social data.

Covariance:

Covariance provides a measure of joint variability between two random variables X, Y:

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] \; . \tag{5.5}$$

A *positive* covariance indicates that the variables tend to *increase together*, while a *negative* value indicates *opposite tendencies*. However, covariance depends on the scale of measurement and is therefore difficult to interpret in isolation.

To illustrate the computation of covariance, consider two short data sets:

$$x = (1, 2, 3), \quad y = (2, 4, 6).$$

Their sample means are $\bar{x} = 2$ and $\bar{y} = 4$. The covariance is obtained by averaging the products of deviations from the means:

$$\mathrm{Cov}(X,Y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}).$$

Substituting the values:

$$\mathrm{Cov}(X,Y) = \frac{1}{2}\big[(-1)(-2) + (0)(0) + (1)(2)\big] = 2.$$

This positive value reflects the fact that the variables increase together. In general, larger covariance indicates stronger joint variation, while values near zero suggest weak or no linear relationship.

Covariance is *scale*—or *unit*—dependent. It is not a bounded quantity, and that can make it not very informative if we are not very careful about the units we used to measure x and y. Correlation is designed to account for this precise issue.

Correlation:

To address the *scale dependence*, we can define the correlation coefficient, which normalizes covariance with the variances of both variables:

$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \; . \tag{5.6}$$

As we discussed in Chapter 1 and Chapter 3 population quantities such as ρ are typically unknown, and must be estimated from data. The most common estimator for the correlation is Pearson's *sample correlation coefficient*:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \; . \tag{5.7}$$

This quantity is then scale independent, and bound between -1 and 1. While Pearson's coefficient measures linear association, alternative measures have been proposed to capture more general forms of dependence. In particular, Spearman [35] introduced a rank-based correlation that is sensitive to monotonic relationships and less affected by outliers.

A striking illustration of the limitations of summary statistics is provided by Anscombe [36], who exhibited datasets with identical means, variances, correlations, and regression lines, yet radically different graphical structures. This example underscores the importance of complementing numerical measures with visualization.

5.5 Multiple linear regression

In many applications, a response variable depends on several factors simultaneously. Extending the linear model to incorporate multiple predictors leads to the multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon.$$

This generalization reflects the complexity of real-world phenomena, where outcomes are rarely determined by a single variable. Each coefficient β_j represents the effect of the corresponding predictor X_j, holding all other variables fixed. In this sense, the model allows us to isolate the contribution of each variable while accounting for the presence of others.

Historically, such models emerged as data collection expanded in scope and complexity during the late nineteenth and early twentieth centuries, particularly in biology, economics, and the social sciences. The work of Fisher [53] was instrumental in formalizing methods for analyzing systems influenced by multiple factors, especially in the context of experimental design.

From a mathematical perspective, this model can be viewed as a natural extension of functions of a single variable to functions of several variables, as studied in multivariable calculus. In that setting, one considers functions of the form $y = f(x_1, x_2, \ldots, x_p)$, where the output depends jointly on several inputs. The linear regression model corresponds to the simplest such case, where the function is linear in each variable.

Concrete examples arise in many domains. In economics, one may model income as a function of education, experience, and age. In physics, the pressure of a gas may depend on temperature and volume. In epidemiology, the risk of disease may be related to multiple environmental and genetic factors. In each case, the goal is to understand how several variables jointly influence an outcome.

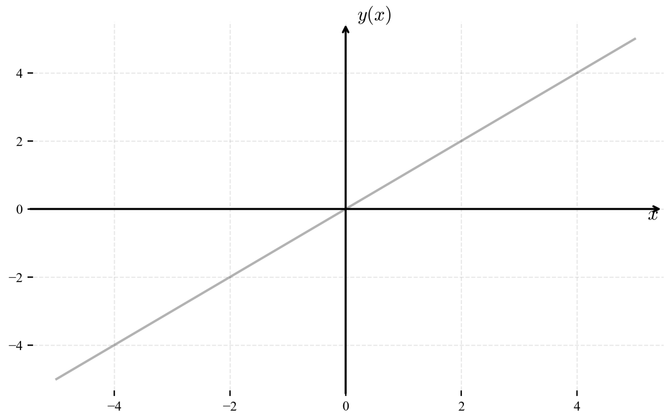
In compact form, the model can be written using matrix notation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

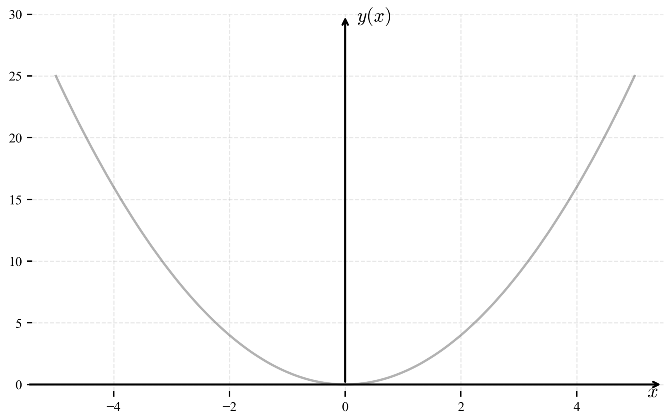
which reveals its connection to linear algebra. In this setting, estimation by least squares corresponds to projecting the observed data onto the subspace spanned by the columns of $\mathbf{X}$ [78]. This geometric interpretation provides additional insight into the structure of the problem, linking statistical estimation with concepts from vector spaces and orthogonality.

A final note, back to correlation. The Pearson sample correlation coefficient r, introduced by Karl Pearson across several works in the 1890s, measures the strength and direction of linear association between two variables, taking values in the interval $[-1, 1]$. In simple linear regression, it is common to also report its square r^2, known as the *coefficient of determination.*

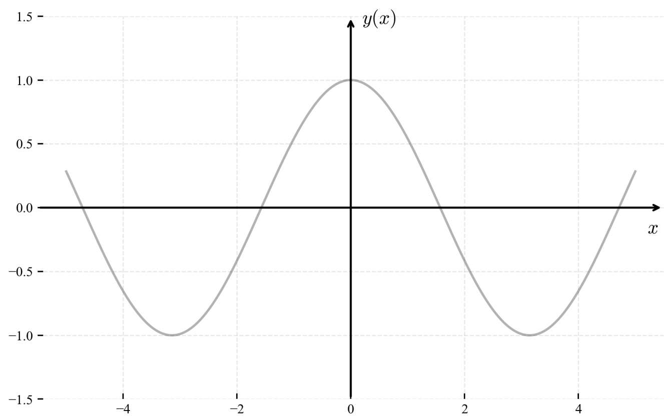
This measure, bounded between 0 and 1, represents the proportion of variance in the response variable explained by the linear relationship with the predictor. Its interpretation in terms of explained variance was developed within the broader regression framework established by Pearson and Fisher, and was later connected to path analysis by Sewall Wright in 1921 [79]. However, a high r^2 does not imply causation and may be misleading if the underlying relationship is non-linear, influenced by outliers, or driven by omitted variables. The frequent confusion between r and r^2 arises because both quantify aspects of linear association, but they carry different information: r captures both the direction and strength of the relationship, while r^2, being always non-negative, discards directional information and is interpreted as a measure of explanatory power. They agree in magnitude only when $r \in \{-1, 0, 1\}$.



(a) Representation of a linear function $y(x) = a\ x + b$. Examples of these are the force and displacement in a spring as given by Hooke's law $F(x) = k\ x$.

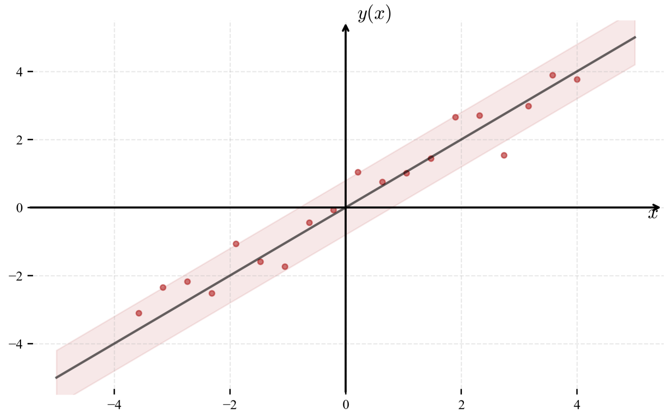


(b) Representation of a quadratic function $y(x) = a\ x^2 + b\ x + c$. Example of these are, for instance, the displacement of a particle under constant acceleration in terms of time $x(t) = x_0 + v_0 t + \frac{1}{2}at^2$.

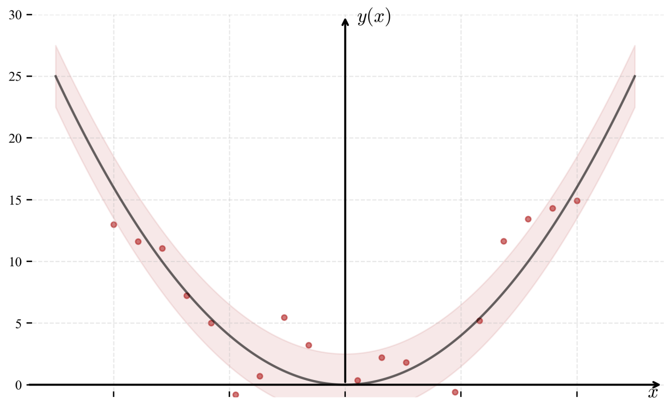


(c) Representation of a sinusoidal function $y(x) = a\ \sin(x) + b\ \cos(x)$. Example of these are, for instance, the position of a wirte with respect to time in a harmonic movement $x(t) = A\cos(\omega t + \phi)$.

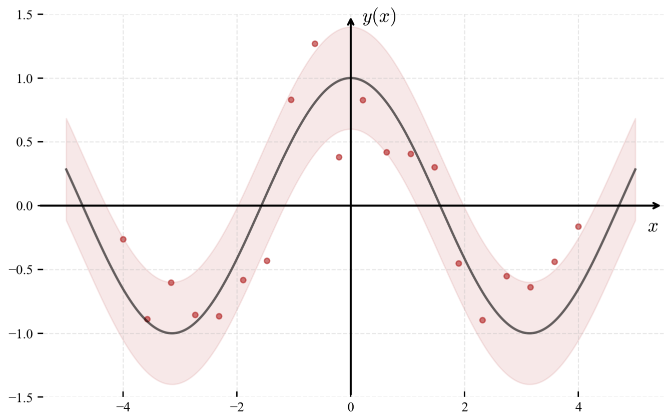
Figure 5.1: Representation of different dependencies between variables, commonly found in physical systems and daily life examples. As examples; linear, quadratic, and the sinusoidal.



(a) Representation of a linear function with some statistical noise $y(x) = a\ x + b \pm \varepsilon$. Examples of these are the force and displacement in a spring as given by Hooke's law $F(x) = k\ x$.



(b) Representation of a quadratic function $y(x) = a\ x^2 + b\ x + c \pm \varepsilon$. Example of these are, for instance, the displacement of a particle under constant acceleration in terms of time $x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$.



(c) Representation of a sinusoidal function $y(x) = a\ \sin(x) + b\ \cos(x) \pm \varepsilon$. Example of these are, for instance, the position of a write with respect to time in a harmonic movement $x(t) = A \cos(\omega t + \phi)$.

Figure 5.2: Representation of different dependencies between variables, with some statistical noise to differentiate from deterministic, exact mathematical models. The red dots represent observations following that correlation plus the statistical fluctuations.

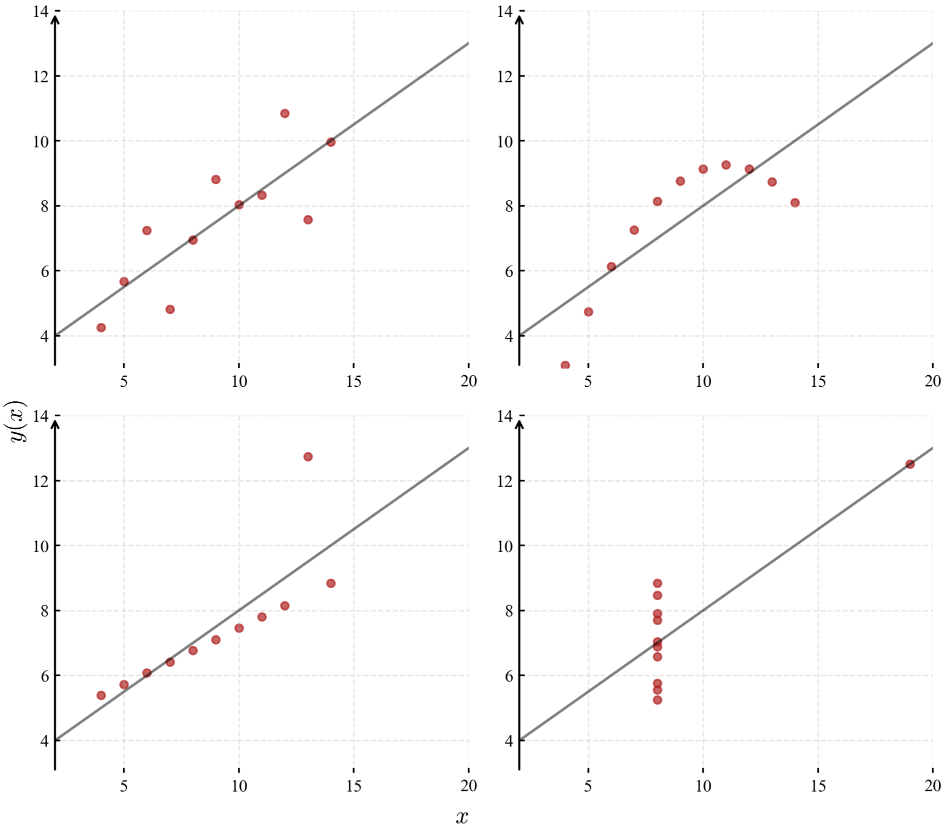


Figure 5.3: Representation of Anscombe's quartet (from English statistician Francis John Anscombe, 1918 – 2001), illustrating the limitations of summary statistics. All four datasets have exact same statistics and correlation values, while having clear different dependencies.

Chapter 6

Introduction to Bayesian probability

Probability statements are just summaries of repeated observations.

— W. V. Quine

Bayesian probability plays a central role in modern statistical reasoning, providing a framework for quantifying and updating uncertainty. As a mathematical framework, it has deep philosophical roots, and it is grounded in the idea that probabilities do not represent only long-run frequencies, but they should be interpreted more generally as a measure of uncertainty about events *and* prior information. This flexibility allows Bayesian methods to address a wide range of inferential problems in a coherent and unified manner, and they are widely used in the scientific community to represent uncertainty, incorporate prior knowledge, and update conclusions as new data become available [3, 45].

As discussed in the previous chapter, classical hypothesis testing is primarily concerned with p-values, and then controlling Type I and Type II errors. While this framework has been highly influential, it introduces important limitations: inference is reduced to binary decisions, and conclusions depend critically on arbitrary significance thresholds, as we saw in previous chapters. Moreover, p-values do not directly quantify the probability of a hypothesis being true, but rather cumulative probabilities of certain intermediate quantities, such as statistic tests, which often leads to misinterpretation in practice [68, 69].

Attempts to address these issues include adjusting p-values (e.g. for multiple comparisons) or returning to Fisher's original interpretation of significance as a measure of evidence rather than a strict decision rule [53, 54]. However, such modifications do not fully resolve concerns about the reliability and interpretation of statistical conclusions.

A more fundamental response is to reconsider the epistemological foundations of statistical inference. Instead of viewing statistics as a tool for hypothesis rejection, one may interpret it as a formal mechanism for updating beliefs in light of evidence. This perspective, rethinking what we mean by probability in the first place, what is a hypothesis, how to quantify or change a degree of belief as new evidence comes available, is what is normally referred to as *Bayesian reasoning*, in which prior information is combined with observed data to produce posterior conclusions.

From a philosophical standpoint, this shift reflects broader debates on the nature of probability, scientific reasoning, and theory of knowledge itself. As we discuss in the first two chapters, the frequentist interpretation, associated with the work of von Mises [42], defines probability through limiting frequencies of repeated measurements. The Bayesian and subjectivist interpretations, as we will see, rooted in the works of Thomas Bayes, and further developed by Ramsey and de Finetti, treat probability as an expression of rational belief under uncertainty [30, 44]. These perspectives are discussed extensively in the philosophy of statistics literature [4].

6.1 Independent and conditional events

In many situations involving uncertainty, we are interested not only in whether events occur, but in how the occurrence of one event influences another. In many situations, observing an outcome may change our expectations about what follows. The language of conditional probability provides a precise way to describe this dependence, forming the basis for reasoning under uncertainty.

A central question is therefore whether two events are related or not. If knowing that one event has occurred does not change our assessment of the other, the events are said to be *independent*; otherwise, we say they are *dependent*. This distinction is fundamental, as it determines whether information can be ignored or must be incorporated into our reasoning. In particular, conditional probability allows us to update probabilities when new information becomes available, a principle that will be essential when we later consider inverse problems and the Bayes' rule.

Two events A and B are said to be independent if the occurrence of one does not affect the probability of the other:

$$P(A \cap B) = P(A)P(B) \ , \tag{6.1}$$

this means that the probability of both events occurring, or the *intersection* $P(A \cap B)$, is just the product of probabilities $P(A)P(B)$.

$$P(A \mid B) = P(A) \ . \tag{6.2}$$

Examples of independent events:

- *Two coin tosses:* Let A be "first toss is heads" and B be "second toss is heads." Then

  $$P(A) = \tfrac{1}{2}, \quad P(B) = \tfrac{1}{2}, \quad P(A \cap B) = \tfrac{1}{4},$$

  and indeed $P(A \cap B) = P(A)P(B) = \frac{1}{4}$, so the events are independent.

- *Two dice:* Let A be "first roll shows a 6" and B be "second shows even number." Then

  $$P(A) = \tfrac{1}{6}, \quad P(B) = \tfrac{3}{6} = \tfrac{1}{2}, \quad P(A \cap B) = \tfrac{1}{12},$$

  which again satisfies $P(A \cap B) = P(A)P(B)$, and hence both dice rolls are also independent.

When this property fails, A and B are what we call *dependent*, and $P(A \mid B)$ is given by:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \tag{6.3}$$

Examples of dependent events:

- *Drawing cards without replacement.* Consider a standard deck of 52 cards. In this case, if we let A be "first card is an Ace" and B be "second card is an Ace." Then

  $$P(A) = \tfrac{4}{52}, \qquad P(B) = \tfrac{4}{52},$$

  but we have to be careful,

  $$P(B \mid A) = \tfrac{3}{51} \neq \tfrac{4}{52},$$

  since one Ace has already been removed from the deck. Hence the events are dependent.

- *Urn example.* Suppose an urn contains 3 red and 2 blue balls. Let A be "first draw is red" and B be "second draw is red," without replacement. Then

  $$P(A) = \tfrac{3}{5}, \qquad P(B \mid A) = \tfrac{2}{4} = \tfrac{1}{2},$$

  while the second ball is again conditional to which one we picked first:

  $$P(B) = \tfrac{3}{5}.$$

  Thus $P(B \mid A) \neq P(B)$, confirming dependence.

6.2 The Bayes rule

The notion of conditional probability introduced in the previous section naturally leads to a fundamental question: how can we reason in the reverse direction? That is, if we observe an event B, how should we update our belief about a possible cause A? Problems of this kind are often referred to as *inverse problems*, and they lie at the heart of statistical inference.

The mathematical solution to this question originates in the work of Thomas Bayes, whose posthumous essay *"An Essay towards Solving a Problem in the Doctrine of Chances"* addressed the problem of inferring causes from observed events [43]. The significance of this result was later recognized and greatly extended by Laplace, who developed it into a general framework for scientific reasoning [16]. Philosophically, Bayes' rule embodies the idea that learning from data consists in updating prior beliefs in a consistent and quantitative manner.

It follows directly from the symmetry of joint probability,

$$P(A \cap B) = P(B \cap A) \ , \tag{6.4}$$

which, together with the definition of conditional probability, leads to the famous *Bayes' rule*

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \ . \tag{6.5}$$

This formula shows how prior beliefs $P(A)$ are updated in light of observed evidence B through the likelihood $P(B \mid A)$, and normalized by the total probability of B, ensuring that the result is a valid probability. The elements of this expression are named as follows:

- $P(A)$ is the **prior** probability, representing initial beliefs about the event A.
- $P(B \mid A)$ is the **likelihood**, describing how probable the observed event B is given A.
- $P(B)$ is the **marginal** probability of observing B.
- $P(A \mid B)$ is the **posterior** probability, representing the updated belief about A after observing B, or *given that* B happened.

Example 1: biased vs. fair die.

Suppose we are given two dice: one fair and one biased, such that $P(6 \mid \text{biased}) = \frac{1}{2}$. One die is chosen at random and rolled once, resulting in a 6. What is the probability that the chosen die is biased?

Hypotheses:

$$H_1: \text{ die is biased}, \qquad H_2: \text{ die is fair}.$$

Priors:

$$P(H_1) = \tfrac{1}{2}, \qquad P(H_2) = \tfrac{1}{2}.$$

Likelihoods:

$$P(6 \mid H_1) = \tfrac{1}{2}, \qquad P(6 \mid H_2) = \tfrac{1}{6}.$$

Marginal probability:

$$P(6) = P(6 \mid H_1)P(H_1) + P(6 \mid H_2)P(H_2) = \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{6} \cdot \tfrac{1}{2} = \tfrac{1}{4} + \tfrac{1}{12} = \tfrac{1}{3}.$$

Posterior:

$$P(H_1 \mid 6) = \frac{P(6 \mid H_1)P(H_1)}{P(6)} = \frac{\frac{1}{2} \cdot \frac{1}{2}}{\frac{1}{3}} = \tfrac{3}{4}.$$

Example 2: medical testing.

Suppose a disease has prevalence $P(H_1) = 0.05$. A test has sensitivity $P(+ \mid H_1) = 0.99$ and false positive rate $P(+ \mid H_2) = 0.01$. Given a positive test result, what is the probability of having the disease?

Hypotheses:

$$H_1: \text{ disease present}, \qquad H_2: \text{ no disease}.$$

Priors:

$$P(H_1) = 0.05, \qquad P(H_2) = 0.95.$$

Likelihoods:

$$P(+ \mid H_1) = 0.99, \qquad P(+ \mid H_2) = 0.01.$$

Marginal probability:

$$P(+) = 0.99 \cdot 0.05 + 0.01 \cdot 0.95 = 0.0495 + 0.0095 = 0.059.$$

Posterior:

$$P(H_1 \mid +) = \frac{0.99 \cdot 0.05}{0.059} \approx 0.84.$$

Example 3: Monty Hall problem.

The Monty Hall problem became widely known through its popularization in the late twentieth century [80]. A contestant faces three doors, behind one of which is a prize. After the contestant makes an initial choice, the host—who knows where the prize is—always opens one of the remaining doors to reveal no prize, and offers the option to switch. The question is: should the contestant switch?

We label the doors 1, 2, 3, and assume without loss of generality that the contestant initially chooses door 1 and the host opens door 3. Define the hypotheses

$$H_i: \text{ prize is behind door } i, \quad i = 1, 2, 3,$$

with equal priors $P(H_1) = P(H_2) = P(H_3) = \frac{1}{3}$, and let E be the event "host opens door 3."

Likelihoods:

$$P(E \mid H_1) = \tfrac{1}{2}, \qquad P(E \mid H_2) = 1, \qquad P(E \mid H_3) = 0,$$

since if the prize is behind door 1 the host may open either remaining door with equal probability; if it is behind door 2 the host must open door 3; and if it is behind door 3 the host cannot open it.

Marginal probability:

$$P(E) = \tfrac{1}{2} \cdot \tfrac{1}{3} + 1 \cdot \tfrac{1}{3} + 0 \cdot \tfrac{1}{3} = \tfrac{1}{6} + \tfrac{1}{3} = \tfrac{1}{2}.$$

Posteriors via Bayes' rule:

$$P(H_1 \mid E) = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \tfrac{1}{3}, \qquad P(H_2 \mid E) = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \tfrac{2}{3}, \qquad P(H_3 \mid E) = 0.$$

The probability of winning by switching to door 2 is therefore

$$P(\text{win by switching}) = P(H_2 \mid E) = \tfrac{2}{3},$$

while staying with the original choice wins with probability only $\frac{1}{3}$. The contestant should always switch.

This result may seem counterintuitive, as one might expect the two remaining doors to carry equal probability $\frac{1}{2}$ each after one is eliminated. The key point is that the host's action is not random: it is conditioned on knowledge of the prize location. This asymmetry means the host's choice carries information, shifting probability mass toward the unchosen door. The Monty Hall problem is a striking illustration of how carefully accounting for the process that generates observations—not just the observations themselves—is essential when applying Bayes' rule.

6.3 Frequentist vs Bayesian

The frequentist interpretation defines probability as long-run relative frequency and treats parameters as fixed but unknown constants, while randomness arises from sampling variation [3, 42]. In this view, probability statements are understood through hypothetical repetition: probabilities describe the behavior of a procedure if it were repeated indefinitely under identical conditions.

In this framework, statements such as *"the probability that the parameter lies in an interval is 0.95"* are avoided, since the parameter is not random; instead, one constructs procedures that produce intervals containing the true parameter in 95% of repeated samples. This grounds better the notion of confidence intervals, which are properties of methods rather than probability statements about specific parameters.

The Bayesian interpretation, by contrast, treats probability as a coherent degree of belief constrained by rational consistency, allowing probabilities to be assigned to hypotheses and parameters themselves [30]. This interpretation extends the use of probability beyond repeatable experiments to situations involving uncertainty about fixed but unknown quantities.

Under this view, *parameters are modeled as random variables* with prior distributions reflecting initial uncertainty, and inference consists of updating this distribution through Bayes' rule when data are observed. The resulting posterior distribution directly answers questions of interest, such as the probability that a parameter lies within a given interval.

Philosophically, the difference concerns meaning rather than mathematics: frequentism anchors probability in hypothetical repetition, whereas Bayesianism interprets probability as rational belief updated by evidence. As a consequence, Bayesian inference provides a unified framework in which learning, uncertainty, and decision-making are naturally integrated within the probabilistic model itself [44].

6.4 Bayesian inference

As we have seen, Bayesian probability provides a general interpretation of probability as a degree of belief under uncertainty. Bayesian *statistics*, in contrast, is the application of this interpretation to *infer unknown quantities from data.* In this setting, unknown parameters are treated as random variables, and inference consists in updating our beliefs about these parameters after observing data.

More precisely, Bayesian statistics combines prior information with observed evidence in a coherent mathematical framework. This represents a conceptual shift from classical approaches: rather than estimating fixed but unknown parameters, we describe uncertainty *about them* through probability distributions and update these distributions as new data become available.

Bayesian inference is based on Bayes' rule applied to parameters θ given some data x:

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\, \pi(\theta)}{p(x)}.$$

This is common notation in inference contexts, and each term has a clear interpretation:

- $\pi(\theta)$ is the **prior**, representing initial beliefs about the parameter θ,
- $p(x \mid \theta)$ is the **likelihood**, describing how probable the observed data x are for a given θ,
- $p(x)$ is the **marginal** normalization constant, ensuring the posterior integrates to one. It is obtained by integrating over all possible values of θ:

$$p(x) = \int_0^1 p(x \mid \theta)\, \pi(\theta)\, d\theta.$$

$\pi(\theta \mid x)$ is the **posterior** distribution, often summarized as

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior},$$

emphasizing that learning consists of combining prior information with evidence.

Example: coin flipping.

We want to estimate the probability θ of obtaining heads when tossing a coin.

i) Prior distribution. Before observing any data, we express our uncertainty about θ through a prior distribution. A natural choice is the Beta distribution,

$$\theta \sim \text{Beta}(\alpha, \beta),$$

since it is defined on $[0, 1]$ and is therefore well-suited for modeling probabilities. The parameters α and β act as effective sample sizes: α encodes prior evidence for heads and β for tails. For example, $\alpha = \beta = 1$ gives a uniform prior, expressing no preference for any value of θ. When $\alpha = \beta = 10$, the distribution peaks at $1/2$, indicating a strong belief that the coin is fair.

ii) Likelihood. Suppose we observe n coin tosses, of which k result in heads. The probability of this outcome, given θ, is binomial:

$$p(k;\, n, \theta) = \binom{n}{k} \theta^k (1-\theta)^{n-k}.$$

Since the binomial coefficient $\binom{n}{k}$ does not depend on θ, the likelihood simplifies to

$$p(k;\, n, \theta) \propto \theta^k (1-\theta)^{n-k}.$$

iii) Marginal likelihood. The marginal likelihood normalizes the posterior and is obtained by integrating over all possible values of θ:

$$p(k;\, n) = \int_0^1 p(k;\, n, \theta)\, \pi(\theta)\, \mathrm{d}\theta.$$

This quantity does not depend on θ and acts purely as a normalizing constant.

iv) Posterior via Bayes' rule. Applying Bayes' rule,

$$\pi(\theta \mid k) = \frac{p(k;\, n, \theta)\, \pi(\theta)}{p(k;\, n)} \propto \theta^{\alpha+k-1} (1-\theta)^{\beta+n-k-1}.$$

This is again a Beta distribution, so the posterior is:

$$\theta \mid k \sim \text{Beta}(\alpha + k,\, \beta + n - k).$$

This property—that the posterior has the same form as the prior—is called *conjugacy*. The posterior mean,

$$\mathbb{E}[\theta \mid k] = \frac{\alpha + k}{\alpha + \beta + n},$$

is a weighted average between prior belief and observed data.

Numerical example. Take $\alpha = \beta = 1$ (uniform prior) and observe $k = 7$ heads in $n = 10$ tosses. Then

$$\theta \mid k \sim \text{Beta}(8,\, 4),$$

reflecting an updated belief that favors larger values of θ.

Example 2: Gaussian mean.

We observe $x_1, \ldots, x_n$ and assume each x_i is drawn independently from a normal distribution with unknown mean μ and known variance σ^2:

$$X_i \sim \mathcal{N}(\mu, \sigma^2).$$

The goal is to infer μ from the data.

i) Prior distribution. We express our initial belief about μ through a Gaussian prior:

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2),$$

where μ_0 is our initial guess and σ_0^2 reflects our uncertainty about it. A large σ_0^2 encodes a vague prior; a small σ_0^2 encodes a confident one.

ii) Likelihood. Given μ, the probability of observing $x = (x_1, \ldots, x_n)$ is:

$$p(x;\, \mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\!\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \propto \exp\!\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right).$$

This expression is maximized when $\mu = \bar{x}$, where $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ is the sample mean, so the data favor values of μ near $\bar{x}$.

iii) Marginal likelihood. The marginal likelihood integrates out μ:

$$p(x) = \int_{-\infty}^{+\infty} p(x;\, \mu)\, \pi(\mu)\, \mathrm{d}\mu,$$

and serves as a normalizing constant that does not depend on μ.

iv) Posterior via Bayes' rule. Applying Bayes' rule:

$$\pi(\mu \mid x) = \frac{p(x;\, \mu)\, \pi(\mu)}{p(x)} \propto p(x;\, \mu)\, \pi(\mu).$$

Multiplying the two Gaussians yields a new quadratic in μ, which is itself a Gaussian. This is why the normal prior is *conjugate* to the normal likelihood. The posterior is:

$$\mu \mid x \;\sim\; \mathcal{N}\big(\mu_n,\, \sigma_n^2\big)\,,$$

where

$$\mu_n = \frac{\dfrac{n}{\sigma^2}\, \bar{x} + \dfrac{1}{\sigma_0^2}\, \mu_0}{\dfrac{n}{\sigma^2} + \dfrac{1}{\sigma_0^2}}, \qquad \sigma_n^2 = \frac{1}{\dfrac{n}{\sigma^2} + \dfrac{1}{\sigma_0^2}}.$$

Interpretation. The posterior mean μ_n is a weighted average of the sample mean $\bar{x}$ and the prior mean μ_0, with weights proportional to their respective *precisions* (inverse variances). As n grows, the data dominate and $\mu_n \to \bar{x}$. If instead the prior is very precise (small σ_0^2), more weight is given to μ_0.

Frequentist comparison. A classical approach would instead test hypotheses such as $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$, compute a z-score from $\bar{x}$, and derive a p-value or confidence interval. However, confidence intervals are defined in terms of repeated sampling and do not assign probabilities to specific values of μ. By contrast, the Bayesian posterior $\pi(\mu \mid x)$ directly quantifies how plausible each value of μ is, given the observed data.

Example 3: Estimating a Poisson rate.

Suppose we count the number of events occurring in a fixed time interval—for instance, the number of emails received per hour, or the number of defects on a production line. A natural model is the Poisson distribution with unknown rate $\lambda > 0$:

$$X_i \sim \text{Poisson}(\lambda),$$

where λ represents the average number of events per interval. The goal is to infer λ from n observed counts $x_1, \dots, x_n$.

i) Prior distribution. Since $\lambda > 0$, we need a prior supported on the positive real line. A natural and convenient choice is the Gamma distribution:

$$\lambda \sim \text{Gamma}(\alpha, \beta),$$

with density

$$\pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda},$$

where $\alpha > 0$ is a shape parameter and $\beta > 0$ is a rate parameter. The prior mean is α/β, so α and β encode our initial belief about the event rate.

ii) Likelihood. Given λ, the probability of observing counts $x = (x_1, \dots, x_n)$ is:

$$p(x;\, \lambda) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \propto \lambda^{\sum_{i=1}^{n} x_i}\, e^{-n\lambda}.$$

Writing $s = \sum_{i=1}^{n} x_i$ for the total count, this simplifies to

$$p(x;\, \lambda) \propto \lambda^{s}\, e^{-n\lambda}.$$

iii) Marginal likelihood. The marginal likelihood integrates out λ:

$$p(x) = \int_0^{+\infty} p(x;\, \lambda)\, \pi(\lambda)\, \mathrm{d}\lambda,$$

and serves as a normalizing constant that does not depend on λ.

iv) Posterior via Bayes' rule. Applying Bayes' rule:

$$\pi(\lambda \mid x) \propto p(x;\, \lambda)\, \pi(\lambda) \propto \lambda^{\alpha+s-1}\, e^{-(\beta+n)\lambda}.$$

This is again a Gamma distribution, confirming that the Gamma prior is *conjugate* to the Poisson likelihood. The posterior is:

$$\lambda \mid x \;\sim\; \text{Gamma}(\alpha + s,\; \beta + n).$$

The posterior mean is

$$\mathbb{E}[\lambda \mid x] = \frac{\alpha + s}{\beta + n},$$

a weighted average between the prior mean α/β and the sample mean $\bar{x} = s/n$.

Numerical example. Suppose we believe the rate is around 2 events per hour, encoded as $\alpha = 4$, $\beta = 2$ (prior mean $= 2$). We then observe $n = 5$ hours with a total of $s = 15$ events. The posterior is

$$\lambda \mid x \;\sim\; \text{Gamma}(19,\; 7),$$

with posterior mean $19/7 \approx 2.71$, reflecting an upward revision of our estimate toward the observed rate of $15/5 = 3$ events per hour.

Chapter 7

Stochasticity and Markov processes

The theory of probabilities is at bottom nothing but common sense reduced to calculation.
— Pierre-Simon Laplace

Much of classical probability and statistics, as we have seen them in previous chapters, rests on the assumption of *independence*. So far, we have generally assumed that samples are drawn independently, errors are uncorrelated, and observations do not influence one another. These assumptions are not merely mathematical conveniences; they also reflect a deeper philosophical picture of how randomness operates in the world. There is, perhaps, a certain comfort in the assumption of independence. If the outcome of one experiment tells us nothing about the next, then the world becomes, in a sense, *memoryless*: the past dissolves, and only the present matters.

The mathematization of uncertainty, as we covered in previous chapters, began precisely from this idealized picture. Bernoulli's law of large numbers established that stable regularities emerge from repeated random trials, revealing that randomness itself could obey precise quantitative laws [13]. Laplace extended this insight far beyond gambling to scientific reasoning more broadly, arguing that probability expresses rational inference under incomplete knowledge, and that apparent randomness often reflects limited information rather than metaphysical indeterminacy [16]. On this view, probability is thus embedded within a wider epistemological framework, and the world is not fundamentally chaotic; it is merely incompletely known, and probability is the language through which we reason about that incompleteness.

Yet independence is far rarer in practice than classical theory initially suggested. A familiar example is the weather forecast: tomorrow's weather is not independent of today's. Similarly, the price of a stock this afternoon is shaped by its price this morning, and the word that follows in a sentence is constrained by the words that came before it. In all these cases, the present state of a system carries information about its future evolution, and ignoring such dependence leads to poor models and unreliable predictions.

During the 19^{th} and early 20^{th} centuries, attention gradually shifted from isolated probabilistic events to sequences of dependent variables evolving through time. The question was no longer merely what the probability of a single event might be, but how a system changes, accumulates information, and forgets as it moves through a succession of states. It was in this context that one of the more unusual disputes in the history of mathematics unfolded—a dispute that was simultaneously about probability theory, theology, and the nature of human freedom.

Among the first to address this problem was Russian mathematician Pavel Nekrasov. As a theologian and conservative intellectual, he occupied a prominent position within the Russian academic establishment of the late 19^{th} century, eventually becoming rector of Moscow University and a close ally of tsarist institutions. Nekrasov argued that the classical law of large numbers fundamentally depended upon assumptions of independence. He did not present this merely as a technical observation. Rather, he claimed that it carried profound metaphysical consequences: if stable social regularities such as crime rates or patterns of collective behavior obey probabilistic laws, and if such laws rely upon independent actions, then the individuals comprising society must each be making genuinely free and unconstrained choices. The law of large numbers, on this reading, became not merely a theorem about coin tosses, but evidence for free will and, by extension, for the moral foundations of Christian society and the legitimacy of the tsarist order. Mathematics, in Nekrasov's hands, became entangled with conservative theology.

It is precisely this dispute that gives rise to the central topic of this chapter, through the works of Andrey Andreyevich Markov (1856–1922). As deeply concerned with pure analysis questions, and firmly against personal beliefs used to found mathematical theories. Markov was, by all accounts, a man of fierce temperament and uncompromising convictions. An atheist, a socialist sympathizer, and an outspoken critic of both the Orthodox Church and the tsarist establishment. When Lev Tolstoy was excommunicated from the Russian Orthodox Church in 1901, Markov publicly requested his own excommunication in protest, a gesture of defiant solidarity that caused considerable scandal. He was not inclined to let a mathematical error stand, least of all one wrapped in theological language and deployed in support of institutions he despised. In Markov's view, Nekrasov's insistence on independence was simply wrong, and he set out to demonstrate this explicitly [21].

Markov's strategy was to construct a concrete counterexample: a sequence of *dependent* events that nevertheless exhibited the stable long-run regularities associated with the law of large numbers. Rather than work with an artificial mathematical construction, he chose a literary one. He took the opening chapters of Pushkin's celebrated novel in verse, *Eugene Onegin*, and analyzed the text with meticulous care, manually counting, letter by letter, the transitions between vowels and consonants across thousands of characters. For each letter, he recorded whether it was a vowel (V) or a consonant (C), and then counted how often a vowel was followed by another vowel, how often by a consonant, how often a consonant was followed by a vowel, and so on. The result was a table of transition counts: the raw material for what we would now call a two-state Markov chain, with states V and C and transition probabilities estimated from data.

The point was both precise and devastating. The letters of a text are manifestly not independent: knowing that the current letter is a vowel changes the probability that the next letter will be a consonant or another vowel. Dependence is everywhere, and yet the long-run proportions of vowels and consonants stabilize perfectly well, much as the law of large numbers predicts in the independent case. Independence, Markov showed, is not a necessary condition for probabilistic regularity; it is merely a *sufficient* one, and Nekrasov's theological argument collapsed along with its mathematical foundation.

What Markov had done, almost as a polemical aside, was inaugurate the systematic study of dependent stochastic sequences. Once the example is understood, the abstraction becomes natural: instead of tracking the actual letters of a text, one retains only their classification into states—V or C—and studies how the system moves—evolves, *transitions*—between those states over time. Each state carries probabilities of transitioning to other states, and the resulting sequence forms what we now call a *Markov chain*. The specific literary content of *Eugene Onegin* disappears entirely; what remains is a mathematical structure of extraordinary generality and power.

Many of these ideas will reappear throughout this chapter in increasingly sophisticated forms. From the simple idea of memoryless processes, to Markov chains, hidden states, Monte Carlo methods, and modern topics such as reinforcement learning and large language models, they all share deep roots in Markovian stochasticity.

7.1 Stochasticity and the Markov property

Few ideas in mathematics are as powerful as a well-chosen simplification. The theory of Markov chains rests on one such simplification: that the future of a system, given its present state, is independent of its past. This is known as the *Markov property*, and while it may seem like a strong assumption, it captures the essential structure of an enormous range of real-world processes. As we saw in the introduction, Markov arrived at this idea by studying sequences of letters in a poem. The abstraction he performed—retaining only the state of the system at each step and the probabilities of moving between states—turns out to be applicable far beyond literary analysis, and it is worth developing carefully.

To make this precise, consider a sequence of random variables $X_0, X_1, X_2, \ldots$, each taking values in some set $\mathcal{S}$ called the *state space*. This sequence represents the evolution of a system over time: X_n is the state of the system at time n. The sequence is called a *Markov chain* if, for all times n and all states $x_0, x_1, \ldots, x_{n+1} \in \mathcal{S}$,

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \ldots, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n) \; . \quad (7.1)$$

In words: given the present state X_n, the probability of the next state X_{n+1} does not depend on how the system arrived at X_n. The past, once the present is known, becomes irrelevant. This is the *Markov property*, also sometimes called the *memoryless property*.

It is worth pausing to appreciate what this assumption does and does not say. It does not claim that the past has no influence on the future—it clearly does, through the current state. What it says is that all the relevant information about the past is *summarized* in the present state. If we know where we are, knowing the path we took to get here adds nothing. In Markov's vowel-consonant chain, for instance, knowing that the current letter is a vowel tells us everything the model needs to know about what comes next—we do not need to remember whether the letter before that was also a vowel, or what the opening line of the poem said. This is a modeling assumption, not a law of nature, and its appropriateness depends entirely on how we define the state of the system. Chosen well, the state summarizes the past; chosen poorly, the Markov property fails and the model misleads.

Transition probabilities and the transition matrix

The essential ingredient of a Markov chain is the collection of probabilities governing how the system moves from one state to another. For a finite state space $\mathcal{S} = \{1, 2, \ldots, k\}$, these are called *transition probabilities*:

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i) \; , \quad (7.2)$$

representing the probability of moving from state i to state j in one step. We assume throughout that these probabilities do not depend on the time n, a property normally called *time homogeneity*: the rules of the game do not change as the system evolves. The transition probabilities satisfy, for each state i,

$$p_{ij} \geq 0 \quad \text{for all } j \; , \qquad \sum_j p_{ij} = 1 \; , \quad (7.3)$$

since the system must transition to *some* state at each step, and probabilities must be non-negative and sum to one.

These probabilities are naturally arranged into a *transition matrix* $\mathbf{P}$, where the entry in row i and column j is p_{ij}. Each row of $\mathbf{P}$ sums to one, which is why such matrices are called *stochastic matrices*. The transition matrix is the complete mathematical description of a Markov chain: given the initial state and $\mathbf{P}$, all probabilistic properties of the chain can in principle be derived. It is a remarkably compact encoding of the system's dynamics.

Example: letters and words

Returning to Markov's original setting, consider a two-state chain with states V (vowel) and C (consonant). Suppose that a vowel is followed by another vowel with probability 0.4 and by a consonant with probability 0.6, while a consonant is followed by a vowel with probability 0.3 and by another consonant with probability 0.7. These numbers are schematic, but they reflect the qualitative structure Markov observed in Russian text: consonants are more likely to follow consonants, and vowels to follow vowels, but neither transition is impossible. The transition matrix is

$$\mathbf{P} = \begin{array}{c} \\ \mathrm{V} \\ \mathrm{C} \end{array} \begin{array}{c} \begin{array}{cc} \mathrm{V} & \mathrm{C} \end{array} \\ \begin{pmatrix} 0.4 & 0.6 \\ 0.3 & 0.7 \end{pmatrix} \end{array} .$$

Reading row by row: from a vowel, we go to another vowel with probability 0.4 and to a consonant with probability 0.6; from a consonant, we go to a vowel with probability 0.3 and stay a consonant with probability 0.7. Every row sums to one.

Example: Weather forecast

A simple weather model with two states, Sunny (S) and Rainy (R), captures the same structure in a different domain. Suppose that a sunny day is followed by another sunny day with probability 0.8 and a rainy day with probability 0.2, while a rainy day is followed by a sunny day with probability 0.4 and another rainy day with probability 0.6. The transition matrix is

$$\mathbf{P} = \begin{array}{c} \\ S \\ R \end{array} \begin{array}{c} \begin{array}{cc} S & R \end{array} \\ \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \end{array} .$$

Despite the completely different subject matter, the mathematical structure is identical to the vowel-consonant chain. This is the power of abstraction: once a process has been encoded as a Markov chain, all the tools of the theory apply regardless of whether the states represent letters, weather patterns, or anything else.

This idea, that *states* are represented as vectors, and *transitions*, or transition probabilities, as matrices, is extremely powerful, and it extends far beyond the further examples we will see (Hidden Markov Models, Monte Carlo, etc). From modern physics and quantum mechanics to modern machine learning algorithms, such as the transformer and GPT models, they all heavily rely in this kind of representation.

Chapman–Kolmogorov equations

A natural question is: what is the probability of moving from state i to state j in exactly n steps, rather than just one? If we want to know the probability that it will be sunny in three days given that it is raining today, we cannot simply read off an entry of $\mathbf{P}$—we need to account for all the possible intermediate states the system might pass through. These are called the *n-step transition probabilities*, written $p_{ij}^{(n)}$, and they satisfy the *Chapman–Kolmogorov equations*:

$$p_{ij}^{(m+n)} = \sum_{k \in \mathcal{S}} p_{ik}^{(m)}\, p_{kj}^{(n)} \ . \tag{7.4}$$

This equation says that the probability of going from i to j in $m+n$ steps equals the sum, over all possible intermediate states k, of the probability of reaching k from i in m steps and then reaching j from k in n steps. All paths through the state space are accounted for, weighted by their probabilities. In matrix notation, this takes a particularly clean form: the n-step transition matrix is simply $\mathbf{P}^n$, the n-th matrix power of $\mathbf{P}$ [81]. To find the probability distribution over states after n steps, starting from an initial distribution $\boldsymbol{\pi}_0$, one computes $\boldsymbol{\pi}_0\mathbf{P}^n$.

Classification of states

Not all states in a Markov chain behave in the same way, and understanding the long-run behavior of the chain requires classifying them. Some states, once left, may never be returned to; others are visited again and again, indefinitely. A state i is called *recurrent* if, starting from i, the chain returns to i with probability one. Intuitively, a recurrent state is one the system cannot escape permanently—it will always come back, however long it wanders. A state is called *transient* if there is a positive probability that the chain never returns to it after leaving. A particularly extreme case is an *absorbing state*: a state i with $p_{ii} = 1$, from which there is no escape at all. In the gambler's ruin problem, bankruptcy and winning are both absorbing states.

A Markov chain is called *irreducible* if every state can be reached from every other state in a finite number of steps. Irreducibility ensures that the chain does not fracture into isolated components that can never communicate. It is a global property of the chain as a whole, and it plays a central role in guaranteeing the existence and uniqueness of a stationary distribution, as we will see shortly. A chain is called *aperiodic* if the system does not get trapped in a fixed cycle, returning to each state only at regular multiples of some period greater than one. Irreducibility and aperiodicity together are the key conditions for the chain to behave well in the long run.

Stationary distributions and detailed balance

One of the most important questions about a Markov chain is what happens as time goes on. If we run the chain for a very long time, does the distribution over states settle into a stable pattern, independent of where we started? The answer is given by the notion of a *stationary distribution.* A probability distribution $\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots, \pi_k)$ over the state space is called *stationary* if

$$\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P} \,, \qquad \sum_i \pi_i = 1 \,. \tag{7.5}$$

In words: if the system starts in distribution $\boldsymbol{\pi}$, one step of the chain leaves it in distribution $\boldsymbol{\pi}$. The distribution is unchanged by the dynamics. The stationary distribution represents the long-run proportion of time the chain spends in each state—in the weather model, it tells us what fraction of days, over a very long period, will be sunny and what fraction rainy, regardless of today's weather.

Example. For the weather chain above, the stationary distribution satisfies $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$, which gives the system of equations

$$\pi_S = 0.8\,\pi_S + 0.4\,\pi_R \,, \qquad \pi_R = 0.2\,\pi_S + 0.6\,\pi_R \,, \qquad \pi_S + \pi_R = 1 \,.$$

Solving gives $\pi_S = 2/3$ and $\pi_R = 1/3$. In the long run, two thirds of days will be sunny and one third rainy, regardless of whether today is sunny or rainy.

A sufficient condition for $\boldsymbol{\pi}$ to be stationary is the *detailed balance* condition:

$$\pi_i\, p_{ij} = \pi_j\, p_{ji} \quad \text{for all } i, j \,. \tag{7.6}$$

This condition says that, in the stationary regime, the flow of probability from state i to state j exactly equals the flow from j to i. The system is in equilibrium not just globally but at every pair of states individually. Chains satisfying detailed balance are called *reversible*, because their dynamics look the same whether run forwards or backwards in time. Reversibility is a stronger condition than stationarity—it implies stationarity but is not implied by it—and it plays a central role in the design of Markov chain Monte Carlo algorithms.

Ergodicity and convergence

The central theorem of Markov chain theory, which justifies much of what follows in this chapter, concerns the long-run behavior of irreducible and aperiodic chains. Such chains are called *ergodic*, and for them the following holds: regardless of the initial state, the distribution over states converges to the unique stationary distribution as the number of steps grows. Formally,

$$\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j \quad \text{for all } i, j \;, \tag{7.7}$$

meaning that after sufficiently many steps, the probability of being in state j approaches π_j no matter where the chain started [81]. This is the Markov chain analogue of the law of large numbers, and it is precisely what Markov needed to answer Nekrasov: even for dependent sequences, long-run averages stabilize, and they stabilize at the values the stationary distribution prescribes. The chain forgets its initial condition, and what remains is the intrinsic geometry of the transition structure.

Example: The random walk

Consider a particle on the integers that, at each step, moves one position to the right with probability p and one position to the left with probability $1 - p$. This is the simplest example of a random walk, and one of the most studied objects in probability theory. The state space is infinite, but the Markov property holds: the next position depends only on the current one. When $p = 1/2$, the walk is symmetric and recurrent in one and two dimensions—the particle will return to any given position with probability one—but transient in three or more dimensions, where it drifts away and never returns. This celebrated result is due to Pólya [82], and it has been described as one of the most surprising facts in all of probability.

Example: The gambler's ruin

A gambler starts with k coins and plays a series of fair games, winning or losing one coin at each step. The game ends when the gambler either reaches N coins (winning) or loses everything (ruin). Both endpoints are absorbing states: once reached, the chain stays there forever. The probability of ruin, starting from k coins, is $(N-k)/N$ for a fair game. This result has a sobering implication: even in a perfectly fair game, a gambler with finite resources playing against an opponent with larger resources will eventually go bankrupt with certainty if the game continues indefinitely. The mathematics offers no comfort to the persistent gambler.

Example: PageRank

Google's original PageRank algorithm, developed by Brin and Page in the late 1990s, models a web surfer who randomly follows hyperlinks from page to page [83]. The web is treated as a directed graph, and the surfer's position is a Markov chain on the pages of the internet. The stationary distribution of this chain assigns to each page a score proportional to how often the random surfer would visit it in the long run. Pages linked to by many important pages receive high stationary probability, capturing the intuition that influence propagates through networks. The algorithm that shaped the modern internet is, at its mathematical core, the computation of a stationary distribution of a very large Markov chain.

Example: The Wright–Fisher model

In population genetics, the Wright–Fisher model describes how the frequency of a gene variant changes from generation to generation due to random reproduction [84]. A population of N individuals is replaced each generation by sampling with replacement from the previous one. The frequency of a variant is the state, and the resulting Markov chain has absorbing states at zero and one, corresponding to the extinction or fixation of the variant. This model provides the mathematical foundation for understanding genetic drift—the random fluctuation of gene frequencies over time—and represents one of the earliest and most influential applications of Markov chains in biology.

7.2 Continuous-time Markov chains

The Markov chains we have studied so far evolve in discrete steps: time advances by integer increments, and at each step the system transitions from one state to another. This framework is natural when the underlying process itself unfolds in discrete time—generations in a population, days in a weather model, moves in a game. But many processes in nature evolve continuously, and forcing them into a discrete-time framework introduces an artificial structure. Consider a radioactive atom that decays at a random moment, or a customer who arrives at a queue after a random waiting period, or a molecule that switches between states at unpredictable instants. In all these cases, the relevant question is not what happens at the next integer time step, but rather how long the system stays in its current state before transitioning, and where it goes when it does. This motivates *continuous-time Markov chains*, in which time is a continuous variable but the state space remains discrete. The Markov property carries over directly: given the current state, the future is independent of the past.

In continuous time, the transition matrix $\mathbf{P}$ is replaced by the *generator matrix* $\mathbf{Q}$, also called the *rate matrix*. Its off-diagonal entries $q_{ij} \geq 0$ represent the instantaneous rates of transitioning from state i to state j, while the diagonal entries are defined by $q_{ii} = -\sum_{j \neq i} q_{ij}$, so that each row sums to zero. The time spent in state i before any transition follows an exponential distribution with rate $|q_{ii}|$—a direct expression of the memoryless property in continuous time—and when the system leaves state i, it moves to state j with probability $q_{ij}/|q_{ii}|$. The evolution of the probability distribution over states is governed by the *Kolmogorov forward equations* [23],

$$\frac{d\boldsymbol{\pi}(t)}{dt} = \boldsymbol{\pi}(t)\,\mathbf{Q}\,, \tag{7.8}$$

and the stationary distribution, when it exists, satisfies $\boldsymbol{\pi}\mathbf{Q} = \mathbf{0}$. The simplest example is the *Poisson process* [46], which counts random events occurring at a constant average rate λ, with exponentially distributed waiting times between them. A natural generalization is the *birth–death process*, where a population grows and shrinks by one individual at a time with state-dependent rates. In the M/M/1 queue of queueing theory, customers arrive at rate λ and are served at rate μ, and the system is stable when $\lambda < \mu$ [85].

Example: physical systems.

Continuous-time Markov chains appear throughout the natural sciences. In physics, a radioactive atom transitions from an unstable to a stable state at a random exponentially distributed time, and at the population level the expected number of undecayed atoms follows $N(t) = N_0 e^{-\lambda t}$, recovering the deterministic decay law of Chapter 5 as the mean of the stochastic process.

Example: clinical sciences.

In epidemiology, the SIR model partitions a population into Susceptible, Infected, and Recovered individuals, and its stochastic formulation as a continuous-time Markov chain captures phenomena—such as the random extinction of a disease—that the deterministic version cannot represent [86]. In molecular biology, proteins switch between conformational states at random times, and these transitions can be modeled as continuous-time chains whose generator entries are inferred from experimental data, providing a quantitative framework for understanding enzyme kinetics and ion channel gating [87]. In each case, the continuous-time framework is not merely a mathematical refinement but a more faithful representation of the underlying process, capturing the randomness of timing rather than forcing it into artificial discrete steps.

7.3 Hidden Markov Models

The study of Markov processes eventually expanded far beyond the simple examples considered by Markov himself. In the 1940s, Stanislaw Ulam and John von Neumann realized that random sequences could be used not only to describe stochastic systems, but also to solve otherwise intractable computational problems [88, 89]. Their Monte Carlo methods estimated complicated probabilities by repeated simulation rather than direct calculation, an idea that later evolved into Markov chain Monte Carlo methods, now central to modern Bayesian statistics and machine learning. At the same time, Kolmogorov and Doob placed stochastic processes on rigorous mathematical foundations, formalizing random evolution over time within measure-theoretic probability [23, 22]. Today, Markov models appear throughout science, from physics and genetics to economics, speech recognition, and artificial intelligence.

In many real systems, however, the underlying state is not directly observable. A doctor may observe symptoms without directly seeing the disease process, or a speech recognition system may hear sounds without directly observing the words being spoken. This leads to the idea of a *hidden Markov model* (HMM), in which the hidden states evolve according to a Markov chain, while the observations are noisy signals generated from those hidden states.

Formally, let

$$X_1, X_2, \ldots, X_T$$

denote the hidden states and

$$O_1, O_2, \ldots, O_T$$

the observations. The hidden states satisfy the Markov property

$$\mathbb{P}(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_1) = \mathbb{P}(X_{t+1} = j \mid X_t = i),$$

while each observation depends only on the current hidden state:

$$\mathbb{P}(O_t = o \mid X_t = i) = b_i(o).$$

A simple example is weather prediction. Suppose the hidden state is either

$$X_t \in \{\text{Sunny}, \text{Rainy}\},$$

but we cannot observe the weather directly. Instead, we only observe whether people carry umbrellas. A rainy day is more likely to produce umbrella observations, while sunny days are less likely to do so. By observing a sequence such as

$$\text{Umbrella}, \ \text{Umbrella}, \ \text{No Umbrella},$$

the goal is to infer the most probable hidden weather sequence that generated the observations.

Three fundamental computational problems arise naturally. First, given the observations, what is the probability that the model generated them? Second, what is the most likely hidden state sequence? Third, how can the transition probabilities and observation probabilities be learned from data? These questions lead respectively to the forward algorithm, the Viterbi algorithm, and the Baum–Welch algorithm, which we will study later in this chapter. Together, these ideas form one of the most important frameworks in modern probabilistic modelling.

7.4 Beyond Markov chains

7.4.1 The Monte Carlo method

The history of Markov chains begins with a literary quarrel in St. Petersburg, the history of Monte Carlo methods begins in a hospital bed in Los Alamos, New Mexico, during the Second World War. The mathematician Stanislaw Ulam, convalescing from encephalitis in 1946, found himself playing solitaire and wondering what the probability was of a particular game succeeding. He could not work it out analytically. Instead, he reasoned, one could simply play the game many times and count the fraction of successes. The idea of using repeated random simulation to estimate quantities that resist exact calculation struck him as both obvious and powerful. He mentioned it to John von Neumann, who immediately saw its potential for numerical problems in neutron diffusion, and a method was born [88]. They named it Monte Carlo, after the famous casino in Monaco, as a knowing tribute to chance itself.

The Monte Carlo method, as Ulam and von Neumann formalized it, rests on a simple principle: if a quantity can be expressed as the expected value of some function of a random variable, then it can be estimated by drawing many samples from that variable and averaging. More precisely, if we wish to compute an integral

$$I = \int f(x)\, p(x)\, dx \;, \tag{7.9}$$

where $p(x)$ is a probability density, we draw N independent samples $x_1, x_2, \ldots, x_N$ from p and estimate I by the sample average

$$\hat{I}_N = \frac{1}{N} \sum_{n=1}^{N} f(x_n) \;. \tag{7.10}$$

By the law of large numbers, $\hat{I}_N \to I$ as $N \to \infty$. The convergence rate is $O(1/\sqrt{N})$, which is independent of the dimension of the problem—a crucial advantage over classical numerical integration methods, whose cost grows exponentially with dimension. This dimension-independence makes Monte Carlo methods indispensable in statistics, physics, and machine learning, where one routinely faces integrals over spaces of hundreds or thousands of dimensions [89].

The practical difficulty, however, is that drawing independent samples from $p(x)$ may itself be difficult, particularly when p is a complicated posterior distribution arising in Bayesian inference, or a Boltzmann distribution in statistical physics. This is where Markov chains enter the picture.

7.4.2 Markov Chain Monte Carlo

The central insight of *Markov chain Monte Carlo* (MCMC) is that, rather than drawing independent samples from a target distribution π, one can construct a Markov chain whose stationary distribution *is* π, and then simulate that chain. After a sufficient number of steps—a *burn-in* period during which the chain approaches stationarity—the states visited by the chain constitute approximate samples from π, which can be used to estimate expectations of interest. One trades the independence of classical Monte Carlo for a chain with memory, but gains the ability to sample from distributions that would otherwise be inaccessible.

The first and most celebrated algorithm of this type is the *Metropolis–Hastings algorithm*, introduced by Metropolis and colleagues in 1953 [90] for problems in statistical physics, and generalized by Hastings in 1970 [91]. The idea is elegant. At each step, the algorithm proposes a move from the current state x to a candidate state x', drawn from a proposal distribution $q(x' \mid x)$. The proposed move is then accepted or rejected according to an acceptance probability

$$\alpha(x, x') = \min\!\left(1,\; \frac{\pi(x')\, q(x \mid x')}{\pi(x)\, q(x' \mid x)}\right) \;. \tag{7.11}$$

If accepted, the chain moves to x'; if rejected, it stays at x. The acceptance ratio compares the probability of the proposed state to that of the current state, corrected for any asymmetry in the proposal. The key observation is that one needs only to evaluate π up to a normalizing constant, since the constant cancels in the ratio $\pi(x')/\pi(x)$. This is precisely the situation in Bayesian inference, where the posterior distribution is known up to the normalizing marginal likelihood.

It is straightforward to verify that this algorithm satisfies detailed balance with respect to π:

$$\pi(x)\,\alpha(x,x')\,q(x' \mid x) = \pi(x')\,\alpha(x',x)\,q(x \mid x')\ , \tag{7.12}$$

and hence, by the discussion of Section 1, the chain leaves π invariant. As long as the chain is irreducible and aperiodic—conditions that are easy to ensure in practice—it converges to π by the ergodic theorem, and time averages converge to expectations under π.

A particularly important special case is *Gibbs sampling*, introduced by Geman and Geman in 1984 in the context of image restoration [92]. When the target distribution is over a high-dimensional vector $(x_1, x_2, \ldots, x_d)$, Gibbs sampling updates one coordinate at a time, sampling each x_i from its conditional distribution given all the others. No explicit acceptance step is needed: the Gibbs sampler is a special case of Metropolis–Hastings in which proposed moves are always accepted. Its practicality depends on the conditional distributions being tractable, which is the case for many models encountered in Bayesian statistics. The probabilistic programming language Stan, widely used in the social and natural sciences, implements more sophisticated variants of this idea, including Hamiltonian Monte Carlo, which uses gradient information to propose moves more efficiently [93].

Example: Bayesian posterior inference:

Suppose we observe data $y_1, \ldots, y_n$ and wish to infer the parameter θ of a model with likelihood $p(y_i \mid \theta)$ and prior $p(\theta)$. The posterior is

$$p(\theta \mid y_1, \ldots, y_n) \propto p(\theta) \prod_{i=1}^{n} p(y_i \mid \theta)\ ,$$

but the normalizing constant is typically intractable. MCMC allows us to draw approximate samples from this posterior and compute posterior means, variances, credible intervals, or any other quantity of interest, without ever evaluating the normalizing constant. This is one of the most widely used tools in modern applied statistics.

Example: The Ising model:

The Ising model imagines a grid of tiny magnets, each pointing either up or down (± 1), where neighboring magnets prefer to align. The probability of any given configuration depends on its energy and temperature through the Boltzmann distribution $\pi(\mathbf{x}) \propto \exp(-E(\mathbf{x})/T)$: low-energy configurations are more likely, and higher temperatures make the distribution more uniform. For a large grid, the number of possible configurations is astronomical, and direct sampling is completely infeasible.

This is precisely where Markov chain Monte Carlo enters. The Glauber dynamics—a Gibbs sampler that updates one spin at a time according to its neighbors—defines a Markov chain whose stationary distribution is exactly π [94]. The connection runs deeper than a convenient trick: thermodynamics and Bayesian inference share, at a mathematical level, the same fundamental structure of distributions proportional to exponentials, and MCMC is the natural computational bridge between them.

7.5 Transformers and modern Machine Learning

There is a certain irony in the fact that Markov introduced his chains as a mathematical response to a debate about free will and social order, and that one of the most direct descendants of his ideas now powers systems capable of generating human-like prose, translating between languages, and reasoning about scientific literature. The path from Markov's vowel-consonant counts in *Eugene Onegin* to the large language models of the early 21$^{\text{st}}$ century is long, but it is a continuous one, and tracing it reveals how deeply the Markovian perspective has shaped modern machine learning.

The connection begins with the most natural application of Markov chains to language: the *n-gram model*. If the states of a Markov chain are words, and the transition probabilities represent how likely each word is to follow the previous one, then the chain generates sequences of words. A first-order chain, in which the next word depends only on the current one, is a *bigram model*; a second-order chain, depending on the two most recent words, is a *trigram model*. Such models were the workhorses of statistical language processing for decades, and their limitations—the inability to capture long-range dependencies, the exponential growth of the state space with order—drove the search for more powerful architectures. Yet the underlying probabilistic formulation, that language is modeled by a distribution over sequences and that the goal is to learn this distribution from data, remains unchanged.

7.5.1 Markov decision processes and reinforcement learning

Before turning to neural language models, it is worth noting a major extension of the Markov chain framework that has proved equally influential: the *Markov decision process* (MDP). Whereas a Markov chain evolves according to fixed transition probabilities, a Markov decision process introduces an *agent* that selects *actions* at each step, with the transitions depending on both the current state and the chosen action. The agent receives a scalar *reward* after each transition and seeks a *policy*—a mapping from states to actions—that maximizes the expected cumulative reward over time [95, 96].

Formally, an MDP is defined by a state space $\mathcal{S}$, an action space $\mathcal{A}$, transition probabilities $p(s' \mid s, a)$, and a reward function $r(s, a)$. At each time t, the agent observes state S_t, selects action $A_t \sim \pi(\cdot \mid S_t)$, transitions to state $S_{t+1} \sim p(\cdot \mid S_t, A_t)$, and receives reward $R_t = r(S_t, A_t)$. The goal is to find the policy π^* maximizing the expected discounted return

$$V^{\pi}(s) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, S_0 = s\right] , \tag{7.13}$$

where $\gamma \in [0, 1)$ is a discount factor that weights immediate rewards more heavily than distant ones.

The field of *reinforcement learning* is devoted to learning good policies from experience, without requiring explicit knowledge of the transition probabilities [97]. The Q-learning algorithm of Watkins and Dayan [98] showed that an agent could learn the optimal policy by iteratively updating estimates of the *action-value function* $Q(s, a)$—the expected return from taking action a in state s and then following the optimal policy. The convergence of Q-learning, like so much else in this chapter, rests on the ergodic theorem for Markov chains: provided all state-action pairs are visited sufficiently often, the estimates converge to their true values.

Reinforcement learning was dramatically thrust into public view with the success of DeepMind's AlphaGo Zero [99], which learned to play Go at superhuman level entirely through self-play, with no human knowledge beyond the rules of the game. AlphaGo Zero combined Monte Carlo tree search—itself a form of simulation-based planning rooted in the Monte Carlo methods of Section 4—with deep neural networks trained by reinforcement learning. The game of Go, which had long resisted computer mastery due to the astronomical size of its state space, was conquered by an algorithm whose mathematical foundations lay, in no small part, in the Markovian framework developed a century earlier.

More recently, the RLHF (Reinforcement Learning from Human Feedback) paradigm has become central to the training of large language models [100, 101]. Human evaluators compare model outputs, their preferences are used to train a reward model, and the language model is then fine-tuned by reinforcement learning to maximize that reward. The optimization problem is again an MDP, with the model's internal state and partial output as the state, each generated token as an action, and human preference as the reward signal. The systems that users interact with as conversational AI are, in a precise technical sense, policies trained by reinforcement learning over Markov decision processes.

7.5.2 Recurrent networks and sequence models

The deep neural network revolution of the 2010s brought a new class of sequence models. *Recurrent neural networks* (RNNs) process sequences one element at a time, maintaining a hidden state h_t that summarizes the history up to time t and is updated at each step:

$$h_t = f(h_{t-1}, x_t) \,, \tag{7.14}$$

where x_t is the current input and f is a learned nonlinear function [102]. The hidden state plays the same conceptual role as the state in a Markov chain: it summarizes the past in a form sufficient for predicting the future. The crucial difference is that h_t is a real-valued vector of potentially large dimension, and the function f is learned from data rather than specified in advance. The relationship to hidden Markov models is direct and instructive: an HMM is a probabilistic model with a discrete hidden state evolving according to a Markov chain; an RNN is a deterministic dynamical system with a continuous hidden state that has been optimized to capture structure in sequences. Both attempt the same fundamental task—compressing the past into a useful summary—by very different means.

The long short-term memory network (LSTM) [103] introduced gating mechanisms that allow the hidden state to retain information over long time intervals, addressing the notorious vanishing gradient problem that made it difficult to train deep RNNs on long sequences. LSTMs became the dominant architecture for sequence modeling tasks—speech recognition, machine translation, time series forecasting—for much of the 2010s. Yet they remained fundamentally sequential: each state h_t depends on h_{t-1}, so processing a sequence of length T requires T sequential steps, which cannot be parallelized.

7.5.3 The transformer and the attention mechanism

The *transformer* architecture, introduced by Vaswani and colleagues in 2017 [104], discarded recurrence entirely. Rather than processing sequences step by step, the transformer processes the entire sequence at once, using a mechanism called *self-attention* to allow every position in the sequence to directly attend to every other position. The result is an architecture that is both more parallelizable than RNNs and, empirically, far more effective at capturing long-range dependencies.

To understand the connection to Markov chains, it is useful to think about what a language model must learn to do. At its core, a language model assigns a probability distribution over the next token given all previous tokens:

$$p(x_t \mid x_1, x_2, \ldots, x_{t-1}) \,. \tag{7.15}$$

This is precisely the conditional distribution that defines a Markov chain, but without the Markov assumption: the full history, not just the most recent state, can influence the prediction. The transformer models this conditional distribution directly, using self-attention to selectively weight the influence of each past token on the current prediction. The Markov property is abandoned as a modeling assumption—the model is permitted to depend on the entire history—but the probabilistic framework of sequences and conditional distributions, established by Markov more than a century earlier, remains entirely intact.

The self-attention mechanism computes, for each position t in the sequence, a weighted sum of representations of all other positions. The weights are determined by the compatibility between the current position and each other position, measured by a dot product in a learned representation space. Formally, given matrices of *queries* Q, *keys* K, and *values* V derived from the input by linear projections, the attention output is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \ , \tag{7.16}$$

where d_k is the dimension of the key vectors. The softmax ensures that the attention weights for each position sum to one—they form a probability distribution over positions—and the $\sqrt{d_k}$ factor prevents the dot products from growing too large in high dimensions. The entire operation is a weighted average over representations, where the weights are themselves learned and data-dependent. The representation of states as vectors and transitions as matrices, foreshadowed in the discussion of Markov chains, is here realized in its most general learned form.

Example: language modeling as a Markov chain, and beyond.

Consider a large language model generating text token by token. At each step, the model maintains, through its transformer layers, a representation of the entire preceding context, and uses this to sample the next token from the conditional distribution $p(x_t \mid x_1, \ldots, x_{t-1})$. The resulting sequence of tokens is the trajectory of a *non-Markovian* stochastic process—each step depends on the full history—but the sequential, probabilistic structure, and the idea of a system transitioning through states according to learned probabilities, is recognizably Markovian in spirit. At a sufficient level of abstraction, a language model is a learned Markov chain with an astronomically large state space, where the state encodes not just the current token but a compressed representation of the entire context.

Example: diffusion models.

A further connection between modern machine learning and Markov chains arises in *diffusion models*, which have become the leading approach to image and audio generation. The forward process of a diffusion model is a discrete-time Markov chain that gradually corrupts a data point x_0 by adding Gaussian noise at each step, producing a sequence $x_0, x_1, \ldots, x_T$ where x_T is approximately pure noise. The generative model learns to reverse this chain: starting from noise and applying a learned denoising step at each stage, it recovers a sample from the data distribution. The mathematics of this reversal is precisely the theory of time-reversed Markov chains, and the training objective—predicting the noise added at each step—can be derived from variational inference and the theory of score functions. The Markovian structure is not a simplifying assumption adopted for tractability, but a deliberate architectural choice that makes the training problem tractable while retaining expressive generative power [105].

The arc traced in this chapter spans more than a century, from a theological dispute in late Imperial Russia to the neural networks that process billions of documents daily. What is perhaps most striking about this arc is not the sophistication of the destination, but the continuity of the ideas throughout. The Markov property, the compression of history into a sufficient present state, the transition matrix, states as vectors, dynamics as linear maps, the convergence of time averages to ensemble averages, etc, all developed in the first decade of the 20^{th} century for the analysis of poetry, regarded today as structural principles in the most advanced systems of the 21^{st}.

Appendix A

A review of linear algebra

Linear algebra is one of the central languages of modern mathematics and science. It provides the formal framework for describing linear relations, symmetry and structure, and it underlies vast areas of analysis, probability, statistics, physics, and computer sciences. Historically, linear algebra did not arise as a single theory, but rather as a collection of methods developed to solve systems of linear equations, study geometry, and understand transformations of space. Only in the late 19^{th} and early 20^{th} centuries was it unified into the abstract theory now taught under the name *linear algebra* [78, 106].

From a philosophical perspective, linear algebra marks a transition from concrete computation to structural thinking. Instead of focusing on individual equations or numerical solutions, algebra emphasizes abstract mathematical objects such as vector spaces, mappings between them, and invariant properties under changes of coordinates. This abstraction allows the same mathematical ideas to apply equally to geometry, differential equations, probability models, and statistical inference.

Historically, the development of linear algebra is associated with several key figures and traditions spanning many centuries. Gauss laid the foundations for the systematic solution of linear systems through elimination methods [15], while Hermann Grassmann introduced abstract vector spaces in his *"Ausdehnungslehre (The Theory of Extension)"* (1844), a work far ahead of its time [107]. Arthur Cayley and James Joseph Sylvester developed matrix theory in the 19^{th} century [108], and David Hilbert and Emmy Noether later contributed decisively to the structural and axiomatic understanding of linear spaces and linear operators.

An often underemphasized contribution comes from the medieval Arabic mathematical tradition, which played a crucial role in the development of symbolic and algorithmic methods for solving equations. In particular, Muhammad ibn Musa al-Khwarizmi, working in 9^{th}-century Baghdad, during the islamic golden era, presented systematic procedures for solving linear and quadratic equations in his treatise *"Al-Kitab al-Mukhtasar fi Hisab al-Jabr wa-l-Muqabala (Compendious Book on Calculation)"* [109]. In the attempt of translating the works of classic Greek philosophers about geometry, and using for that already developed tools from indian atirhmetic, modern algebra started to shape.

Although formulated rhetorically rather than using modern symbolic notation, this work introduced rule-based manipulation of equations and classification of solution types. These methods strongly influenced later European mathematics through Latin translations [110] and helped establish algebra as a distinct discipline. The term *algebra* itself derives from the word *al-jabr* in the title of al-Khwarizmi's treatise, reflecting the historical continuity between equation solving, algorithmic reasoning, and the later emergence of linear algebra as a formal theory.

A solid background in linear algebra typically includes the following core topics, which are either required or strongly recommended for further study in probability, statistics, and applied mathematics:

Vectors and vector spaces:
Vectors are elements of a vector space over a field (typically $\mathbb{R}$ or $\mathbb{C}$), equipped with addition and scalar multiplication. Linear combinations, span, linear independence, bases, and dimension formalize the idea of degrees of freedom and coordinate representations. Formally, a vector space V satisfies

$$\alpha\mathbf{v} + \beta\mathbf{w} \in V \quad \text{for all } \mathbf{v}, \mathbf{w} \in V \text{ and } \alpha, \beta \in \mathbb{F}.$$

Matrices and systems of linear equations:
Matrices arise naturally as representations of linear maps between finite-dimensional vector spaces. Central topics include matrix operations, rank, inversion, row reduction, and the solution of linear systems. A system of linear equations can be written compactly as

$$A\mathbf{x} = \mathbf{b},$$

where A is a matrix, $\mathbf{x}$ the vector of unknowns, and $\mathbf{b}$ the observation or constraint vector.

Linear transformations:
Linear transformations are functions between vector spaces that preserve addition and scalar multiplication. For a linear map $T : V \to W$,

$$T(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha T(\mathbf{v}) + \beta T(\mathbf{w}).$$

Key concepts include kernel and image, injectivity and surjectivity, change of basis, and composition of linear maps. This viewpoint emphasizes structure over computation and clarifies the geometric meaning of matrices.

Inner products and geometry:
Inner product spaces introduce notions of length, angle, and orthogonality. An inner product, often written as $\langle \cdot, \cdot \rangle$, induces a norm via

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Important concepts include orthogonal projections, orthonormal bases, and the Gram–Schmidt process. These ideas are foundational for least squares methods, regression, and statistical estimation.

Eigenvalues and eigenvectors:
Eigenvalues describe intrinsic directions and scaling properties of linear transformations. A scalar λ is an eigenvalue of a matrix A if there exists a nonzero vector $\mathbf{v}$ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Diagonalization and spectral decomposition, especially for symmetric matrices, play a central role in applications ranging from differential equations to principal component analysis.

Determinants and volume:
Although less central in modern abstract treatments, determinants provide geometric insight into volume, orientation, and invertibility. For a square matrix A,

$$\det(A) \neq 0 \quad \Longleftrightarrow \quad A \text{ is invertible.}$$

Determinants also appear in change-of-variables formulas and multivariate analysis.

In modern mathematics, linear algebra serves both as a computational toolkit and as a conceptual foundation. Mastery of its basic structures is essential not only for solving concrete problems, but also for understanding more advanced theories where linear spaces provide local or approximate descriptions of complex phenomena. As such, linear algebra is best learned not merely as a collection of techniques, but as a coherent framework for reasoning about structure, symmetry, and linearity.

Appendix B

A review of functions and derivatives

Calculus begins with the study of how quantities depend on one another and how they change. At its foundation lies the concept of a *function*, which formalizes the idea that one quantity is determined by another. Functions provide the language through which variation, motion, and growth are described in mathematics, physics, and the natural sciences.

Historically, the notion of a function evolved gradually. Early uses appear implicitly in the work of Rene Descartes, who introduced coordinate geometry and expressed curves through algebraic equations [37]. The explicit concept of a function as a mapping between quantities was later clarified in the 18^{th} century by Leonhard Euler, whose writings established much of the notation and terminology still in use today [111].

A decisive step toward modern calculus was taken in the 19^{th} century by Augustin-Louis Cauchy, who placed limits and derivatives on a rigorous analytical foundation. Cauchy replaced intuitive notions with rigurous and precise definitions based on limiting processes, thereby clarifying continuity, convergence, and differentiation [74]. This shift marked the transition from classical calculus to modern mathematical analysis.

Earlier traditions also contributed important ideas related to change and approximation. In the Indian mathematical tradition, members of the Kerala school (14^{th} to 16^{th} centuries, while the Copernican revolution happened in modern Europe) developed infinite series expansions for trigonometric functions and sophisticated approximation techniques motivated by astronomical computation [112]. In the medieval Arabic world, mathematicians refined algebraic methods, studied polynomial behavior, and developed geometric approaches to tangents and maxima, providing conceptual tools later absorbed into European mathematics through translation and transmission [113].

From a philosophical standpoint, calculus represents an effort to make sense of continuous change using finite reasoning. The introduction of limits resolved long-standing paradoxes about infinity and infinitesimals by replacing informal and intuitive arguments with precise definitions. Modern calculus, as taught today, builds on this foundation by emphasizing clarity, rigor, and conceptual understanding rather than purely mechanical computation [114].

A solid background in differential calculus typically includes the following core topics, which are either required or strongly recommended for further study in probability, statistics, and applied mathematics:

Functions:

The development of differential calculus is closely associated with Isaac Newton and Gottfried Wilhelm Leibniz, who independently formulated its fundamental principles in the late 17^{th} century [72, 73]. Newton emphasized motion and physical interpretation, while Leibniz introduced a symbolic notation that proved especially flexible and influential.

Informally, a function assigns to each input value exactly one output value. Formally, a function f from a set A to a set B is written as

$$f : A \to B, \qquad x \mapsto f(x).$$

This abstract definition allows the same concept to describe physical laws, statistical models, and mathematical relationships.

A central idea in calculus is that of a *limit*. Limits capture the behavior of a function as its input approaches a given value, even if the function is not defined or not well behaved at that point. Informally, limits allow us to reason about processes that involve approaching, rather than reaching, a value. This concept is essential for making precise sense of continuity, instantaneous change, and accumulation.

Limits:

Historically, the idea underlying limits emerged from attempts to reason about motion, change, and infinite divisibility. Ancient Greek philosophers, most famously in Zeno's paradoxes, highlighted conceptual difficulties surrounding infinite processes and the passage to a limit. Early mathematicians addressed these issues using geometric intuition rather than formal definitions, as seen in the method of exhaustion developed by Archimedes.

A decisive conceptual shift occurred in the nineteenth century, particularly through the work of Augustin-Louis Cauchy. Cauchy replaced infinitesimal arguments with definitions based on limiting processes, making explicit the dependence of approximation on arbitrarily small errors [74]. This transition laid the groundwork for the modern, rigorous treatment of limits and continuity, later refined through precise definitions, commonly referred to as the ε–δ definitions.

From a philosophical perspective, the concept of a limit provides a way to reason about idealized processes without requiring that they ever be completed. It allows mathematics to describe continuous change using finite statements, reconciling infinite processes with logical precision. Limits thus form the conceptual bridge between discrete approximation and continuous behavior, underpinning the modern formulation of calculus [114].

Given a function f, we say that the limit of $f(x)$ as x approaches a is L if

$$\lim_{x \to a} f(x) = L.$$

This notation expresses that the values of $f(x)$ can be made arbitrarily close to L by choosing x sufficiently close to a, regardless of the value of $f(a)$ itself.

Derivatives:

The derivative arises from the study of limits and provides a precise definition of instantaneous rate of change. Geometrically, the derivative of a function at a point corresponds to the slope of the tangent line at that point. Physically, it describes quantities such as velocity or growth rate.

The derivative of a function f at a point x is defined as the limit

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists. This definition links algebraic computation with geometric intuition.

Appendix C

A review of integral calculus

Integral calculus is concerned with accumulation and total change. It provides a way to answer questions such as how much area lies under a curve, how far an object travels when its speed changes over time, or how a quantity distributed across a range adds up to a whole. Whereas differential calculus focuses on how quantities change at a single point, integral calculus asks how these small changes combine over an interval. Together, differentiation and integration offer a unified way to reason about continuous change.

At an intuitive level, the basic problem of integration is simple: when something varies from place to place or from moment to moment, how can we compute its total effect? Long before modern calculus, mathematicians confronted this problem in concrete settings such as measuring land, volumes, or the paths of moving objects. In ancient Greece, Archimedes developed the method of exhaustion, in which areas and volumes were approximated by adding up many small pieces. Although expressed geometrically rather than algebraically, this approach captured the core idea behind integration: a total quantity can be understood as the limit of increasingly accurate approximations [115].

From a philosophical standpoint, integral calculus addresses the challenge of understanding the whole as arising from infinitely many parts. By replacing informal geometric intuition with limit-based reasoning, it provides a reliable framework for reasoning about continuous quantities. At an introductory level, integral calculus offers both practical tools and conceptual insight into how local behavior combines to produce global effects [114].

Earlier mathematical traditions also contributed important ideas related to integration. In the Indian mathematical tradition, the Kerala school developed infinite series and summation techniques for computing areas and trigonometric quantities, motivated by astronomical problems [112]. In the medieval Arabic world, mathematicians refined methods for computing areas and volumes and developed geometric techniques closely related to integral reasoning, which later entered Europe through translation movements [116].

The systematic development of integral calculus is closely associated with Isaac Newton and Gottfried Wilhelm Leibniz, who independently formulated its fundamental principles in the late 17^{th} century [72, 117]. Leibniz introduced the integral symbol $\int$, motivated by the idea of summation, a notation that remains standard today.

As with differential calculus, rigor was substantially strengthened in the 19^{th} century. Augustin-Louis Cauchy clarified the role of limits and convergence in integration, while Bernhard Riemann introduced a precise definition of the integral based on partitions of intervals, providing a foundation suitable for modern analysis [74, 118].

A solid background in integral calculus typically includes the following core topics, which are either required or strongly recommended for further study in probability, statistics, and applied mathematics:

Accumulation and integrals:

In modern calculus, integration formalizes this idea of accumulation. Suppose a quantity changes continuously and its local contribution at position x is described by a function $f(x)$. If we divide an interval $[a, b]$ into small pieces of width Δx, the total effect can be approximated by adding up the contributions from each piece,

$$\sum f(x_i)\,\Delta x.$$

As the pieces are made smaller and more numerous, this approximation becomes more accurate. The definite integral represents the idealized value obtained in the limit,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x.$$

Rather than thinking of this formula technically, it can be read as a precise way of saying: *the whole is obtained by adding up many very small parts.*

A fundamental insight of calculus is that accumulation and change are closely connected. The *Fundamental Theorem of Calculus* shows that the total accumulated effect over an interval can be computed using an antiderivative, linking integration directly to differentiation. In practical terms, this means that knowing how a quantity changes locally allows us to determine its overall impact.

This viewpoint is especially important in probability and statistics. When a function represents a *probability density*, integration provides the total probability over a range of values. For example, the probability that a continuous random variable lies between a and b is obtained by integrating its density over that interval. In this way, integrals translate local likelihoods into global probabilities, making integral calculus a foundational tool for statistical reasoning and inference.

Fundamental Theorem of Calculus:

If f is continuous on $[a, b]$ and F is an antiderivative of f, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Earlier mathematical traditions also contributed important ideas related to integration. In the Indian mathematical tradition, the Kerala school developed infinite series and summation techniques for computing areas and trigonometric quantities, motivated by astronomical problems [112]. In the medieval Arabic world, mathematicians refined methods for computing areas and volumes and developed geometric techniques closely related to integral reasoning, which later entered Europe through translation movements [116].

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Appendix D

Some special functions

In mathematics and statistics, certain functions appear so frequently and play such a foundational role that they are regarded as *special functions*. Rather than being introduced for their own sake, these functions typically emerge from attempts to solve concrete problems involving error, accumulation, normalization, or symmetry. In probability and statistical inference, they often serve as canonical models or normalizing constants.

The Gaussian distribution:

The Gaussian (or normal) distribution originated in practical problems of measurement and observation. When astronomers and physicists attempted to combine repeated measurements of the same quantity, they observed that small errors were common while large deviations were rare, and that positive and negative errors tended to balance symmetrically. In the early nineteenth century, Carl Friedrich Gauss showed that assuming such errors followed a specific bell-shaped curve led to mathematically optimal estimation procedures under reasonable symmetry assumptions [15]. Closely related ideas were also developed independently by Pierre-Simon Laplace in his analytic theory of probability [16].

In modern probability theory, the Gaussian distribution occupies a central position because of its stability under aggregation. The Central Limit Theorem explains that when many small, independent effects are added together, their combined outcome tends to follow a Gaussian distribution, regardless of the details of the individual components [23]. This explains why Gaussian models arise so widely in natural and social phenomena.

A real-valued random variable X is said to follow a Gaussian distribution with mean μ and variance $\sigma^2 > 0$ if its probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The exponential distribution:

The exponential distribution arises from the study of waiting times and the occurrence of events over time. Historically, it is closely connected to early work on stochastic processes and the modeling of random events, particularly in contexts where events occur continuously and independently, such as radioactive decay or arrivals to a queue. In the nineteenth century, Siméon-Denis Poisson's work on rare events laid the groundwork for such models, while later developments by Andrey Markov clarified the role of memoryless processes [46, 21].

Intuitively, the exponential distribution describes situations in which the probability of an event occurring in the next instant does not depend on how long one has already waited. This *memoryless* property distinguishes it from most other continuous distributions and makes it a natural model for lifetimes, waiting times, and durations until the next random event.

A nonnegative random variable X is said to follow an exponential distribution with rate parameter $\lambda > 0$ if its probability density function is

$$f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

In probability and statistics, the exponential distribution plays a foundational role as the continuous analogue of the geometric distribution and as a building block for more complex stochastic models, including Poisson processes and reliability theory [3].

The Euler Gamma function:

The Euler Gamma function arose from a simple but far-reaching question: how can the factorial function be extended beyond whole numbers? While the factorial $n!$ counts arrangements for positive integers, many problems in analysis and probability require a smooth function that behaves like a factorial even when its argument is not an integer. Leonhard Euler addressed this problem in the eighteenth century by introducing what is now called the Gamma function, providing a continuous extension of the factorial concept.

It arises in the evaluation of improper integrals, solutions of differential equations, and complex analysis through its analytic continuation and functional equation. In probability theory, the Gamma function appears systematically as a normalization factor ensuring that probability densities integrate to one

For $x > 0$, the Gamma function is defined by the improper integral

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} \, dt.$$

For positive integers n, it satisfies $\Gamma(n) = (n-1)!$. In probability theory, the Gamma function appears naturally in the normalization of continuous distributions and in the study of waiting times and scale parameters [3].

The Euler Beta function:

The Euler Beta function is closely related to the Gamma function and arose historically from the study of definite integrals involving powers of variables constrained to a finite interval. Euler showed that many such integrals could be expressed compactly using a special two-parameter function, now known as the Beta function.

The Beta function arises naturally when integrating products of powers over bounded intervals, making it a fundamental object in integral calculus and special-function theory. Through its relationship with the Gamma function, it encodes how probability mass is distributed between competing components constrained to sum to one. In probability theory, it serves as the normalizing constant for distributions on the unit interval and simplex, playing a key role in Bayesian modeling of proportions and probabilities.

For $x > 0$ and $y > 0$, the Beta function is defined by

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} \, dt.$$

A key identity links the Beta and Gamma functions:

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

In probability and statistics, the Beta function appears as the normalizing constant of the Beta distribution, which is widely used to model proportions, probabilities, and prior beliefs in Bayesian inference [45]. Together, the Gamma and Beta functions illustrate how integral calculus underlies the mathematical structure of many statistical models.

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